
Supplementary information

Martini 3: a general purpose force field for coarse-grained molecular dynamics

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Supporting information for: Martini 3: A General Purpose Force Field for Coarse-Grained Molecular Dynamics

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Supplementary Notes A) Description of beads and Martini 3 model

A1) Blocks, chemical types and sizes

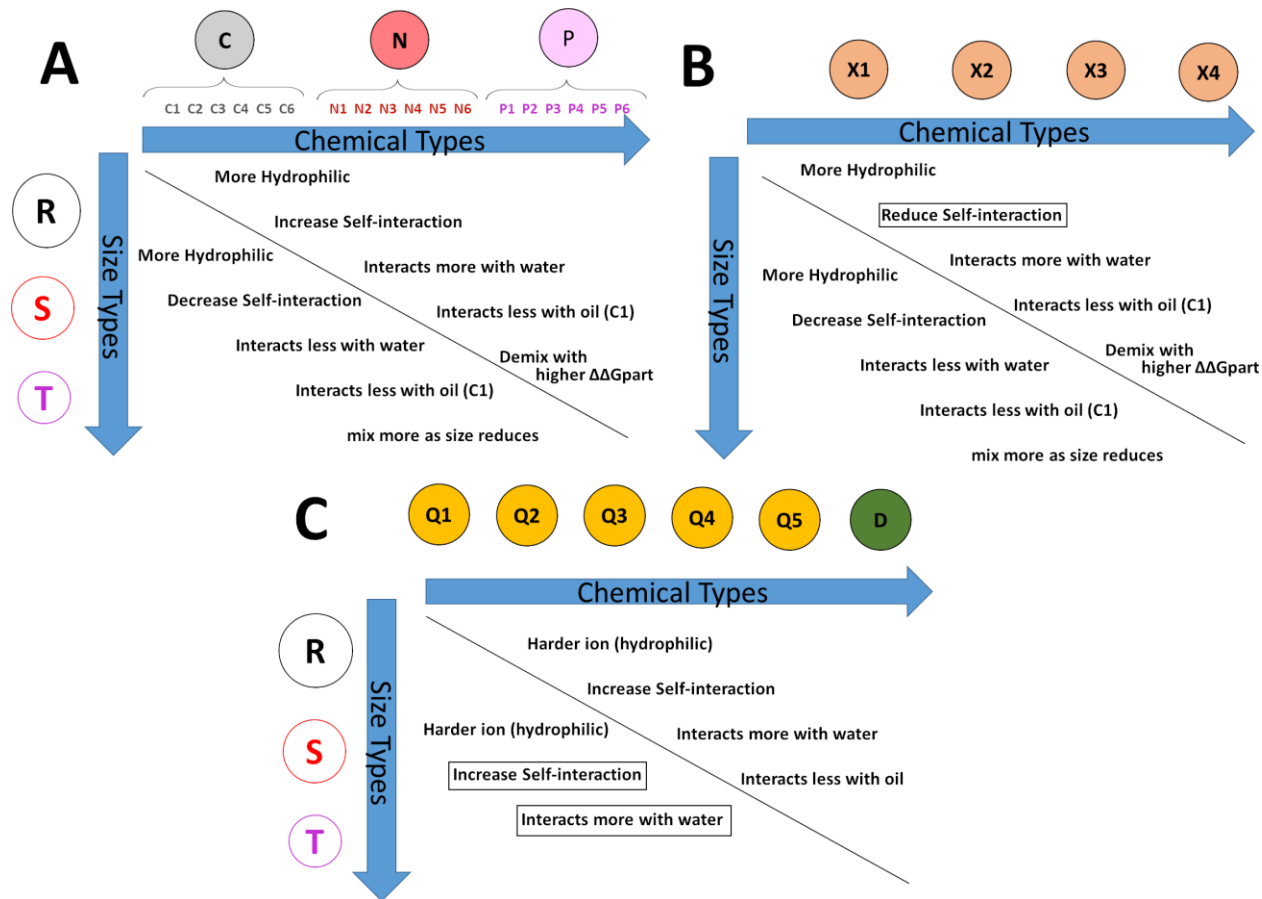
Martini 3 is composed of seven chemical bead types: polar (P), intermediate/non-polar (N), apolar (C), halo-compounds (X), monovalent ions (Q), divalent ions (D) and water (W). All of these bead types come in three different sizes: regular (R-bead, assumed default), small (S-bead, denoted with prefix S), and tiny (T-bead, denoted with prefix T). X, D, and W are completely new chemical types that did not exist in the previous version. Except for W and D beads, all beads have subtypes that are distinguished by a number indicating the relative degree of polarity (from 1, low polarity, to a maximum of 6, high polarity). These subtypes allow for a more accurate representation of the chemical nature of the underlying atomistic structure. Other chemically-specific subtypes are described in **section A4**. In comparison with the previous version, N and Q beads underwent the largest expansion, to increase coverage of the chemical space. From the oil-water partition perspective, N-beads involve most of the compounds¹, which naturally demands for more options to allow for the development of accurate models.

The chemical bead types are separated in blocks, each with independent sets of parameters. P-, N-, C-, and X-beads are part of the organic block while Q- and D-beads are part of the ion block. For now, W is the only bead that is part of the solvent block. Each block was parametrized to follow certain trends that depend on chemical types and sizes. Here is a short description of the properties of the organic and the ion blocks. For a more elaborate description of the parametrization strategy, including the solvent block, see **sections B2-B4**. **Supplementary Figure 1** summarizes the key trends of chemical types and sizes while quantitative data about the beads is presented in **sections A5-A6**.

Organic block: the self-interaction and the interaction with water beads are gradually reduced, as the beads of the same size are more hydrophobic ($P > N > C$, see **Supplementary Figure 1A**). As the difference in hydrophobicity increases, the beads also tend to form heterogeneous biphasic systems. In organic molecules, an important exception for these trends are halo-compounds (**Supplementary Figure 1B**). In their case, as the molecules are more hydrophobic, the self-interaction actually increases. This is the reason why these compounds have a dedicated chemical type (X-beads) in Martini

3. The trends of the organic block are also different regarding size compared to the previous version of Martini. Smaller beads represent organic fragments with fewer carbon atoms, which reduces their self-interaction and their interaction with the environment but makes them more hydrophilic².

Ion block: the key difference for the trends involving ions in Martini are related with changes in properties of different sizes (see **Supplementary Figure 1C**). Contrary to the organic block, smaller Q/D beads have stronger interactions due to stronger electrostatic interactions. At the same time, the smaller sizes increase their interactions with the environment, in particular with water and polar solvents, which is also opposite compared to the organic block.



Supplementary Figure 1: Summary of trends in terms of different chemical bead types: (A) organic block, except the X-beads; (B) X-beads, which is a new chemical class, dedicated to halo-compounds; (C) ion block. Main differences with respect to the organic block are highlighted.

In the next section, the bead pair interactions, defined by the interaction levels and interaction matrix, are presented. See **chapter B** for a detailed discussion about bonded and non-bonded potentials and parametrization strategy of the pair interactions.

A2) Interaction matrix

The interaction matrix (**Supplementary Table 1**) is the core of the Martini force field, with all the assumptions for the behavior of the different chemical types encoded in the choice of the interaction levels. As in Martini 2, a limited number of ϵ_{ij} and σ_{ij} values is used; such values are classified as different levels of attractive, intermediate, and repulsive interactions. In fact, all interactions are attractive, and this naming is only used to indicate the net effect of the interaction balance between the pair of beads. The number of the levels is indicative of these trends, with lower numbers representing higher ϵ_{ij} values.

More levels are now defined in Martini 3, which allows a smoother transition between the different chemical bead types. For the organic and ion-organic block interactions, a total of 22 levels are defined now, while Martini 2 had only 10 levels for these interactions. In addition, W, D, and Q beads have their own set of levels in Martini 3. Although different size combinations have different definitions for the interaction levels, they all follow the same trend defined by the interaction matrix, which is an assumption of the model. The interaction matrix is symmetric regarding the default LJ parameters used. However, because of the p-factor approach (see **section B3**), the interactions between ions and neutral molecules are not symmetric for cross-interactions of different sizes.

Supplementary Table 1: Martini 3 interaction matrix. Colors indicate blocks with different sets of interaction levels.

Blocks with different balance when we reduce the bead size:

organic

ion

water

ion-others

	D	Q5	Q4	Q3	Q2	Q1	P6	P5	P4	P3	P2	P1	N6	N5	N4	N3	N2	N1	C6	C5	C4	C3	C2	C1	X4	X3	X2	X1	W	
D	0	0	1	2	3	4	1	2	3	3	3	4	4	5	5	6	7	12	16	18	19	20	21	21	15	16	17	18	0	D
Q5	0	0	1	3	8	11	2	3	3	4	4	5	5	6	6	7	7	8	13	17	18	19	21	21	14	15	16	17	1	Q5
Q4	1	1	2	4	8	11	3	4	4	4	5	6	6	6	7	8	8	8	12	15	16	17	20	20	13	14	15	16	2	Q4
Q3	2	3	4	4	7	11	5	5	5	5	6	7	7	7	7	8	8	8	12	13	14	15	19	19	12	13	14	15	3	Q3
Q2	3	8	8	7	7	10	6	5	5	6	6	7	7	7	7	8	8	8	10	12	13	14	19	19	11	12	13	14	4	Q2
Q1	4	11	11	11	10	10	7	6	6	6	6	6	6	7	7	7	8	9	11	12	13	18	18	11	11	12	13	5	Q1	
P6	1	2	3	5	6	7	3	4	5	6	6	7	8	10	11	12	13	14	14	15	16	17	17	19	13	13	14	15	4	P6
P5	2	3	4	5	5	6	4	5	6	7	7	8	9	10	10	12	13	14	14	14	15	16	16	17	12	13	13	15	4	P5
P4	3	3	4	5	5	6	5	6	6	7	7	8	9	10	10	11	12	13	13	14	15	16	16	17	12	13	13	15	5	P4
P3	3	4	4	5	6	6	6	7	7	7	7	8	9	10	10	11	11	12	12	13	14	16	16	17	12	12	13	14	6	P3
P2	3	4	5	6	6	6	6	7	7	7	7	8	8	9	10	11	11	11	12	13	14	15	15	16	12	12	13	14	6	P2
P1	4	5	6	7	7	6	7	8	8	8	8	8	8	9	9	11	11	11	12	13	14	15	15	16	12	12	13	14	7	P1
N6	4	5	6	7	7	6	8	9	9	9	8	8	8	8	8	10	11	11	11	12	13	14	14	15	11	12	13	13	7	N6
N5	5	6	6	7	7	7	10	10	10	10	9	9	8	9	9	10	10	11	11	12	13	13	14	15	10	11	12	13	8	N5
N4	5	6	7	7	7	7	11	10	10	10	10	9	8	9	9	10	10	11	11	12	12	13	14	15	9	10	12	13	9	N4
N3	6	7	8	8	8	7	12	12	11	11	11	11	10	10	10	10	10	11	11	12	12	13	13	14	10	11	12	12	9	N3
N2	7	7	8	8	8	8	13	13	12	11	11	11	11	10	10	10	10	11	11	12	12	12	13	14	10	11	12	12	11	N2
N1	12	8	8	8	8	9	14	14	13	12	11	11	11	11	11	11	11	10	10	11	11	12	13	14	11	11	12	12	12	N1
C6	16	13	12	12	10	9	15	14	13	12	12	12	11	11	11	11	11	10	10	11	11	12	13	14	11	11	11	11	13	C6
C5	18	17	15	13	12	11	16	14	14	13	13	13	12	12	12	12	12	11	11	11	11	12	13	13	12	12	12	11	14	C5
C4	19	18	16	14	13	12	16	15	15	14	14	14	13	13	13	12	12	12	11	11	11	11	12	13	12	12	12	11	15	C4
C3	20	19	17	15	14	13	17	16	16	16	15	15	14	13	13	13	13	12	12	12	12	11	11	11	12	12	12	11	15	C3
C2	21	21	20	19	19	18	17	16	16	16	15	15	14	14	14	13	13	13	13	13	12	11	11	10	14	12	12	12	16	C2
C1	21	21	20	19	19	18	19	17	17	17	16	16	15	15	15	14	14	14	14	13	13	11	10	11	14	12	12	12	18	C1
X4	15	14	13	12	11	11	13	12	12	12	12	12	11	10	9	10	10	11	11	12	12	12	14	14	11	11	11	11	13	X4
X3	16	15	14	13	12	11	14	13	13	12	12	12	11	10	11	11	11	11	11	12	12	12	12	12	11	11	10	10	14	X3
X2	17	16	15	14	13	12	15	14	13	13	13	13	12	12	12	12	12	11	12	12	12	12	12	12	11	10	10	10	14	X2
X1	18	17	16	15	14	13	15	15	15	14	14	14	13	13	13	12	12	12	11	11	11	11	12	12	11	10	10	9	16	X1
W	0	1	2	3	4	5	4	4	5	6	6	7	7	8	9	9	11	12	13	14	15	15	16	18	13	14	14	16	0	W
D		Q5	Q4	Q3	Q2	Q1	P6	P5	P4	P3	P2	P1	N6	N5	N4	N3	N2	N1	C6	C5	C4	C3	C2	C1	X4	X3	X2	X1	W	

levels	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
	hyper attractive	super attractive	attractive	almost attractive	semi attractive	intermediate	almost intermediate	semi repulsive	almost repulsive	repulsive	super repulsive											

A3) Interaction levels

All final parameters used in Martini 3 are described in the **Supplementary Tables 2 to 10** below. Units of σ_{ij} and ε_{ij} are in nm and kJ/mol, respectively. Each block and size combination has a specific set of parameters. Interactions between neutral and charge-beads make use of the parameters of the organic block combined with the p-factor approach, which is described in **section B3**.

Supplementary Table 2: Size and interaction levels for the organic block: $\sigma(\text{O-O})$ and $\epsilon(\text{O-O})$

level	R-R		R-S		S-S		R-T		S-T		T-T	
	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ
0	0.470	5.790	0.430	5.510	0.410	5.130	0.395	5.270	0.365	4.860	0.340	4.300
1	0.470	5.490	0.430	5.210	0.410	4.840	0.395	4.870	0.365	4.510	0.340	4.020
2	0.470	5.240	0.430	4.920	0.410	4.550	0.395	4.540	0.365	4.220	0.340	3.750
3	0.470	4.990	0.430	4.680	0.410	4.290	0.395	4.230	0.365	3.930	0.340	3.470
4	0.470	4.730	0.430	4.400	0.410	4.040	0.395	3.910	0.365	3.640	0.340	3.220
5	0.470	4.480	0.430	4.170	0.410	3.820	0.395	3.670	0.365	3.370	0.340	2.990
6	0.470	4.250	0.430	3.970	0.410	3.550	0.395	3.430	0.365	3.100	0.340	2.740
7	0.470	4.060	0.430	3.770	0.410	3.310	0.395	3.190	0.365	2.830	0.340	2.500
8	0.470	3.880	0.430	3.590	0.410	3.070	0.395	2.940	0.365	2.570	0.340	2.280
9	0.470	3.690	0.430	3.380	0.410	2.840	0.395	2.700	0.365	2.340	0.340	2.020
10	0.470	3.520	0.430	3.160	0.410	2.600	0.395	2.500	0.365	2.110	0.340	1.770
11	0.470	3.390	0.430	2.920	0.410	2.350	0.395	2.310	0.365	1.910	0.340	1.510
12	0.470	3.240	0.430	2.770	0.410	2.200	0.395	2.110	0.365	1.750	0.340	1.230
13	0.470	3.070	0.430	2.530	0.410	2.040	0.395	1.850	0.366	1.600	0.352	1.050
14	0.470	2.960	0.430	2.320	0.410	1.910	0.395	1.670	0.378	1.450	0.366	0.935
15	0.470	2.790	0.430	2.160	0.410	1.750	0.398	1.450	0.394	1.290	0.385	0.855
16	0.470	2.620	0.430	1.930	0.414	1.560	0.399	1.180	0.404	1.130	0.411	0.801
17	0.470	2.440	0.430	1.730	0.418	1.400	0.401	0.983	0.409	1.080	0.424	0.776
18	0.485	2.240	0.443	1.480	0.421	1.250	0.404	0.830	0.423	0.981	0.438	0.731
19	0.520	2.120	0.473	1.280	0.453	1.150	0.465	0.703	0.484	0.890	0.505	0.680
20	0.570	2.000	0.514	1.110	0.496	1.050	0.511	0.659	0.537	0.808	0.561	0.630
21	0.620	1.960	0.552	1.010	0.535	0.986	0.524	0.629	0.588	0.763	0.599	0.620

Supplementary Table 3: Size and interaction levels for the solvent block: $\sigma(\text{W-W})$ and $\epsilon(\text{W-W})$

level	W-W		W-SW		W-TW		SW-SW		SW-TW		TW-TW	
	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ
-	0.470	0.465	0.440	0.375	0.410	0.340	0.410	0.320	0.380	0.295	0.350	0.280

Supplementary Table 4: Size and interaction levels for the regular water and organic block: $\sigma(\text{W-O})$ and $\epsilon(\text{W-O})$

level	W-R		W-S		W-T		level	W-R		W-S		W-T	
	σ	ϵ	σ	ϵ	σ	ϵ		σ	ϵ	σ	ϵ	σ	ϵ
0	0.465	6.090	0.425	5.750	0.385	5.710	10	0.465	3.500	0.425	3.000	0.385	2.510
1	0.465	5.640	0.425	5.370	0.385	5.540	11	0.465	3.290	0.425	2.740	0.385	2.180
2	0.465	5.260	0.425	5.090	0.385	5.180	12	0.465	3.040	0.425	2.510	0.385	1.860
3	0.465	4.990	0.425	4.810	0.385	4.860	13	0.465	2.780	0.425	2.260	0.385	1.550
4	0.465	4.750	0.425	4.530	0.385	4.630	14	0.465	2.560	0.425	2.020	0.387	1.360
5	0.465	4.520	0.425	4.240	0.385	4.300	15	0.465	2.420	0.425	1.800	0.393	1.120
6	0.465	4.330	0.425	4.030	0.385	3.730	16	0.465	2.300	0.425	1.690	0.395	0.920
7	0.465	4.140	0.425	3.830	0.385	3.420	17	0.465	2.180	0.431	1.560	0.401	0.750
8	0.465	3.950	0.425	3.600	0.385	3.100	18	0.470	2.060	0.437	1.420	0.415	0.550
9	0.465	3.700	0.425	3.270	0.385	2.780							

Supplementary Table 5: Size and interaction levels for the small water and organic block:
 $\sigma(\text{SW-O})$ and $\epsilon(\text{SW-O})$

level	SW-R		SW-S		SW-T		level	SW-R		SW-S		SW-T	
	σ	ϵ	σ	ϵ	σ	ϵ		σ	ϵ	σ	ϵ	σ	ϵ
0	0.435	4.910	0.405	4.840	0.375	5.040	10	0.435	2.670	0.405	2.340	0.375	1.870
1	0.435	4.650	0.405	4.550	0.375	4.670	11	0.435	2.480	0.405	2.120	0.375	1.620
2	0.435	4.390	0.405	4.270	0.375	4.320	12	0.435	2.290	0.405	1.900	0.375	1.370
3	0.435	4.130	0.405	4.000	0.375	3.960	13	0.435	2.100	0.405	1.680	0.375	1.130
4	0.435	3.880	0.405	3.730	0.375	3.650	14	0.435	1.920	0.405	1.460	0.375	0.950
5	0.435	3.640	0.405	3.460	0.375	3.340	15	0.435	1.740	0.405	1.280	0.388	0.830
6	0.435	3.440	0.405	3.220	0.375	3.030	16	0.435	1.570	0.405	1.100	0.395	0.710
7	0.435	3.240	0.405	3.000	0.375	2.720	17	0.435	1.410	0.414	0.930	0.408	0.610
8	0.435	3.050	0.405	2.780	0.375	2.430	18	0.440	1.270	0.423	0.800	0.432	0.550
9	0.435	2.860	0.405	2.560	0.375	2.150							

Supplementary Table 6: Size and interaction levels for the tiny water and organic block:
 $\sigma(\text{TW-O})$ and $\epsilon(\text{TW-O})$

level	TW-R		TW-S		TW-T		level	TW-R		TW-S		TW-T	
	σ	ϵ	σ	ϵ	σ	ϵ		σ	ϵ	σ	ϵ	σ	ϵ
0	0.395	4.290	0.378	4.310	0.345	4.460	10	0.395	2.350	0.378	2.150	0.345	1.910
1	0.395	4.060	0.378	4.000	0.345	4.170	11	0.395	2.190	0.378	1.960	0.351	1.680
2	0.395	3.830	0.378	3.730	0.345	3.880	12	0.395	2.030	0.378	1.780	0.351	1.470
3	0.395	3.610	0.378	3.480	0.345	3.590	13	0.395	1.880	0.378	1.610	0.353	1.330
4	0.395	3.410	0.378	3.290	0.345	3.300	14	0.395	1.730	0.378	1.440	0.355	1.230
5	0.395	3.210	0.378	3.100	0.345	3.060	15	0.395	1.590	0.378	1.290	0.372	1.130
6	0.395	3.020	0.378	2.910	0.345	2.840	16	0.395	1.450	0.386	1.160	0.384	1.030
7	0.395	2.840	0.378	2.720	0.345	2.600	17	0.395	1.330	0.390	1.030	0.410	0.930
8	0.395	2.670	0.378	2.530	0.345	2.370	18	0.400	1.240	0.397	0.920	0.440	0.830
9	0.395	2.510	0.378	2.340	0.345	2.140							

Supplementary Table 7: Size and interaction levels for monovalent ions: $\sigma(\text{Q-Q})$ and $\epsilon(\text{Q-Q})$

level	R-R		R-S		S-S		R-T		S-T		T-T	
	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ
0	0.470	6.450	0.438	6.020	0.410	4.620	0.405	6.620	0.373	3.600	0.354	1.180
1	0.470	6.200	0.438	5.770	0.410	4.320	0.405	6.360	0.373	3.400	0.357	1.080
2	0.470	5.950	0.438	5.550	0.410	4.080	0.405	6.130	0.373	3.200	0.361	0.950
3	0.470	5.700	0.438	5.350	0.410	3.850	0.405	5.930	0.373	3.000	0.362	0.900
4	0.470	5.450	0.438	5.170	0.410	3.690	0.405	5.730	0.373	2.750	0.364	0.850
5	0.470	5.200	0.438	4.940	0.410	3.510	0.405	5.500	0.373	2.500	0.366	0.818
6	0.470	4.950	0.438	4.710	0.410	3.310	0.405	5.280	0.373	2.220	0.366	0.786
7	0.470	4.700	0.438	4.530	0.410	3.100	0.405	5.080	0.375	1.900	0.371	0.748
8	0.470	4.450	0.438	4.250	0.410	2.800	0.405	4.880	0.377	1.500	0.373	0.725
9	0.470	4.200	0.438	4.020	0.410	2.520	0.405	4.680	0.378	1.350	0.380	0.650
10	0.470	3.980	0.438	3.720	0.411	2.240	0.405	4.400	0.384	1.300	0.383	0.525
11	0.470	3.790	0.438	3.320	0.413	1.960	0.405	4.150	0.385	1.250	0.385	0.500
12	0.470	3.600	0.438	3.020	0.414	1.670	0.405	3.950	0.394	1.200	0.393	0.450

Supplementary Table 8: Size and interaction levels for divalent ions: $\sigma(\text{D-D})$ and $\epsilon(\text{D-D})$

level	D-D		D-SD		D-TD		SD-SD		SD-TD		TD-TD	
	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ
-	0.500	4.090	0.471	6.560	0.430	10.000	0.475	1.980	0.455	2.400	0.438	1.100

Supplementary Table 9: Size and interaction levels between divalent and monovalent ions:
 $\sigma(D-Q)$ and $\epsilon(D-Q)$

level	R-R		R-S		R-T		S-S		S-T		T-T	
	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ	σ	ϵ
0	0.470	5.170	0.447	4.500	0.425	4.700	0.419	6.200	0.414	4.000	0.379	2.150
1	0.470	4.150	0.448	3.500	0.427	3.650	0.420	5.650	0.415	3.300	0.382	1.500
2	0.470	3.350	0.452	2.400	0.433	3.350	0.421	5.000	0.420	1.800	0.388	1.050
3	0.480	2.750	0.462	2.300	0.438	1.750	0.421	4.300	0.429	1.200	0.389	1.000
4	0.490	2.370	0.471	1.200	0.449	1.100	0.422	3.400	0.433	1.100	0.392	0.950

Supplementary Table 10: Asymmetric (size) levels between divalent and monovalent ions:
 $\sigma(D-Q)$ and $\epsilon(D-Q)$

level	S-R		T-R		T-S	
	σ	ϵ	σ	ϵ	σ	ϵ
0	0.430	11.040	0.402	15.850	0.380	9.050
1	0.430	9.670	0.402	15.100	0.380	8.420
2	0.431	8.850	0.402	14.600	0.380	7.950
3	0.437	8.300	0.402	13.750	0.380	7.250
4	0.438	7.650	0.403	13.200	0.380	6.650

A4) Labels

Specific subtypes (generically called “labels” here) are an easy way to fine-tune the CG models in Martini 3. The concept already existed in the previous versions but was only applied to N and Q types to take into account different hydrogen bond capabilities in molecules with a similar degree of hydrophobicity. In Martini 3, labels were expanded and can be applied to all beads, slightly changing their behaviors, which is translated into small changes in the interaction matrix. Three of these labels (hydrogen bonding, electron polarizability, positive/negative) are chemically specific, which means they are only applied to certain chemical types. The other two labels (self-interaction, partial charge) are more generic, and can be applied on top of all beads of the organic block, including beads which have already been modified with some of the chemically specific labels. A detailed description of each type of label is given below.

Supplementary Tables 11 to 16 indicate the modifications in the interaction matrix between beads containing labels. For instance, if the level of interaction between regular size P5 and P4 is 6, the interactions between P5d-P4d will use level 8 (*i.e.* the interaction will be reduced by +2 levels; see **Supplementary Table 11**). It should also be noted that the labels introduce changes that are size dependent. The idea behind the size dependent changes is to increase the specificity of the interactions as the resolution of the model is increased. Such behavior is essential to capture specific interactions in, for

instance, nucleobases and aedamers (see **Figures 1D and 2A**, respectively). Labels based on hydrogen bonding are better validated in Martini 3, as a lot of molecules and systems were tested using them. Other labels may need further testing and hence the rules for their application may need refinements. Together, all possible combinations of chemical types, sizes and labels lead to a total of 843 bead types which correspond to around 15 times more options than in the previous version of Martini.

Hydrogen bonding label (d/a):

Hydrogen donor (“d”) and acceptor (“a”) character can be added to all P and N chemical types of Martini 3. The hydrogen bonding capability of ions is covered by the *p/n* labels discussed later.

Supplementary Table 11: Hydrogen bonding labels – d/a				
Size pair		RR/RS	SS/ST/RT	TT
Label pair	d-d/a-a*	+2	+1	+1
	d-a	0	-1	-3
	W-d/W-a	+1	+1	+1

* new level cannot be higher than 12

This label reduces the self-interaction of beads with the same hydrogen bond capability and, for smaller bead sizes, increases the interactions between “d” and “a” (**Supplementary Table 11**). The interaction with water is also reduced by one level which affects oil-water partitioning: labeled beads are around 2-3 kJ/mol more hydrophobic than the corresponding non-labeled beads. The miscibility with other neutral organic beads is increased because of the reduced self-interaction. The miscibility with water is the same for regular size beads, while slightly reduced for smaller bead sizes.

Electron polarizability label (v/e):

Just like P and N chemical types, C and X chemical type can have preferential interactions, but induced by neighboring chemical groups.

Supplementary Table 12: Electron polarizability labels – v/e				
Size pair		RR/RS	SS/ST/RT	TT
Label pair	v-v/e-e*	+1	+1	+1
	e-v	-1	-2	-3
	e/v - others	-1	-1	-1

* new level cannot be higher than 14

Electron donor or acceptor character of aromatic π -electrons (that can be inspected with quantum chemically calculated electrostatic potential surfaces) are good examples which can induce preferential interactions. To mimic such effects, “v” (electron poor, vacancy) and “e” (electron richer) labels were created. Similar to the hydrogen bonding

labels, the self-interaction is reduced while the “e-v” cross-interaction is increased, in a size dependent way (**Supplementary Table 12**). The interaction with other non-labeled beads is increased by 1 level to mimic the extra polarity of these groups. Partitioning and miscibility should not be affected in relation to the solvents using non-labeled beads.

Positive and negative labels (p/n):

Some charged molecules can show hydrogen bonding capabilities, which also depends on the sign of their charge. Positive charges are usually hydrogen donors while negative charges are usually hydrogen acceptors.

Supplementary Table 13: Positive and negative labels – p/n				
Size pair		RR/RS	SS/ST/RT	TT
Label pair	<i>p-p/n-n*</i>	+3	+2	+1
	<i>p-n</i>	0	0	0
	<i>p-C/X beads</i>	-1	-1	-1
	<i>n-P/N beads</i>	-1	-1	-1

* new level cannot be higher than 12

The sign of the charge is also related with other properties. Organic hydrogen bond donor cations usually show stronger ion- π interactions while anions show stronger hydrogen bonds with neutral polar molecules. These differences in behavior explain some exceptions in relation to the classical Hofmeister series³. These aspects were encoded in “p” and “n” labels which were designed to be applied to Q-beads and to fine-tune the properties of organic cations and anions, respectively (**Supplementary Table 13**). For now, stronger “p-n” interactions were not included as this behavior is already mainly covered by the electrostatic interactions.

Cross-interactions between labels:

All beads with chemically specific labels can also interact between themselves which demands defined rules as well.

Supplementary Table 14: Cross-interactions between labels				
Size pair		RR/RS	SS/ST/RT	TT
Label pair	<i>d-e/a-v</i>	-1	-1	-1
	<i>d-v/a-e</i>	+2	+1	+1
	<i>p-d/n-a/n-e/p-v</i>	+2	+1	+1
	<i>p-a/n-d/p-e/n-v</i>	-1	-2	-3

Since hydrogen bonding, electron polarization, and charges can also promote specificity in the interactions, more attractive and repulsive interaction were also defined (**Supplementary Table 14**).

Self-interaction labels (*h/r*):

As the beads in Martini are generic, they do not necessarily have fully optimized self-interactions. Different chemical groups can have similar oil/water transfer free energies (and be modeled by the same bead), but, at the same time, have different self-interactions, which are not covered by the chemically specific labels.

Supplementary Table 15: Self-interaction labels – <i>h/r</i>				
Size pair		RR/RS	SS/ST/RT	TT
Label pair	<i>h-h</i>	-1	-1	-1
	<i>r-r</i>	+1	+1	+1
	<i>h-r</i>	0	0	0
	<i>h/r</i> - others	0	0	0

Slight differences in packing, dipole moment, and dispersive interactions are examples that are not covered by the other labels. As a more advanced way to solve these problems, we also introduced the self-interaction labels (**Supplementary Table 15**). They can be used to classify higher (“*h*”) or reduced (“*r*”) self-interaction, but they are limited to certain chemical types. In the case of P/N beads, they mimic small differences in dipole moments, while for C/X beads they could represent differences in the dispersive interactions. These labels do not have any effect on interactions between P/N and C/X beads, only internally in these two subgroups. They do not affect oil/water partitioning, but affect the miscibility: “*h*” reduces the miscibility with non-labeled beads while “*r*” increases it.

Partial charge labels (*q*):

Partial charges were not commonly used in the previous non-polarizable versions of Martini, except for some specific applications⁴. In Martini 3, partial charges can be used via the label “*q*”. Dipoles inside a single bead should be captured (if necessary) in polarizable Martini models^{5,6}. They are not included here.

Supplementary Table 16: Partial charge labels - <i>q</i>				
Size pair		RR/RS	SS/ST/RT	TT
Label Pair	<i>q-q</i>	-3	-2	-1
	<i>q-W/Q</i>	-3	-3	-3
	<i>q-others*</i>	-2	-2	-2

* D beads are not included here.

Rules for usage of partial charges:

Charge ≥ 0.75 : Q-bead

Charge 0.25- 0.75: organic beads with q-labels.

Charge ≤ 0.25 : organic beads without q-labels.

The “*q*” labels modify the LJ potential of the non-labeled beads in the organic block considering that this particle has a partial charge now (**Supplementary Table 16**). Beads with *q*-labels interact more with themselves and with the environment, in particular with W and Q beads. Ideally, the values of the bead partial charges should be computed, redistributing the atomic partial charges and dipoles calculated quantum chemically over the bead centers in a manner that preserves the local dipole moments optimally, according to methods such as *Dipole Preserving Charge*, introduced by Thole and van Duijnen⁷. It is worth mentioning that we only model the increased polarizability due to the charge delocalization using this “*q*” label. So, its usage is restricted to large charged molecules represented by more than one bead (as in *e.g.*, a charged imidazole). In the future, the usage of partial charges may be expanded to other cases as polar molecules/fragments with permanent and high dipole moments.

A5) Bead hydration, vaporization and oil/water transfer free energies

Supplementary Tables 17 to 19 present the main thermodynamic data for the single beads in Martini 3. For oil/water transfer free energies, six nonpolar organic phases were considered: hexadecane (HD), octanol (OCO), chloroform (CLF), ether (ETH), benzene (BENZ) and cyclohexane (CHEX). Except for beads with hydrogen donating labels, all other subtypes are not shown in **Supplementary Table 17 to 19**, as the oil/water transfer free energies only have small variations (~1 kJ/mol) with respect to standard (non-labeled) beads. All Gibbs free energies (ΔG) were computed at 300 K and 1 bar, using the protocol described in **section D1**. Beads without vaporization free energy indicated in the Tables are not in the liquid phase at the temperature and pressure simulated. The most stable phase is highlighted, except for Q-beads for which a definition of vaporization free energy is not applicable.

Supplementary Table 17. Regular (R) bead oil-water transfer, hydration, and vaporization Gibbs free energies (ΔG).

Bead	Oil-water transfer ΔG (kJ/mol)						Hydration ΔG (kJ/mol)	Vaporization ΔG (kJ/mol)
Regular	HD $\rightarrow$ W	OCO $\rightarrow$ W	CLF $\rightarrow$ W	ETH $\rightarrow$ W	BENZ $\rightarrow$ W	CHEX $\rightarrow$ W		
C1	17.2	18.9	17.2	14.5	14.9	20.0	-8.6	10.4
C2	15.4	14.8	14.3	12.9	12.1	16.9	-5.7	10.2
C3	13.5	13.8	13.3	13.3	15.1	16.3	-4.8	10.3
C4	10.6	13.4	13.6	13.7	18.0	16.3	-5.1	10.2
C5	8.8	11.2	12.1	11.9	16.7	13.2	-3.6	10.4
C6	5.4	10.1	11.4	11.4	14.6	11.3	-1.6	11.1
N1	2.9	8.1	7.6	9.0	11.9	8.6	1.0	10.9
N2	0.5	5.6	5.1	7.4	6.6	6.1	3.6	10.9
N3	-3.6	1.8	1.1	3.5	2.6	2.1	7.6	11.1
N4	-5.0	2.2	1.1	3.5	2.3	-0.5	7.6	12.0
N5	-7.5	0.0	-1.4	1.0	0.3	-2.6	9.9	11.8
N6	-9.8	-1.1	-5.1	-2.2	-2.0	-7.1	12.0	12.5
P1	-11.1	-2.0	-4.8	-2.0	-5.2	-8.5	12.0	12.7
P2	-13.2	-3.8	-6.9	-3.8	-7.5	-10.5	13.7	13.8
P3	-15.0	-5.1	-6.9	-4.0	-7.6	-13.0	13.9	13.8
P4	-17.1	-7.4	-9.3	-7.5	-12.2	-15.3	16.2	14.9
P5	-19.2	-9.1	-10.9	-11.2	-14.0	-16.9	18.0	16.0
P6	-27.2	-9.2	-11.9	-11.2	-17.0	-19.0	18.0	18.8
X1	12.9	14.3	17.1	14.5	18.9	17.2	-5.9	11.9
X2	10.3	12.7	14.5	12.1	13.5	13.1	-3.4	10.9
X3	10.7	13.9	14.8	13.6	13.8	13.4	-3.7	10.3
X4	5.8	8.7	11.5	12.9	11.9	11.6	-1.7	10.7
N1d/a	5.6	10.7	10.3	11.7	14.6	11.3	-1.7	9.7
N2d/a	3.0	7.8	7.6	9.9	9.0	8.5	1.1	9.7
N3d/a	-1.9	3.8	3.2	5.6	4.7	4.2	5.6	9.5
N4d/a	-3.0	4.3	3.1	5.5	4.3	1.5	5.6	10.4
N5d/a	-5.1	2.2	1.0	3.4	2.6	-0.2	7.5	10.5
N6d/a	-7.6	1.0	-2.9	0.01	0.2	-4.9	9.9	11.1
P1d/a	-9.1	0.2	-2.9	-0.1	-3.3	-6.6	9.9	10.9
P2d/a	-11.2	-1.9	-4.9	-1.8	-5.5	-8.5	11.7	11.8
P3d/a	-13.2	-3.5	-5.1	-2.2	-5.8	-11.3	12.1	11.9
P4d/a	-14.7	-5.1	-6.8	-5.1	-9.8	-12.9	13.8	13.0
P5d/a	-16.8	-7.0	-8.8	-8.8	-11.8	-15.1	15.8	13.7
P6d/a	-25.2	-7.4	-10.0	-8.8	-15.0	-16.9	15.8	15.9
Q1	-24.2	-10.9	-9.5	-2.6	-4.7	-12.5	23.2	-
Q2	-32.7	-15.1	-13.9	-5.1	-10.4	-17.1	25.5	-
Q3	-35.6	-17.4	-17.9	-7.9	-17.0	-21.9	28.0	-
Q4	-48.3	-18.8	-22.3	-10.9	-26.6	-29.8	31.0	-
Q5	-64.1	-23.0	-28.3	-13.1	-37.7	-41.4	35.2	-
D	-78.8	-26.8	-42.3	-24.3	-51.9	-60.3	49.2	-

Supplementary Table 18: Small (S) bead oil-water transfer, hydration, and vaporization
Gibbs free energies (ΔG).

Bead	Oil-water transfer ΔG (kJ/mol)						Hydration ΔG (kJ/mol)	Vaporization ΔG (kJ/mol)
Small	HD $\rightarrow$ W	OCO $\rightarrow$ W	CLF $\rightarrow$ W	ETH $\rightarrow$ W	BENZ $\rightarrow$ W	CHEX $\rightarrow$ W		
C1	15.3	14.2	14.9	11.7	12.3	14.9	-9.4	gas
C2	13.9	9.9	11.6	9.8	9.0	11.4	-6.1	gas
C3	11.0	9.2	10.9	10.9	9.9	10.8	-5.4	gas
C4	7.5	8.4	10.6	10.8	11.6	10.8	-5.3	gas
C5	6.0	6.3	9.3	9.4	10.2	7.9	-3.9	gas
C6	2.7	5.3	8.7	8.5	8.5	5.9	-2.0	7.6
N1	1.0	3.6	5.5	6.7	6.6	4.0	-0.1	7.6
N2	-1.0	2.1	3.8	6.6	2.8	2.3	1.8	7.6
N3	-5.1	-1.8	-0.4	2.6	-1.3	-2.1	5.9	7.6
N4	-6.6	-0.9	-0.3	2.3	-1.5	-3.3	5.9	8.7
N5	-9.2	-3.6	-3.2	-0.1	-4.1	-6.1	8.5	8.7
N6	-11.1	-4.2	-6.9	-3.8	-6.1	-9.0	10.6	10.0
P1	-13.1	-5.2	-6.7	-3.9	-7.6	-10.7	10.6	10.0
P2	-14.8	-6.9	-8.9	-5.8	-9.5	-12.5	12.4	11.2
P3	-16.2	-7.7	-8.5	-5.5	-9.2	-14.0	12.2	11.2
P4	-18.7	-9.8	-11.0	-9.1	-13.6	-16.7	14.6	12.5
P5	-20.4	-11.8	-12.7	-12.9	-15.5	-18.4	16.6	13.9
P6	-28.5	-12.0	-14.4	-12.8	-18.0	-20.1	16.6	16.5
X1	10.3	9.4	14.6	11.7	12.8	11.5	-6.1	gas
X2	7.9	7.2	12.1	9.2	8.3	7.8	-3.7	gas
X3	8.1	8.0	12.3	10.3	8.4	7.7	-3.8	gas
X4	2.9	4.3	8.6	10.3	6.5	5.9	-1.9	gas
N1d/a	3.0	6.0	7.5	8.7	8.6	6.0	-2.1	gas
N2d/a	0.9	3.8	5.7	8.5	4.8	4.2	-0.1	gas
N3d/a	-2.9	0.2	1.9	4.8	0.9	0.1	3.8	gas
N4d/a	-4.5	1.1	1.8	4.5	0.7	-1.1	3.8	gas
N5d/a	-6.6	-1.0	-0.6	2.6	-1.4	-3.5	5.9	gas
N6d/a	-9.2	-2.5	-5.0	-1.9	-4.2	-7.1	8.6	7.6
P1d/a	-11.3	-3.7	-4.8	-2.1	-5.8	-8.8	8.6	7.6
P2d/a	-13.0	-5.4	-7.1	-4.0	-7.7	-10.6	10.6	8.7
P3d/a	-14.5	-6.1	-6.8	-3.8	-7.6	-12.3	10.5	8.7
P4d/a	-16.4	-7.8	-8.7	-6.8	-11.3	-14.4	12.3	10.0
P5d/a	-18.0	-9.5	-10.4	-10.4	-13.1	-16.0	14.1	11.2
P6d/a	-26.0	-9.6	-11.6	-10.4	-15.3	-17.5	14.1	12.5
Q1	-33.8	-10.6	-15.7	-6.0	-5.2	-18.0	37.0	-
Q2	-43.3	-14.3	-21.1	-9.1	-11.9	-23.5	39.4	-
Q3	-47.8	-18.0	-26.2	-12.3	-18.6	-29.5	42.1	-
Q4	-59.2	-18.2	-31.1	-15.2	-27.1	-37.8	44.9	-
Q5	-70.8	-18.2	-36.0	-16.3	-35.0	-46.0	47.5	-
D	-167.8	-36.4	-105.4	-64.7	-84.6	-127.4	141.5	-

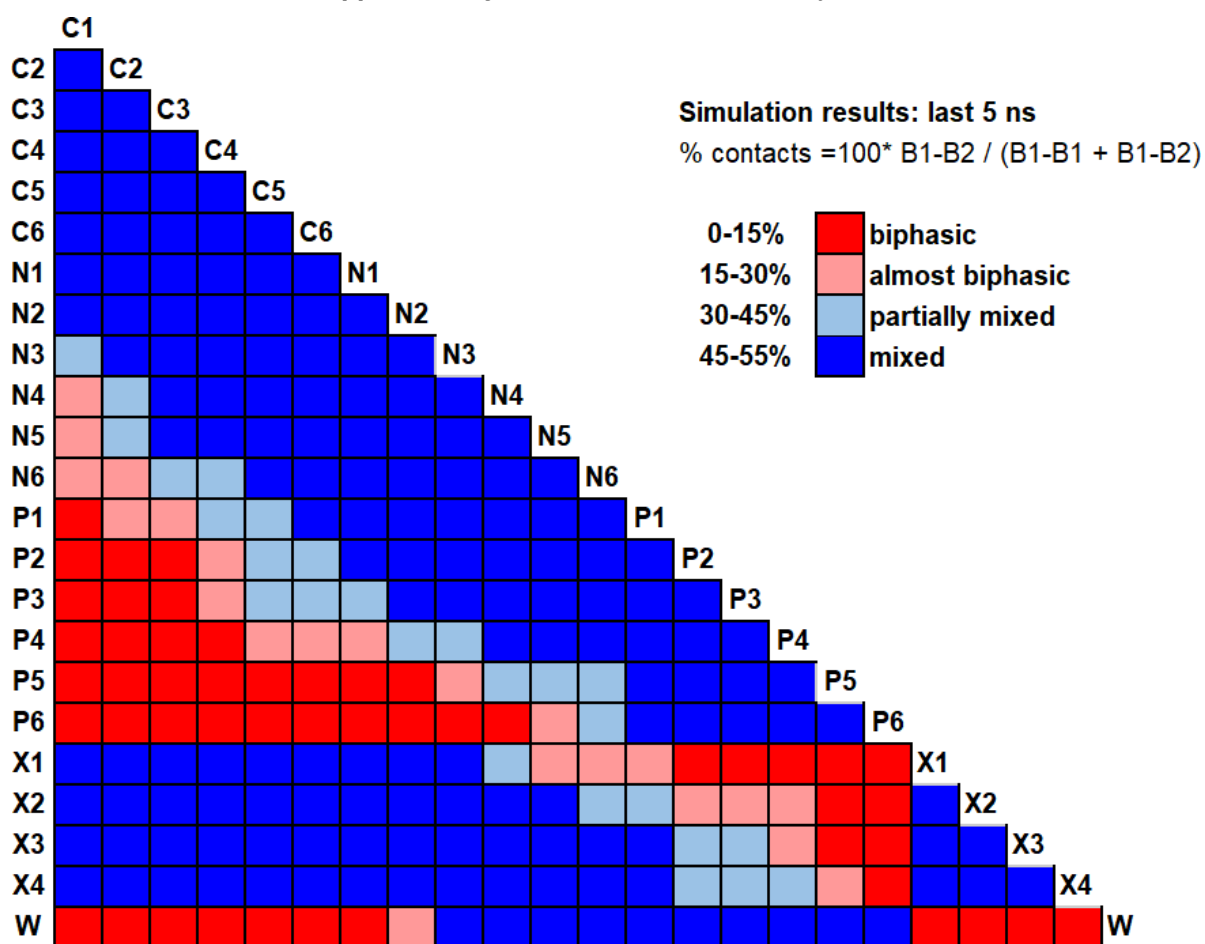
Supplementary Table 19: Tiny (T) bead oil-water transfer, hydration, and vaporization
Gibbs free energies (ΔG).

Bead	Oil-water transfer ΔG (kJ/mol)						Hydration ΔG (kJ/mol)	Vaporization ΔG (kJ/mol)
Tiny	HD $\rightarrow$ W	OCO $\rightarrow$ W	CLF $\rightarrow$ W	ETH $\rightarrow$ W	BENZ $\rightarrow$ W	CHEX $\rightarrow$ W		
C1	13.2	12.0	12.9	10.5	8.2	13.2	-10.9	gas
C2	11.1	7.8	9.3	7.9	4.8	9.6	-7.3	gas
C3	8.6	6.7	8.3	8.2	5.5	8.5	-6.2	gas
C4	5.6	6.4	8.2	8.1	8.0	8.5	-6.1	gas
C5	3.6	4.5	6.5	6.4	6.3	5.7	-4.4	gas
C6	1.3	3.6	6.3	6.4	5.0	4.3	-3.1	gas
N1	-0.3	2.3	3.4	4.6	3.3	2.5	-1.4	gas
N2	-2.4	0.3	1.4	3.7	-1.3	0.6	0.6	gas
N3	-6.1	-3.1	-2.2	0.1	-5.0	-3.1	4.4	gas
N4	-7.7	-2.9	-2.3	0.0	-5.1	-4.1	4.4	gas
N5	-9.7	-4.9	-4.5	-2.2	-7.2	-6.4	6.5	gas
N6	-11.9	-6.1	-8.0	-5.4	-9.2	-9.8	8.5	gas
P1	-13.7	-7.2	-8.0	-5.4	-11.2	-11.7	8.5	gas
P2	-15.6	-8.8	-9.9	-7.3	-13.2	-13.6	10.5	7.7
P3	-16.8	-9.8	-10.0	-7.3	-13.2	-15.7	10.6	7.7
P4	-19.6	-12.1	-12.6	-11.1	-17.3	-18.0	13.1	9.1
P5	-23.2	-15.2	-16.3	-16.3	-21.0	-21.6	16.8	10.3
P6	-31.6	-14.8	-17.2	-16.3	-22.9	-22.2	16.8	13.1
X1	8.4	7.6	11.6	9.3	9.1	9.6	-7.3	gas
X2	5.3	5.2	8.6	6.4	3.7	5.5	-4.4	gas
X3	5.4	5.4	8.6	7.5	3.6	5.5	-4.3	gas
X4	1.2	2.7	6.4	7.4	2.5	4.3	-3.2	gas
N1d/a	1.4	3.9	5.2	6.4	5.1	4.3	-3.1	gas
N2d/a	-0.4	2.3	3.3	5.7	0.7	2.5	-1.3	gas
N3d/a	-4.4	-1.4	-0.5	1.7	-3.3	-1.4	2.6	gas
N4d/a	-5.9	-1.2	-0.6	1.7	-3.3	-2.4	2.6	gas
N5d/a	-7.6	-2.8	-2.4	0.0	-5.0	-4.2	4.3	gas
N6d/a	-9.8	-4.1	-6.0	-3.3	-7.1	-7.8	6.4	gas
P1d/a	-11.4	-5.0	-5.8	-3.1	-8.9	-9.5	6.4	gas
P2d/a	-13.6	-6.8	-7.9	-5.3	-11.2	-11.6	8.5	gas
P3d/a	-14.5	-7.8	-7.8	-5.1	-10.9	-13.4	8.4	gas
P4d/a	-16.9	-9.5	-10.0	-8.4	-14.7	-15.4	10.5	7.7
P5d/a	-20.8	-13.2	-13.9	-13.9	-18.6	-19.3	14.4	9.1
P6d/a	-29.3	-12.7	-15.0	-13.9	-20.7	-19.9	14.4	11.7
Q1	-46.3	-14.2	-25.2	-13.4	-7.2	-31.2	58.7	-
Q2	-59.5	-14.5	-32.0	-17.4	-17.6	-37.2	61.4	-
Q3	-65.8	-18.7	-37.3	-20.7	-24.6	-43.6	63.2	-
Q4	-76.6	-16.3	-42.4	-23.5	-32.9	-51.3	65.6	-
Q5	-87.4	-17.0	-48.6	-24.5	-40.7	-62.8	68.8	-
D	-225.0	-36.8	-143.4	-89.7	-128.3	-179.7	201.2	-

A6) Martini miscibility table

In Martini 3, miscibility trends are as important as oil/water transfer free energies. **Supplementary Table 20** presents the overall miscibility results of regular neutral beads, based on 50:50 mixtures at 300 K and 1 bar. Further details are described in **section D4**. The overall trends reflect the choice of levels in the interaction matrix (**Supplementary Table 1**). Mixtures are classified in four different degrees of miscibility, according to the percentage of contacts between the beads.

Supplementary Table 20: Martini Miscibility Table



Beads of smaller sizes show the same trends, with a slightly increased miscibility as the size is reduced ($RR > RS > SS \approx RT > ST > TT$). Labels can affect the miscibility following the expected trends (based on **Equation 4** of **section B2**). The expected changes are summarized in **section A4**.

Supplementary Notes B) Parametrization strategy of the beads

First of all, the reparametrization strategy of a complete CG force field like Martini needs to define two key aspects of the new model: (i) which potentials will be used; and (ii) which properties and systems will be considered for the calibration and validation. Both aspects are linked, as the potentials will limit the properties which can be quantitatively obtained. In Martini 3, the functional forms for bonded (bonds, angles, and dihedrals) and non-bonded interactions (Lennard-Jones (LJ) and Coulombic) are the same as in previous versions. For non-bonded interactions (**Equation 1**), a cutoff of 1.1 nm is the default choice, as defined by de Jong et al.⁸ Although the LJ potential is not the most accurate possibility for CG models^{9,10}, this option allows Martini to be easily implemented in atomistic molecular dynamics (MD) simulation programs. In this way, Martini can benefit from decades of development in MD codes like GROMACS^{11,12}, including their fast highly-parallel performances and their efficient implementations of modern enhanced sampling techniques. The choice of the LJ potential is therefore justified by the speed of computation, that is the main interest of a CG force field.

$$V_{LJ}(r_1, \dots, r_n) + V_{Coul}(r_1, \dots, r_n) = \sum_{i,j} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \sum_{i,j} \frac{q_i q_j}{4\pi\epsilon_0\epsilon_r r_{ij}} \quad \text{Equation 1}$$

The main disadvantage of a hard potential such as the LJ is the limited range of the liquid phase⁹, which implies that the overall interaction strength of Martini needs to be underestimated in relation to atomistic models, otherwise single beads could easily end up in the solid phase. This problem was already observed for certain systems with Martini 2 water, which was avoided by the non-optimal use of antifreeze waters^{13,14}. Only certain very polar compounds (as primary amides) and ionic salts formed by single beads should be solid at room temperature (around 298 to 310 K). As a consequence of the use of the LJ potential, some properties, such as free energies of solvation, free energies of vaporization, or gas-liquid surface tension, cannot be used as references for the parametrization (although good trends are desirable). On the other hand, properties that involve $\Delta\Delta G$ measurements, like the oil/water transfer free energy or the excess free energy of mixing, can still be quantitatively obtained by such CG models. As partitioning

and miscibility are important to reproduce a wide range of phenomena, they were chosen to be the main targets for the parametrization of bead-bead interactions in Martini 3.

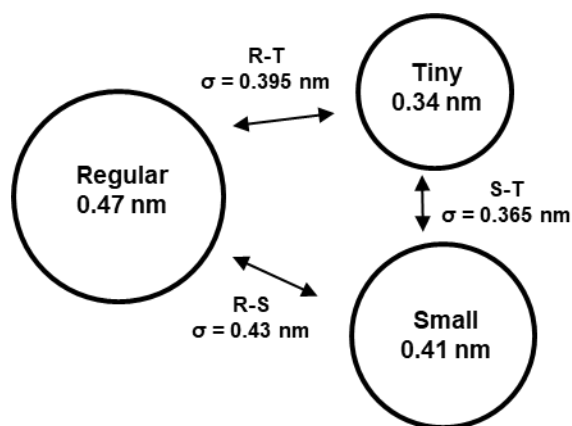
One aspect that differentiates Martini from other CG force fields, and that needed to be considered in the reparametrization as well, is the generic building block approach. The Martini model contains generic beads that can represent different chemical moieties. They are classified in chemical and size classes that follow general trends in hydrophobicity, miscibility, and self-interaction. The key advantage of this approach is the transferability of the model, which makes Martini very versatile and user-friendly. The main disadvantage is that the quality of the models does not only depend on the optimized LJ parameters, but also on four other interdependent ingredients of the model: atom-to-bead mapping of the molecules, bonded parameters, number of bead types available, and bead type assignment. The impact of LJ parameters and bonded parameter choices, in terms of bead sizes and bond distances, was recently discussed by Alessandri *et al.*² Mapping, number of bead types, and bead type assignment were also recently explored by Kanekal and Bereau¹.

With so many variables, problems such as too strong protein-protein interactions^{15,16} are very hard to pinpoint and cannot be fully solved with one simple modification. All aspects of the model need to be reviewed, with an iterative parametrization strategy starting from the LJ parameters of single beads and ending with validation in complex target systems, such as lipid bilayers, proteins, or polymers. In the next sections, we describe all essential aspects of the new parametrization. As already mentioned in **section A1**, the beads are now divided in 3 blocks: organic, ion, and solvent. Each block has its unique set of interaction levels, including different parameters and approaches for the self and cross-interactions between bead sizes. This choice facilitates the implementation of future improvements in the model, as the blocks of parameters are largely independent of one another. For instance, future improvements in the polarizable water^{5,6} based on Martini 3 will mainly require changes in water (solvent block) and charged beads (ion block).

The next **sections (B1 to B4)** describe how we defined the bead sizes, masses and interaction strengths, including self-interactions (R-R, S-S, and T-T) and cross-interactions (R-S, R-T, and S-T).

B1) Bead sizes and masses

The bead sizes available represent, in general, the mapping of different number of non-hydrogen atoms: regular (“R”, with 4 atoms-to-1-bead mapping, or simply 4-1), small (“S”, 3-1) and tiny (“T”, 2-1). Rules and exceptions for mappings using different bead sizes are described in **section C1**. In relation to Martini 2, the sizes (σ_{ij} of the LJ potential) of S- and T-beads were redefined to get a better overall volume of the mapped fragments. Besides, different sizes for cross-interactions are now also included. A summary of σ_{ij} values used in organic neutral beads for attractive and intermediate levels in Martini is shown in **Supplementary Figure 2**.



Supplementary Figure 2: Bead sizes in Martini 3. In contrast to Martini 2, cross-interactions between regular, small, and tiny beads now also use specific sizes.

The σ_{ij} values chosen for S- and T-beads have certain biases for molecular systems in which smaller beads are more suitable than the regular ones in order to reproduce the symmetry and the overall shape of the modelled molecules. For instance, T-beads have $\sigma_{TT} = 0.34$ nm, which is tuned to get optimal stacking distances in aromatic rings. On the other hand, S-beads have $\sigma_{SS} = 0.41$ nm, which is more accurate to represent fragments in 3-1 mapping of aliphatic cyclic molecules. The σ_{ij} of cross-interactions are obtained using the arithmetic average as combination rule (**Equation 2**). Moreover, a softness parameter is also included; this parameter is defined as $s_{ij} = 0.01$ nm for most of the beads, except water. The s_{ij} parameter assumes that bigger beads should be softer than smaller ones. The use of combination rules for different sizes helps to solve the issue of artificial free energy barriers observed in dimerization PMFs of rings

in Martini 2^{2,17,18}. It is important to notice that the σ_{ij} of certain repulsive interactions is increased in relation to the standard values. This is especially used to extend the repulsion between very polar/charged and very apolar beads, as implemented in Martini 2 for the super repulsive level¹³.

$$\sigma_{ij} = \left(\frac{\sigma_{ii} + \sigma_{jj}}{2} \right) - s_{ij} \quad \text{Equation 2}$$

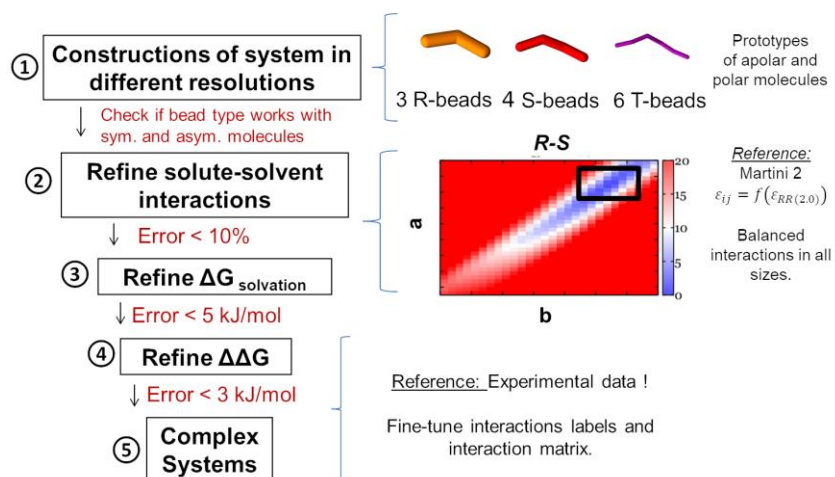
Usually, Martini CG models use default size-dependent masses for the beads. In Martini 3, the default masses of R-, S- and T-beads are 72, 54, and 36 amu, respectively. Note that the dynamics of the system is affected by the mass assignment, so the correct ones (based on the sum of the masses of the atoms mapped by the bead) should be used in case trends in dynamic properties of the system are important. Masses can also be important for simulation stability as discussed in **section C4**.

B2) Organic block

The organic block makes up most of the beads used in Martini 3. They can model organic molecules, or fragments of them, mainly consisting of atoms of the 2nd period of the periodic table. Sulfur, phosphorus and halogen atoms are also included. The main idea here was initially to balance the interaction levels used between all bead sizes as well as possible for Martini models to be used in multiresolution way, with further optimization at the end towards the experimental calibration targets. The parametrization workflow is represented in **Supplementary Figure 3**.

The step 1 was to build prototype models of polar and nonpolar molecules in all three possible resolutions. For example, prototype models of dodecane were generated considering models with three R-beads, four S-beads, and six T-beads (see **Figure 1A**). Following the building block approach, the same beads need to be used in apolar fragments that are part of amphiphilic polar molecules, such as alcohols with a long aliphatic chain. From this point, the step 2 was to explore the LJ parameter space, with initial predefined sizes (σ_{ij} , discussed in the previous section) and variable interaction strengths (ϵ_{ij}). We built simulation boxes containing one solute molecule immersed in a solvent, with all combinations of different sizes. MD simulations were performed in the

NPT ensemble, at 300 K and 1 bar. As initial reference, we used systems containing molecules mapped only with regular beads and using interactions from Martini 2 ($\epsilon_{RR(2.0)}$). The new interactions were defined as function of $\epsilon_{RR(2.0)}$, with truncated Taylor series containing up to 3 terms, as described in **Equation 3**.



Supplementary Figure 3: Parametrization workflow to obtain the interaction levels in the organic block of Martini 3. After construction of molecular models in different resolutions (step 1), solute-solute interactions are scanned (step 2) in relation to the reference (Martini 2 systems modelled by regular beads), as exemplified by the plot in the center of the figure. In this case, the average absolute solute-solute interaction error in systems containing regular and small beads is shown, with the best set of parameters (highlighted in the plot) selected for step 3, where solvation free energies are the target. As final steps, $\Delta\Delta G$ measurements (step 4) and complex systems (e.g. bilayer and proteins) are tested to fine-tune the interaction levels and the choices of levels in the interaction matrix.

$$\epsilon_{ij} = f(\epsilon_{RR(2.0)}) = a + b\epsilon_{RR(2.0)} + c(\epsilon_{RR(2.0)})^2 \quad \text{Equation 3}$$

Using a brute-force approach, the polynomial coefficients were scanned by grid search for each size combination (R-R, R-S, R-T, S-S, S-T and T-T). As constraints, we selected parameters that kept the systems in the expected liquid phase at the temperature and pressure used in the simulations (except for very polar systems, as already mentioned). We found an initial set of parameters with good agreement in all resolutions tested for the average solute-solute interaction energy (with maximum 5% of absolute error in relation to the reference $\epsilon_{RR(2.0)}$).

In step 3, we performed thermodynamic integration calculations (as described in **section D1**) to get the solvation free energy of the solutes (ΔG_{solv}). At this point, entropic differences of models in different resolution were considered, with the best matches (with absolute differences lower than 5 kJ/mol) being selected from the pool of parameters. We found that for certain small values of ε_{ij} (e.g. involving interactions between polar and nonpolar molecules in the extremes of the hydrophobicity scale), a further reduction of the value of ε_{ij} does not increase ΔG_{solv} (i.e. makes the solvation less favorable). This effect takes place as the solute-solvent ε_{ij} starts to be too “soft” in relation to the cohesive interactions of the solvent. These are the cases in which increasing σ_{ij} was deemed to be necessary to mimic the repulsive interaction between very polar/charged and nonpolar molecules.

Steps 1 to 3 were performed to guarantee that the parameters obtained are balanced across the different sizes. It is important to notice that Martini 3 is based on the use of specific pair interactions for ε_{ij} and not on combination rules. The whole procedure until here only provided a few interaction levels used to define the pair interactions. It is important to use enough prototype models that allow the procedure to generate ε_{ij} values for points near to the extremes of the interaction matrix. The rest of the parameters are defined by interpolation and (if necessary) extrapolation, forming the interaction levels of Martini. The procedure described until step 3 naturally leads to reasonable trends in self-interaction and hydrophobicity for beads of different sizes that represent the same chemical fragment at different resolutions (like propylamine, ethylamine, and methylamine). Smaller beads are more hydrophilic and interact less with the environment (and with themselves). This behavior mimics the expected trends observed in organic molecules with different number of aliphatic carbons². Trends of Martini 3 beads in different sizes and chemical types were described in **section A1**.

Finally, steps 4 and 5 were used to fine-tune the ε_{ij} in relation to experimental target properties, which were presented in **section A4**. In these steps, the levels used in the interaction matrix are finally defined. In the previous Martini version, all points in the matrix which were not validated by any experimental data were selected based on the expected ε_{ij} trends. It is worth mentioning that the fraction of validated points for the

regular bead size corresponds to around 20% of the matrix. In Martini 3, despite more experimental validation being considered, a lot of points still do not have any direct reference. For these cases, the levels in Martini 3 were selected based on miscibility, which guarantees that proper trends in phase separation based on the hydrophobicity of the beads are maintained. The free energy of mixing (ΔG_{mix}) of binary mixtures composed of single particles (called here bead A and bead B) can be predicted based on lattice-models (as in Flory-Huggins solution theory)¹⁹ according to **Equation 4**:

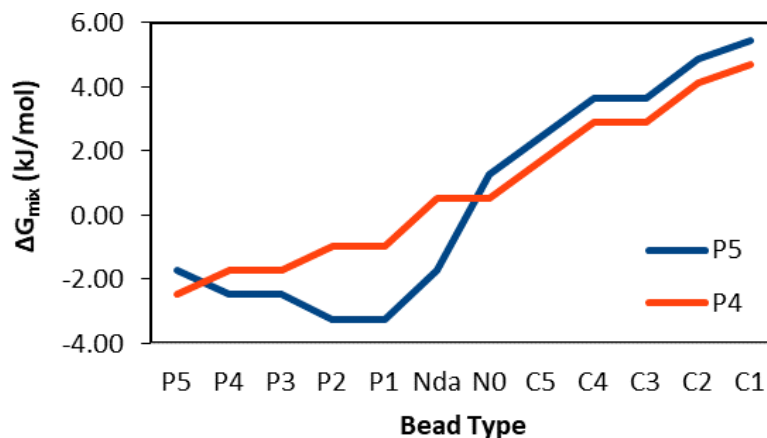
$$\Delta G_{mix} = RT(\chi_A \ln \chi_A + \chi_B \ln \chi_B) + z \left(\epsilon_{AB} - \frac{\epsilon_{AA} + \epsilon_{BB}}{2} \right) \chi_A \chi_B \quad \text{Equation 4}$$

The sum of the first two $\chi \ln \chi$ terms corresponds to the ideal entropy of mixing ($\Delta S_{ideal\ mix}$), while the third term is the enthalpy of mixing (ΔH_{mix}) of the lattice model. χ_A and χ_B are the molar fractions of A and B in the mixture. z corresponds to the number of contacts that each particle can have in the lattice grid. In three-dimensional systems, z can be considered to be 12 (that is the number of neighbors of each bead in a closed-packing lattice model). ϵ_{AA} , ϵ_{BB} , and ϵ_{AB} are the contact energy of each possible pair of interaction. If we consider ϵ_{ij} as the LJ epsilon of the beads in the Martini model, it is possible to predict the expected free energy of mixing of the beads. The approximation works reasonably well for systems that are composed of beads of the same size and that are in the liquid phase at the temperature investigated. In real systems, ΔH_{mix} corresponds to the excess free energy of mixing (ΔG_{ex}), as it also includes other entropic terms like the hydrophobic effect or conformational entropy. The advantage of using miscibility instead of using infinite-dilution properties to define the levels in the interaction matrix is that it additionally takes into account the self-interaction.

For example, **Supplementary Figure 4** shows the interaction levels and miscibility predictions of P5 and P4 beads in Martini 2 for a 50:50 mixture at 300 K. The lattice model predicts that P5 should mix more than P4 with beads P4, P3, P2, P1, and Nda, which would not be the expected behavior based on the hydrophobicity of the beads. It would be expected that P4, as it is less hydrophilic than P5, mixes more with them. It is worth mentioning that P5 actually shows higher-than-ideal miscibility with these beads ($\Delta H_{mix} < 0$ kJ/mol) as ΔG_{mix} is lower than -1.73 kJ/mol (corresponding to the ideal entropy

contribution, $\Delta S_{ideal\ mix}$, for the mixture ratio and temperature considered). Such behavior is not so common for most of the real binary mixtures²⁰. The main possible problem in the miscibility trend of P5 is the assignment of the interaction levels for cross-interactions between this bead and P3, P2, P1, and Nda, which should have been reduced by one interaction level in Martini 2 (i.e. reduced by 0.5 kJ/mol). Another complementary option would have been to increase the self-interaction of P5. Combined, both changes would correct the trend in miscibility. It is important to reinforce here that the optimization of the pair interactions using miscibility trends as target is only reasonable for points in the interaction matrix that are not validated by experimental evidence. The miscibility of all single beads in Martini 3 is presented in **section A6**. The comparison with experimental miscibility data for CG models composed of one or more beads is discussed in **sections E3, E4, and E5**.

	P5	P4
P5	5.50	5.50
P4	5.50	5.00
P3	5.50	5.00
P2	5.50	4.50
P1	5.50	4.50
Nda	5.00	4.00
N0	3.50	3.50
C5	3.10	3.10
C4	2.70	2.70
C3	2.70	2.70
C2	2.30	2.30
C1	2.10	2.10



Supplementary Figure 4: Impact of the Martini 2 interaction levels chosen for P5 and P4 beads in their miscibility with other beads. Mixtures with a 50:50 ratio at 300 K were considered for the prediction of ΔG_{mix} according to **Equation 4**. The table in the left panel shows the LJ epsilon values (in kJ/mol) for each pair interaction.

B3) Ion block

The main difference between neutral and charged (Q and D) beads in Martini is that the latter ones include not only LJ but also Coulomb interactions. These electrostatic

interactions take into account a default explicit screening, with a relative permittivity $\epsilon_r=15$. The parametrization of ions in CG force fields is *per se* challenging, which makes the definition of the “best” strategy very hard. The main reasons for these difficulties are:

- a) Limited amount of available experimental reference data. Moreover, for very hydrophilic ions, partitioning data (logP) are also not very reliable, as it is difficult to experimentally measure their concentration in the hydrophobic phase. For instance, octanol/water partitioning measurements are usually limited to hydrophilic molecules with a minimum logP of -2.5²¹.
- b) For cases where reliable experimental data is available, the partitioning data usually reflects the behavior of ion pairs (cation and anion), which are difficult to decouple^{22,23}.
- c) Co-existence of different protonation states can be important for charged organic molecules. In water, the degree of protonation is determined by easily measurable pKa and pH values; but in other environments, pKa values depend on the hydration level of the organic molecule, its polarity, and the availability of protons.
- d) Atomistic models are still not very accurate for ions^{24,25}. Currently, most atomistic force field developers are trying to improve the behavior of ions by adding non-bonded fixes to the LJ pair interactions²⁶, rescaling charges^{27–29} or adding polarization explicitly³⁰, but there is still no standardized method applicable to any kind of system.
- e) Solubility and osmotic coefficients^{29,31} can be used to estimate the self-interaction of ionic species, but they are mainly available for charged molecules in water and not in other solvents.
- f) The chemical diversity of relevant ions is higher than in the organic block. For instance, biologically relevant ions can be inorganic (e.g. metals, halides, or phosphates) or organic (such as trimethyl ammonium or acetate). This aspect makes ions hard to be grouped in generic beads, including difficulties in predicting the expected changes with size.

Given all these problems, the previous version of the Martini model opted for a simple strategy, which was to only include differences in hydrogen bonding capabilities

of ions. The interaction levels were based on the LJ parameters used for the neutral beads. Besides, the ions were mainly modeled by regular size beads. For small ions, such as sodium or chloride, this choice meant that the ions somehow include their hydration shells in the model, which limits their application.

The Martini 3 parametrization strategy of ions differs from the one used in Martini 2 in several aspects. First of all, as already mentioned, in Martini 3 the interaction levels of charged beads (ion block) are different from the neutral ones. Moreover, Q-Q self-interactions take now into account the effective position of the minimum in the total potential which in this case is the sum of Coulomb and LJ interactions. The LJ σ_{ij} is calibrated to keep similar effective bead sizes as defined for the neutral ones. In particular, for small and tiny beads, the σ_{ij} usually needs some adjustments to compensate for the stronger electrostatic interactions between opposite charges which would otherwise allow too much spatial overlap of the beads.

Another important consideration is the cross-interaction between Q and neutral beads. As neutral beads do not have partial charges in standard (*i.e.* non-polarizable) Martini, the ion-dipole and/or ion-quadrupole interactions need to be incorporated into the ε_{ij} of the LJ potential. More importantly, these ε_{ij} parameters need to rely more strongly on trends of solvation than $\Delta\Delta G$ measurements, as partitioning data is limited for ions. To this end, we introduced the polarization parameter (p) in the LJ ε_{ij} between the Q beads and all neutral beads (denoted by B), as defined by an empirical relationship (found to be useful for this specific purpose) described by **Equation 5**.

$$\varepsilon_{QB,final} = \varepsilon_{QB} + p_{QB} = \varepsilon_{QB} + p_{QW} \frac{\Delta G_{solv,B}}{\Delta G_{solv,W}} \quad \text{Equation 5}$$

ε_{QB} is an interaction level from the organic block and used as the base interaction. It represents the interaction term between Q and a neutral bead B that does not involve the net charge of the ion. p_{QW} is a free parameter (to be optimized) between water and Q-beads. $\Delta G_{solv,B}$ and $\Delta G_{solv,W}$ are the solvation free energy of ion Q in a solvent composed of beads B and water, respectively. If the solvent B is water itself, **Equation 5** gives directly the correction for Q-water interactions ($\varepsilon_{QW,final} = \varepsilon_{QW} + p_{QW}$).

$\Delta G_{solv,B}$ was defined by the Generalized Born (GB) model, an approximation commonly used to treat electrostatics in implicit solvents. GB models are accurate enough to obtain correct trends of solvation for ions in different sizes and solvents²². For bead B with charge q , **Equation 6** defines the GB model as:

$$\Delta G_{solv,B} = -\frac{1}{8\pi\epsilon_0} \left(\frac{1}{\epsilon_{B,inside}} - \frac{1}{\epsilon_B} \right) \frac{q^2}{r_B} \quad \text{Equation 6}$$

where ϵ_B is the experimental relative permittivity for solvent B, and $\epsilon_{B,inside}$ is the relative permittivity within the charged bead. The effective size (r_B) is defined as $r_B = \sigma_{QB} * f_{QB}$, where σ_{QB} is the LJ cross-interaction size between solvent B and the Q beads (defined by the interaction level). f_{QB} represents the correction in size commonly used in GB models²². Using **Equation 6**, the ratio $\Delta G_{solv,B}/\Delta G_{solv,W}$ in **Equation 5** is given by **Equation 7** as:

$$\frac{\Delta G_{solv,B}}{\Delta G_{solv,W}} = \frac{\epsilon_W \epsilon_{W,inside}}{\epsilon_B \epsilon_{B,inside}} \left(\frac{\epsilon_B - \epsilon_{B,inside}}{\epsilon_W - \epsilon_{W,inside}} \right) \gamma_{QB} \frac{\sigma_{QW}}{\sigma_{QB}} \quad \text{Equation 7}$$

with γ_{QB} being the effective size ratio ($\gamma_{QB} = f_{QW}/f_{QB}$). This ratio is assumed to be constant for a specific Q bead type.

As the relative permittivity is well-correlated with the ion solubility^{32,33}, which is related to the hydrophobicity of the solvents, an average value of ϵ_B can be attributed to each neutral bead, considering the most common chemical groups represented by the bead. It is worth to remember that this ϵ_B is not the relative permittivity used to compute the Coulomb interactions of the MD simulation, but only one of the constants used to get the LJ parameter between Q and neutral beads (as defined in **Equation 7**). A list of ϵ_B attributed to each neutral chemical bead type is provided in **Supplementary Table 21**.

Supplementary Table 21: Default relative permittivity defined for each neutral chemical bead type. The values are used as constants in Equation 7 for each bead.																							
Bead	W	P6	P5	P4	P3	P2	P1	N6	N5	N4	N3	N2	N1	C6	C5	C4	C3	C2	C1	X4	X3	X2	X1
ϵ_r	80	50	40	30	25	22	20	18	16	14	12	10	8	6	5	4	3	2.5	2	7	6	4.5	3

$\epsilon_{B,inside}$ values were defined by interpolation (**Equation 8**), considering the ϵ_B values and the extremes of the values of relative permittivity inside of the bead, which are in water ($\epsilon_{W,inside}$) and in the most apolar bead: C1 ($\epsilon_{C1,inside}$).

$$\epsilon_{B,inside} = \epsilon_{W,inside} + \left(\frac{\epsilon_B - \epsilon_W}{\epsilon_{C1} - \epsilon_W} \right) (\epsilon_{C1,inside} - \epsilon_{W,inside}) \quad \text{Equation 8}$$

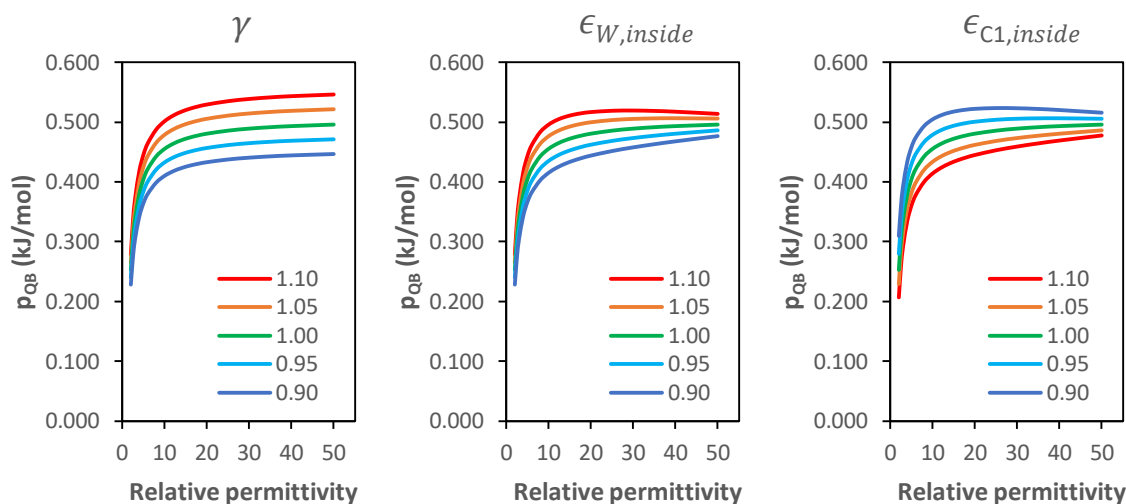
Together, the four parameters p_{QW} , $\epsilon_{W,inside}$, $\epsilon_{C1,inside}$, and γ_{QB} are the only variables to be adjusted for each combination of Q and neutral beads sizes. These parameters are provided in **Supplementary Tables 22 and 23**.

Supplementary Table 22: p-factor parameters of the three organic bead sizes and Q/D beads							
Organic size		Q1	Q2	Q3	Q4	Q5	D
γ	R	0.9643	1.0161	1.0702	1.1130	1.1425	1.3300
	S	0.9390	0.9940	1.0595	1.1109	1.1278	1.3150
	T	0.8742	0.9653	1.0201	1.0899	1.1025	1.2900
$\epsilon_{W,inside}$	R	1.1050	1.2000	1.2600	1.3100	1.4580	1.1000
	S	1.1900	1.1910	1.2590	1.2780	1.4100	1.0100
	T	1.0700	1.0700	1.1550	1.2500	1.5400	0.9500
$\epsilon_{C1,inside}$	R	1.0400	1.2200	1.3800	1.5000	1.6800	1.9900
	S	1.0570	1.1710	1.3550	1.5010	1.6900	1.9800
	T	0.8800	1.0200	1.2200	1.5100	1.8900	1.9050

Supplementary Table 23: p_w (kJ/mol) parameters for each combination of neutral and charged beads				
Sizes		Neutral beads		
		R	S	T
Q-beads	R	0.70	0.48	0.41
	S	2.56	2.47	2.25
	T	5.92	5.10	4.95
D-bead	R	1.51	1.15	0.90
	S	11.50	10.80	10.10
	T	21.00	17.50	13.50

To a certain degree, the so-called “*p-factor*” approach is analogous to some atomistic implementations of GB models, where ϵ_{inside} and f_{QB} in certain solvents (usually water) are variables that need to be adjusted based on explicit solvent models. In practice, the approach followed here reduces the number of variables that would be used if all pair interactions between Q and neutral beads would need to be redefined. In addition, it ensures that the interactions follow the expected trends going from very polar to non-polar beads. **Supplementary Figure 5** shows how p_{QB} values change depending on γ_{QB} ,

$\epsilon_{W,inside}$, and $\epsilon_{C1,inside}$. Using default values for $\epsilon_{W,inside}$, $\epsilon_{C1,inside}$, σ_{QW}/σ_{QB} , and γ_{QB} (all variables equal to 1) in **Equation 7**, polar beads (relative permittivity higher than 20) would have almost a constantly shifted p_{QB} correction (*viz.*, the same value applied to water - p_{QW}). By contrast, for intermediate and apolar ones this correction in the LJ is gradually reduced. $\sigma_{QW}/\sigma_{QB} \neq 1$ only for a few C-beads which use super-repulsive levels (so with $\sigma_{QB} > \sigma_{QW}$), which end up reducing the correction for these cases even more.



Supplementary Figure 5: Changes in p_{QB} factor as function of with relative permittivity defined for each bead, from C1 ($\epsilon_B = 2$) to P6 ($\epsilon_B = 50$). The different plots illustrate how p_{QB} changes varying γ_Q , $\epsilon_{W,inside}$, and $\epsilon_{C1,inside}$. The green line in all graphs represents p_{QB} value with $\epsilon_{W,inside} = \epsilon_{C1,inside} = \gamma_{QB} = \sigma_{QW}/\sigma_{QB} = 1.00$ and $p_{QW} = 0.50$. Other curves were obtained varying one of the parameters (indicated at the top of each graph). Different charged beads have different γ_Q , $\epsilon_{W,inside}$, and $\epsilon_{C1,inside}$ values (see **Supplementary Table 22**), which are fine-tuned to mimic differences in hydrophilicity.

It is worth mentioning that the shape of the function p_{QB} added to ϵ_{ij} of the LJ potential would not be easily achieved with a brute-force approach where ϵ_{ij} would be scanned based on **Equation 3** which was used to fine-tune the levels in the organic and solvent blocks. The parameters $\epsilon_{W,inside}$, $\epsilon_{B,inside}$, and γ_{QB} depend on the specific Q-bead modeled and can complement the interaction levels in terms of how ions interact with different neutral beads. They were fine-tuned in Martini 3 to differentiate ions based on a kind of “Martini Hofmeister series”.

The p-factor approach combined with balanced Q-Q interactions allowed for the parametrization of Q beads in terms of charge, size, and chemical type. Small and tiny Q-beads interact more with themselves via Coulomb interactions, but also more with polar solvents; the latter interactions were fine-tuned with the variable p_{QW} . The same concept was used for divalent charged beads, called D-beads. With a higher net charge, it is expected that $p_{DW} \gg p_{QW}$. An important consequence of this approach is that the interaction between Q/D and neutral beads are not symmetric for different sizes (as they are for neutral beads). For example, an interaction Q4-TP5 is weaker than TQ4-P5. For similar reasons, interactions between D- and Q-beads are also not symmetric for the cross-interactions of different sizes. The chemical types of Q- and D-beads and their trends are discussed in **section B1**.

B4) Solvent block

The solvent block represents a class of models in which more than one molecule is grouped into a single bead. This could be especially important for solvents consisting of very small molecules like water, methanol, or ammonia. In these cases, the representation of each molecule by a single bead would negate the advantage of using a CG force field. Besides, given the peculiar situation of grouping molecules, the properties for calibration, such as transfer free energies and miscibility, need to be carefully defined by clever choices and not only by experimental references. For instance, oil/water transfer free energy of solvents beads were compared in relation to one water molecule, as it is not easy to define a good experimental or theoretical reference for dynamic clusters of molecules. On the other hand, as miscibility depends on the collective interactions between the molecules that compose the mixture, the number of molecules represented by the bead needs to be considered for this validation. For instance, a 50:50 (molar fraction) binary mixture of regular water bead (which represents 4 molecules of water) and benzene Martini CG model (1 molecule) should actually use a ratio 20:80.

Currently, the only bead parametrized in the solvent block is water (called “W”). Given its importance, this W bead has a specific parametrization. With respect to Martini 2, the bead has a reduced self-interaction which avoids the necessity of anti-freeze waters. The interactions with other beads from the organic block were also adapted to

recover the solvation free energies of the water model from Martini 2 (using **Equation 3**). The softness parameter of water in relation to the organic and ion blocks (s_{wj}) is 0.02 nm (to be compared to the standard $s_{ij} = 0.01$ nm), which mimics the softer character of a group of water molecules¹⁰. Finally, as for all the other blocks, we also generated smaller beads for water (called SW and TW, representing three and two water molecules respectively).

The interactions between water beads of different sizes were tuned to have an ideal miscibility (see **Figure 1B**). Besides, the solvation free energies of the organic block beads were tuned to be as similar as possible in all water resolutions. SW and TW mix nearly ideally with RW and with themselves. They are useful for special situations, where hydration of small cavities (like a ligand binding pocket in a protein) is necessary.

B5) Parameter calibration, tests and validation

As Martini is based on pair interactions, it is hard to find simple systems for parameter calibration that cover enough points in the interaction matrix for all bead size combinations (which correspond to 326,028 pair interactions in Martini 3). So, complex systems are not only used for validation but are part of the calibration as well. In Martini 2, the calibration of the LJ parameters mainly focused on oil/water transfer free energies of single beads, with a limited number of solvents fully tested¹³. These solvents were water, hexadecane, chloroform, ether, and octanol. For certain solvents, interfacial tension and densities were also checked. Regular Q-beads were mainly parametrized to improve lipid bilayer properties, which were extensively checked using different structural properties. Lipid simulations also helped to fine-tune the choices of levels in the interaction matrix. Although a few additional beads were further added with the parametrization of new biomolecular topologies^{17,18,34–36}, the majority of beads stayed unchanged since the calibration of the parameters in 2007.

In Martini 3, the overall approach was also not based on a rigorous separation of calibration and validation groups. However, there are some differences. First, we expanded the calibration and validation targets. Second, the protocol was separated in “tiers”, which represent groups of tests with different level of complexity. Every time parameters failed to reasonably reproduce the reference data in advanced tiers, the

choices of mapping, bead assignment, and bonded parameters were reviewed. If, after exploring the possible options, the problems persisted, the procedure went back to the first tier (called “0”), where the specific parameters in the interaction matrix and/or labels were retuned. In the early iterations, even new bead chemical types and labels were generated. This iterative process was repeated until a satisfactory agreement between the simulated properties and the target data was achieved across the whole range of tested systems. Each tier is described below, with most of the results obtained distributed in the main text and in **sections E** and **F** of this Supporting Information.

Tier 0: Tests performed with single beads, small molecules, and solvents. The goal was to have the beads following the expected trends and compare simple small-molecule models with experimental data. So, this tier was mainly used for calibration of LJ parameters. The tests performed include the computation of free energies (solvation, hydration, transfer) and miscibility tests. In case of Q particles, trends were compared to available experimental osmotic coefficient values²⁹.

Tier 1: In this intermediate step, initial tests with bilayers and proteins were performed. So, Martini 3 prototype models for these biomolecules were already necessary at this tier. In addition, benchmarks including other important applications were included. Most of the tests were designed as “yes-or-no” questions to evaluate the overall quality of the force field. These “quality control tests” included negative and positive controls for protein-protein and sugar-sugar interactions, as well as checking the performance of reproducing bilayer structural properties for common lipid types. Points in the interaction levels and matrix not covered by tier 0 were also tested and fine-tuned.

Tier 2: Quantitative tests involving complex systems, e.g. aedamers, ionic-liquids, phase coexistence in ternary lipid membranes, dimerization PMFs of membrane proteins, etc. Usually performed at the end of the parametrization workflow, including quantitative comparisons with experimental or atomistic simulation data. Here most of the tests are considered validation, but still with a few refinements in the interaction matrix and interaction levels.

Supplementary Notes C) Concepts and rules for parametrization of Martini 3 models

After the definition and parametrization of the bead chemical types, sizes and labels, the next step is to generate Martini 3 CG models of complex and bigger molecules using more than one bead. For a quick guide of how to build them, the user can (initially) follow these 3 steps:

1) *Mapping of the atomistic model to the CG model*: split the molecule in reasonable chemical groups (the building blocks) in accordance with the Martini bead sizes: R (4-atoms-to-1-bead), S (3-atoms-to-1-bead), and T (2-atoms-to-1-bead). Higher resolution allows for a better reproduction of symmetry, shape, and chemical details. Lower resolutions allow for simpler models which are more computationally efficient.

2) *Assignment of chemical bead types*: based on the chemical group underlying the bead, one selects a similar fragment in the “Martini Bible” (**Supplementary Tables 24 and 25**, with examples of chemical types based on different groups for neutral and charged fragments, respectively). In case no matching chemical group is present, a suitable bead type can be selected using the partitioning data of the isolated fragment (**Supplementary Tables 17, 18 and 19**).

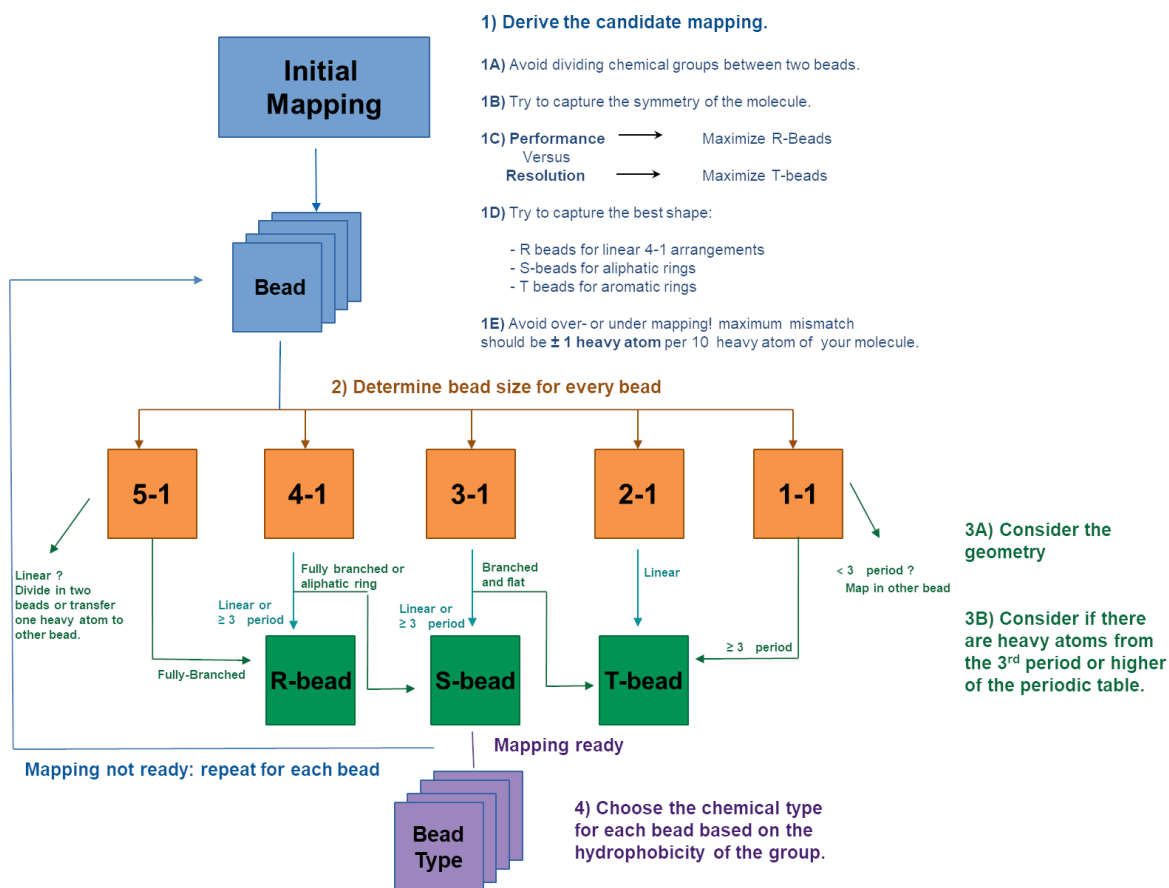
3) *Get the bonded parameters from atomistic data*: bond distances should follow the center-of-geometry “rule” (**Section C3**) including the underlying hydrogen atoms. Appropriate force constants and additional bonded terms such as angles or dihedrals should be calibrated with respect to structural data obtained from either reference atomistic simulations or a structural databank.

For advanced parameterization of Martini 3 CG models, it is important to consider what we call *interaction density*: the interaction energies and the bead density (i.e., the number density of beads) in the model². This concept may be considered analogous to the invariant degree of polymerization concept in polymer field theories, which is a measure related to the number of intermolecular interactions a given polymer chain experiences³⁷. A balanced interaction density guarantees that the Martini CG models do not only have reasonable partitioning and miscibility properties, but also follow correct trends in self-interaction and solvation free energies. In the following section, we describe

in more detail the guidelines summarized above. These guidelines are mainly based on the main class of molecules used to parametrize the neutral beads: organic molecules composed of atoms of the second period of the periodic table, and including sulfur, chloride, and bromide. Some comments about charged molecules and exceptions to the rules are also discussed. Following the guidelines presented here should provide more stable, accurate, and compatible models.

C1) Mapping and choice of bead sizes

Ideally, when generating a Martini CG model, one should avoid dividing chemical groups (e.g., amides or carboxylates) between two beads. Under- or over-mapping should also be avoided whenever possible, with the consequences recently described in Alessandri and Souza *et al.*². In addition, it is desirable to retain as much as possible the volume, shape, and symmetry of the original atomistic molecule. A workflow for mapping CG models in the Martini 3 framework is presented in **Supplementary Figure 6**.



Supplementary Figure 6: General rules for mapping and choice of bead sizes.

Default mapping of linear and ring fragments: Like in the previous version, only non-hydrogen atoms define the mapping. The default option for 4-1, 3-1, and 2-1 mappings are R-, S- and T-beads, respectively. T-beads are usually better to represent the flatness of aromatic rings while S-beads better mimic the “bulkier” shape of aliphatic rings. R-beads are the best option for computational performance, with the bead size reasonably good to represent 4-1 linear molecules with internal flexibility (e.g., aliphatic chains). As a generic guideline, the number of beads should be optimized such that the maximum mismatch in mapping is ± 1 non-hydrogen atom per 10 non-hydrogen atoms in the atomistic structure.

Mapping of branched fragments: Fully branched fragments (such as fragments with quaternary carbon atoms or tertiary amine groups) should usually use beads of smaller size, as the central atom is not exposed to the environment, reducing its influence on the interactions. For instance, a neopentane group contains 5 non-hydrogen atoms but, as it is fully branched, can be safely modeled as an R-bead. Note that neopentane is, indeed, more similar to butane than to pentane in terms of boiling point and oil/water transfer free energies^{38–40}.

The choice of the bead size for partially branched fragments of molecules is not straightforward; it is desirable to validate the choice by comparison to experimental data and/or in relation to other Martini 3 CG models. The main danger of using the “full-branching rule” for partially branched fragments is to reduce the self-interactions too much. Such a reduction can be especially problematic in polymeric molecules. Some other known and useful exceptions for the mapping rules are described below:

Organic compounds with fluorine and iodine: Organofluorines and organoiodines are two extreme mapping exceptions for halo compounds (modelled by X-type beads). As fluorine atoms in halo compounds are very small and have weak VdW interactions, we consider that two fluorine atoms are equivalent to one non-hydrogen atom in these cases. On the other hand, iodine is big and has very strong dispersive interactions which increases the self-interaction in most of the compounds containing this heavy atom. So, in this case, each iodine atom should count as two non-hydrogen atoms in the mapping. Some choices of chemical bead types for all halogens are presented in **section C4**.

Heavy atoms: Atoms from the 4th or higher periods of the periodic table should be carefully considered for the mapping and bead size (as they have not been tested), with the choice validated against experimental data. As default option, we suggest that any atom in periods higher than 3rd one should count as two non-hydrogen atoms, like in the case of iodine. Fragments or molecules containing atoms from the 3rd period that are not well-validated (e.g., silicon) may follow the rules defined in **Supplementary Figure 6**.

Ions and charged groups in general: A strict mapping rule for charged molecules or fragments can also be challenging given that such systems follow different trends compared to the neutral molecules. Localized charged organic groups attached to neutral fragments could be represented by Q-beads. In these cases, the rules summarized in **Supplementary Figure 6** work reasonably well. For small, spherically symmetric soluble ions, mapping can vary from 1-1 (e.g. metals) to 7-1 (e.g., hexafluorophosphate). Thus, having only three different bead size options is, at best, sub-optimal. For these cases, a precise definition for the best bead sizes and types is described in **section C3**.

The rules described here, in particular the workflow illustrated in **Supplementary Figure 6**, should be used as standard procedure. They could also be used as starting point for automated mapping codes such as Auto-Martini⁴¹ (for now only working with Martini 2) or Cartographer (under development in Marrink's group)⁴². Further refinement can always be performed based on the interaction density concept and proper validation against experimental data.

C2) Chemical groups and bead assignment

After getting reasonable mappings, the next step is to select the chemical bead types. Ideally, each chemical group would be represented by a unique and fixed bead, allowing automatic assignment based on any proposed mapping. Unfortunately, such an idea is not the most accurate choice, not only in Martini, but in any approach used to computationally predict thermodynamic properties. Prediction schemes for partition coefficients (logP values) usually use connectivity and proximity parameters as complementary variables in their algorithms⁴³. Even all-atom force-fields have different LJ parameters and/or partial charges for atoms of the same element and chemical group

that are attached to different fragments. For instance, the CHARMM force field uses different LJ parameters for carbon atoms in carbonyl groups present in proteins⁴⁴, nucleobases⁴⁵ and phospholipids⁴⁶.

Different from the previous version, the bead assignment in Martini 3 is based not only on partitioning data of isolated beads but also on partitioning and miscibility data for all the systems tested (around 300 different molecules). The main bead types and labels used for the most common chemical groups in neutral molecules are described in **Supplementary Table 24**. The selected chemical bead types reflect mostly the behavior of isolated chemical groups in carbon-based molecules. They represent a good initial guess for chemical bead types and labels that could be implemented in high-throughput schemes.

Supplementary Table 24. Default bead assignments for neutral molecules/fragments based on the underlying chemical groups mapped.

Bead	Chemical Group	Bead	Chemical Group
C1	linear alkane	N6d	primary amine
C2	branched alkane	N4	secondary amine
C3	cyclic alkane	N3a	tertiary amine
C4	alkene	P1	alcohol
C5	dienes	P2	enol
C5	aromatics	P2	carboxylic acid
C5e	aromatic (no hydrogens)	P2	hemiacetal/hemiketal
C6r	alkynes	P4	diol
C6	thiol	P5	primary amide
C6	sulfide	P3	secondary amide
N1	methyl pyrrole	P3a	tertiary amide
N2	primary imine	P6	sulfoxide
N1a	secondary imine	P6	amino acid
N2ah	acyl chloride	W	water
N2a	nitrile	X1	primary organoiodine
N3r/N3a*	ether	X2	primary organobromide
N4a	acetal/ketal	X3	primary organochloride
N4a	ester	X4e	primary organofluorine
N5a	ketone	X3h	dichloroethane
N5a	aldehyde	X2	trichloromethane
N6	Phenol	X1	tetrachloromethane

* both options could work for ethers depending on the fragments attached.

For a more precise and accurate definition of bead types, proximity with different fragments needs to be considered, especially for smaller bead types (i.e. higher resolution). Usually, around the default solution for a certain bead assignment (defined in **Supplementary Table 24**), small perturbations are allowed depending on proximity/connectivity with other beads. The main aspects that should be considered to refine the types of choices considering proximity are:

1) *Inductive and/or conjugated effect* due to neighboring chemical groups. Changes in chemical type and/or labels “v/e” applied to neighbor beads can be used in these cases. Aadamers are good examples of how methoxy and carbonyl groups can affect the bead assignment of aromatic carbons. Charged fragments can have even stronger effects.

2) *Change in local dipoles* due to proximity between chemical groups, affecting the polarity of the default bead choices. Nucleobases are clear examples of such cases. Another potential change is in the self-interaction, which can be fine-tuned by the labels “h/r”.

3) *Internal hydrogen bonds*, which can change how available protons and electron pairs are in the molecule. These effects can change hydrogen donor/acceptor capabilities (so the best choice of non-labeled or “d/a” labels) and also the oil/water transfer free energies.

4) *Resolution of the model*, which is defined by the choices in mapping and bead size. Illustrative examples are linear alkanes. While low resolution models are based on regular size C1 beads, higher resolution models can differentiate the beads types placed in the middle (C1 type) or the terminals (C2 types) of the carbon chain. These differences are also present in atomistic models, which have different LJ parameters for carbon atoms bound to different number of hydrogens. The difference in bead types used for linear and branched alkanes is also a reflection of the different number of terminal and middle carbon atoms mapped into the bead.

In complex cases, multiple proximity effects may be present in the same molecule, which makes the best bead choice non-trivial. In such situations, we recommend the users to get inspiration from the current CG models of Martini 3 (available at <http://cgmartini.nl/>). Further expansion of the Martini 3 small-molecule database could

help to better refine such ideas, improving the assignment of the best chemical bead types and labels.

Given the diversity of charged groups represented by Q-beads and the sub-optimal representation of all different sizes that ions can have, a more precise bead assignment is required. **Supplementary Table 25** shows the current best options for ions, including chemical types, sizes, and net charge.

Supplementary Table 25. Default bead assignments and net charge for cations and anions.

Bead (charge)	Cation	Bead (charge)	Anion
TQ5 (+1)	sodium	TQ5 (-1)	chloride
SQ4 (+1)	potassium	SQ4 (-1)	bromide
TQ5p (+1)	ammonium	SQ2(-1)	iodine
Q4p/SQ4p (+1)	alkyl ammonium	Q2(-1)	tetrafluoroborate
Q3p/SQ3p (+1)	alkyl methyl ammonium	Q1 (-1)	hexafluorophosphate
Q2p/SQ2p (+1)	alkyl dimethyl ammonium	SQ1 (-1)	thiocyanate
Q2 (+1)	tetramethyl ammonium	SQ3 (-1)	nitrate
Q1 (+1)	choline (lipid head))	Q2 (-1)	perchlorate
Q1 (+1)	tetramethyl phosphonium	Q5n/SQ5n (-1)	carboxylate
SQ3p (+1)	guanidinium	Q4n/SQ4n (-1)	sulfonate
TD (+2)	magnesium	Q5 (-1)	phosphate (lipid head)
SD/TD* (+2)	calcium	D (-2)	phosphate

* Depending of the application, TD could be useful.

C3) Bond distances based on molecular volume and shape

Angles and dihedrals impact the quality of CG models as they control the conformational flexibility of the molecule. In Martini 3, bond distances are equally important, since they will define the volume, shape, and bead density of CG models, which have important consequences for the overall behavior of the force field. Here, we first describe a general rule to get reasonably good models, followed by a description for volume-shape based refinement.

Center-of-geometry mapping: In general, bond distances obtained from center of mass (COM)-mapped atomistic structures proved unsatisfactory, as they often led to overly high packing densities. This is due to the geometrical mismatch enforced by neglecting the lower resolution of CG models (e.g., triangular versus hexagonal shape of benzene) and

by not taking into account the volume of hydrogen atoms. Instead, using a center of geometry (COG)-based mapping of atomistic structures, taking hydrogen atoms into account, allows to better reproduce the molecular volume and mass densities. Therefore, bond distances obtained from COG-mapped atomistic structures constitute the standard procedure for obtaining bonded parameters in Martini 3.

COG-mapping is especially important the closer the mapping is to atomistic resolution, *i.e.* when using T-beads. COM- and COG-based bond distances are indeed almost equivalent in the case of a 4-to-1 mapped alkane chain. In contrast, they differ considerably in the case of 2-to-1 mapped rings. For instance, benzene has a COM bond length of 0.21 nm, while a COG provides 0.29 nm. COG-based, like COM-based, bond distances can be directly extracted from atomistic structures. Thus, bond distances extracted from COG-mapped atomistic structures should be the standard for automated parametrization schemes, such as the Auto-Martini⁴¹ or PyCGtool⁴⁷ algorithms.

Refinement based on volume and shape: Given the CG resolution of Martini models, perfectly matching densities or packing cannot be obtained for all cases simply following a COG-based mapping scheme. Therefore, bond distances can be fine-tuned further if higher accuracy is required. To this end, the solvent accessible surface area (SASA), molecular volume and shape can be used as targets to optimize bond distances.

To compute the SASA, one can use the GROMACS^{11,12} tool `gmx sasa`. For atomistic references, we recommended the van der Waals (vdW) radii from the work of Rowland and Taylor⁴⁸. For Martini CG molecules, the vdW radii for R-, S-, and T-beads are 0.264, 0.230, and 0.191 nm, respectively. These are computed as the position of the minimum of the LJ potential, given the LJ σ values of the various beads:

$$r_{\text{vdW}} = r_{\text{min}} = \frac{d_{\text{min}}}{2} = \frac{\sqrt[6]{2} \sigma}{2} \quad \text{Equation 9}$$

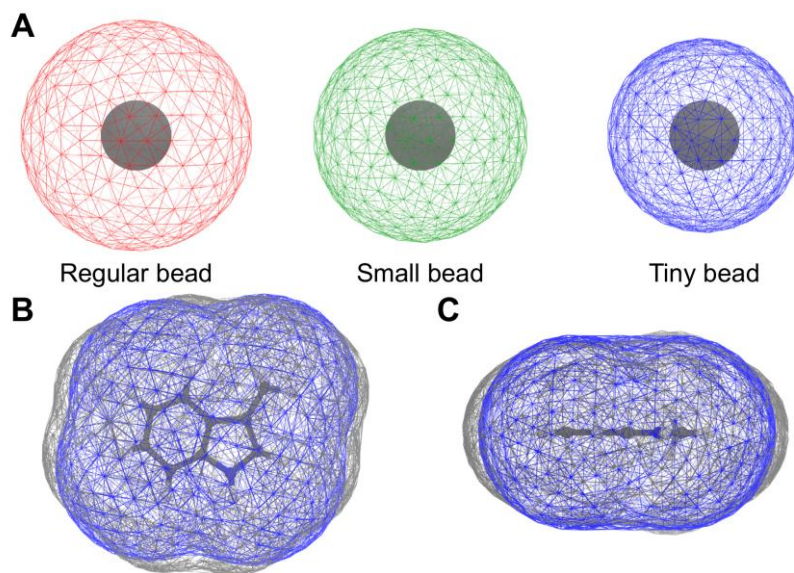
At both AA and CG levels, we used the command:

```
gmx sasa -s mol.pdb -o sasa.svg -probe 0.191 -ndots 4800
```

where: mol.pdb (or mol.gro or trajectory.xtc) is the file containing the molecular geometry (see also below); sasa.xvg is the output file; the flag -probe specifies the size of the probe, for which we chose the T-bead size. The probe size does not necessarily need to be this one, but it is essential to use the same probe size when comparing SASA values for different molecules and models, including the comparison of atomistic and CG results. The -ndots flag, which specifies the accuracy of the SASA calculation, should be set to at least 4800 points for accurate SASA values.

For rigid molecules, such as a small aromatic rings, SASA values have been computed for energy-minimized AA and CG geometries. For more flexible molecules, distributions of SASA values along AA and CG trajectories may be compared. Finally, a Connolly surface (surf.pdb) can also be generated to be able to visualize the molecular volume in 3D. This can be done with the following command:

```
gmx sasa -s mol.pdb -o sasa.xvg -probe 0.191 -ndots 240 -q surf.pdb
```



Supplementary Figure 7: Connolly surfaces for Martini beads and models. A) Connolly surfaces for the three Martini 3 bead sizes: regular (red), small (green), and tiny (blue). A comparison of the Connolly surfaces of atomistic (grey) and Martini 3 (blue) 3-methylindole is also shown: B) top view, C) side view. Tiny beads are required to describe atom-thick structures such as this fused aromatic ring.

Note that for generating a surf.pdb file which is more easily processed by a program for visualization such as VMD⁴⁹, the number of points (-ndots) for the calculation can be reduced to, e.g., 240. Such surfaces can be computed for both AA and CG models, as done for the molecule of 3-methylindole in **Supplementary Figures 7B-C**. The aligned surfaces help to identify the bond distances which can be optimized based on discrepancies between the AA and CG surfaces. A tutorial with more and up-to-date details and examples is provided at <http://cgmartini.nl/>.

Bonded parameters, force constants, and exclusions: In Martini 3, non-bonded interactions between connected beads are always excluded, both in the default bonded terms as in elastic network of proteins. As demonstrated by Alessandri *et al.*², in such condition, weak bond force constants (lower than 700 kJ/mol) should be avoided, as they may allow the CG beads within a molecule getting too close, resulting in too high interactions with the environment and causing artifacts (e.g., particles sitting on top of each other, and excessive aggregation)².

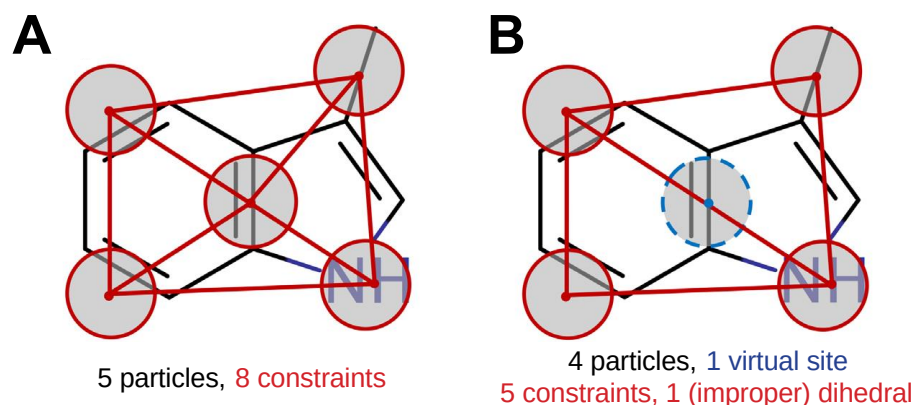
Exclusion can also be applied to 2nd or 3rd neighbors to help the representation of angle and dihedral distributions in certain cases. For rings, even higher orders of exclusion are used, which can help to keep their conformation numerically stable. In particular, exclusion always should be used between non-dummy virtual particles (see **section C4**) and the reference beads used for their construction.

C4) Numerical stability of simulations

A last aspect that needs to be carefully considered for Martini 3 models is their stability in MD simulations. The target is to get models stable using 20 fs as time step, to optimally profit from the computational speedup while keeping energy drifts within acceptable boundaries¹⁴. This, however, can be a challenge. For instance, DNA and RNA models based on the previous version of Martini^{17,18} were usually only stable with a time step of 10 fs. The following sections describe the three main approaches that should be considered for numerically stable simulations: (i) number of constraints and usage of virtual particles; (ii) dihedrals and 3-body angles around 0°/180°; and (iii) bead masses. As the choices of mapping and bonded potential can have an influence on (i) and (ii), we

advise users to always consider potential problems in numerical instability at the beginning of the parametrization.

Use of virtual sites to reduce the number of constraints: For stiff molecular fragments, such as polycyclic compounds requiring more than four Martini particles to be described, virtual (interaction) sites (VSs)⁵⁰ are employed to improve the numerical stability of the models. Martini aromatic ring structures are held together by constraints. As more extended stiff molecular structures need to be described, the number of constraints that need to be used in their Martini representations grows. However, an extended network of constraints leads to numerical instabilities. The use of VSs allows to save on the number of constraints thereby increasing the numerical stability of models. Besides, it also improves the computational performance⁵¹.



Supplementary Figure 8: Use of virtual sites to improve the numerical stability of Martini models: the case of the 3-methylindole molecule. Two possible models for 3-methylindole: **(A)** a “full-constraints” model, susceptible to numerical instabilities, and **(B)** a model leveraging the numerically stable “hinge” construction.

As an example, we describe the proposed Martini 3 model for 3-methylindole, which is used in the description of the amino acid tryptophan. The molecule consists of 10 non-hydrogen atoms, which translates into a 5 T-beads model given the aromaticity of the molecule. Straightforwardly interconnecting the 5 beads with constraints leads to a network of 8 constraints (**Supplementary Figure 8A**). Such a network of constraints is hard to satisfy: a single molecule in vacuum is still numerically stable with a time step of 20 fs, but a condensed phase containing many copies of such a model readily leads to

numerical instabilities with time steps as low as 10 fs, thus requiring even smaller time steps⁵¹. This makes large-scale simulations impracticable. The idea to overcome this limitation is inspired by the “hinge” construction used in the latest Martini models of sterols and hopanoids proposed by Melo and co-workers⁵². Indeed, another model which can be devised for 3-methylindole exploits the “hinge” construction for the 4 external beads (**Supplementary Figure 8B**) while the central bead is described as a VS, *i.e.*, a particle whose position is completely defined by its constructing particles (in this case, the other four beads). Due to the reduction on the number of constraints used (from 8 to 5), the use of the hinge model also accelerates the calculation: in the case of the same 3-methylindole condensed phase, the simulation runs ~1.5 times faster using the same time step. Note also that an improper dihedral is applied in order to keep the hinge model flat. More details and further constructions of ring compounds will be described in a future publication⁵¹.

Numerical instabilities caused by dihedral potentials: As described by Bulacu *et al.*⁵³, traditional torsion angle potentials (*i.e.* proper dihedrals) used in atomistic MD simulations may not always be suitable for modeling CG systems. This is true, for example, in very flexible CG molecules, for which conformations leading to numerical instabilities of the dihedral potentials are often encountered.

Specifically, given a dihedral potential between particles i - j - k - l , numerical instabilities are encountered when either of the angles i - j - k and j - k - l are 0° or 180° . Bulacu *et al.* proposed two solutions⁵³: 1) the use of a newly developed Restricted Bending (ReB) potential (function type 10 in GROMACS^{11,12} 2016.x and newer releases) to prevent the i - j - k and j - k - l from exploring conformations at 0° or 180° ; 2) the use of virtual, or dummy, particles in a so-called dummy-assisted dihedral (DAD) to redefine the dihedral in a way that, for simple geometrical reasons, the angles i - j - k and j - k - l cannot explore the numerically unstable regions around 0° or 180° .

Bulacu *et al.* showcased the use of these approaches in the PEG model, which by using ReB potentials is numerically stable with a time step of 20 fs, unlike the original Martini PEG model of Lee *et al.*⁵⁴ for which numerical instabilities are often encountered with a time step of 10 fs⁵³. The ReB potentials prevent three consecutive PEG beads from

being colinear, i.e. to reach an angle $i-j-k$ of 180° . The approach was also suggested to be used in combination with side-chain dihedral corrections in Martini proteins⁵⁵. In the DAD approach, the use of virtual sites allows to define dihedrals which would otherwise not be possible to define based simply on the structure of the CG model⁵³. The position of the virtual sites is chosen in a way that will restrict the conformations of the $i-j-k$ and $j-k-l$ angles to numerically stable regions of the conformational space, i.e. away from 0° and 180° . The protein backbone⁵³ and polythiophenes⁵⁶ are examples of models that can employ dihedrals which use virtual sites.

Masses of the beads in Martini 3: By default, we use standard masses for the three Martini 3 bead sizes (described in **section B1**), like in Martini 2¹³. These masses were chosen for simplicity and their use guarantees no numerical instabilities with a time step of 20 fs, as validated by all the tests carried out in this work. For analysis of trends in kinetic properties, simulations should be carried out with more realistic masses. However, if the realistic mass of a certain molecule or fragment is much lower compared to when standard masses were used, numerical instabilities may arise also with time steps of 20 fs. This was not extensively tested and we wish to warn users about this possibility.

Supplementary Notes D) Additional Methods

D1) Oil/water transfer free energies

Thermodynamic integration⁵⁷ was used to compute free energies of solvation ($\Delta G_{S \rightarrow 0}$). The solute was solvated in a pre-equilibrated solvent box of size of at least 5x5x5 nm³. Four solvents were considered: pure water, pure hexadecane, pure chloroform and water-saturated octanol (~0.25:0.75 water/octanol molar ratio^{58,59}). A minimum of 11 simulations with equally spaced points going from 0 to 1 were performed, where solute-solvent interactions were scaled from full ($\lambda=1$) to zero solute-solvent interactions ($\lambda=0$). A stochastic integrator⁶⁰ was employed for the simulations that were equilibrated for 2 ns, and each λ point was run for a minimum of 10 ns. A soft-core potential (as implemented in GROMACS^{11,12}) was used to avoid singularities due to solute-solvent particle overlap as interactions were switched⁶¹. The soft-core parameters used were $\alpha=0.5$ and $p=1$ (for the soft-core power). The free energies and corresponding errors were computed using the Multistate Bennett Acceptance Ratio (MBAR)⁶². The transfer free energy associated with transferring a solute from a solvent S1 to a solvent S2 ($\Delta G_{S1 \rightarrow S2}$) is then computed as the difference $\Delta G_{S1 \rightarrow S2} = \Delta G_{S1 \rightarrow 0} - \Delta G_{S2 \rightarrow 0}$. Results in water/hexadecane, water/octanol and water/chloroform were compared with experimental data^{21,63–70}.

D2) Pure liquids: densities and enthalpies of vaporization

A liquid phase was approximated as an equilibrated box of dimensions of about 5x5x5 nm³. Densities were extracted with the GROMACS^{11,12} tool *gmx energy*, based on the volume of the simulation box and the real masses of the molecules being simulated (note that otherwise usually the CG model for a given molecule uses the default, size-dependent masses of the beads, described in **Supplementary Notes, section B2**, rather than the exact mass of the atoms represented by the beads). For the enthalpy of vaporization, an empty simulation box (7x7x7 nm³) containing one molecule was also built to mimic the gas phase. Simulations of the liquid phase were equilibrated in the NPT ensemble at 298 K and 1 bar, while gas phase simulations were simulated in the NVT ensemble at 298 K. The enthalpy of vaporization (ΔH_{vap}) was computed as $\Delta H_{\text{vap}} \approx U_{\text{gas}} -$

$U_{liq} + RT$, where U_{gas} and U_{liq} are the total energies per mole of the gas and liquid phase in the NVT and NPT ensembles, respectively.

D3) Multi-resolution mixtures: dodecane and water

CG models of dodecane and water were simulated in 3 resolutions as a proof of concept to show that the interaction levels in Martini 3 are well-balanced. Simulation boxes with dimensions around $7 \times 7 \times 7 \text{ nm}^3$ were filled with the pure solvents, which were represented by ternary mixtures of three different levels of Martini representation in equal molar ratio. CG molecules were placed in random initial configurations using Packmol⁷¹. Energy minimization (500 steps of steepest decent) and equilibration (100 ns trajectory with a 10 fs time step) were performed, followed by 1 μ s of production simulation in the NPT ensemble at 1 bar and 300 K. To evaluate how close to an ideal solution behavior the multi-resolution system were, we computed the difference in Kirkwood–Buff (KB) integrals ($\Delta G_{ij} = G_{ii} + G_{jj} - 2G_{ij}$) for all i, j pairs. As detailed by Ben-Naim^{72,73}, $\Delta G_{ij} = 0$ for ideal solutions including up to four components. KB integrals (G_{ij}), in units of volume per mol, were computed according to **Equation 10**:

$$G_{ij}(R) = \frac{4\pi}{n_R} \int_0^R [g_{ij}(r) - 1] r^2 dr \quad \text{Equation 10}$$

with g_{ij} corresponding to the radial distribution function of the center of mass of the species i and j , r their intermolecular separation, R a cutoff distance (defined as the distance where G_{ij} approaches a plateau value), and n_R the number of moles at distance R (i.e. the average number of molecules j contained in the volume of the sphere of radius R centered on the molecule i divided by Avogadro's number). g_{ij} was computed with the GROMACS^{11,12} tool *gmx rdf*. For the sake of clarity, ΔG_{ij} values were compared with results of real ideal mixtures using experimental and atomistic simulation data. Atomistic force fields based on Kirkwood–Buff integrals usually show errors in ΔG_{ij} around $\pm 10 \text{ cm}^3/\text{mol}$ ⁷³.

D4) Binary mixtures: interfacial tension and excess free energy of mixing

Binary mixtures of single regular beads were simulated to obtain a qualitative view of their miscibility. An equal number of beads A and B was placed in simulation boxes of dimensions of $\sim 9 \times 9 \times 9 \text{ nm}^3$. The box was then energy-minimized, simulated at a higher pressure (100 bar) for 1 ns and then subsequently relaxed for 1 ns in the NP_zT ensemble (with x and y dimensions fixed and z the normal dimension on the interface) at 1 bar and 300 K. Then, simulations were run in the same ensemble for at least 400 ns. Miscibility was monitored by visual inspection and also based in the number of A-B contacts determined with the GROMACS^{11,12} tool *gmx mindist* (see **Supplementary Notes, section A6**). This method was proposed by Barnoud *et al.*⁷⁴ to measure the degree of phase separation in bilayers. A similar procedure was used to obtain a qualitative view of the miscibility of CG solvent models (that may consist of single or of multiple beads). In this case, the results were directly compared with experimental data. Around 400 binary mixtures in equal molar ratio were evaluated (see **Supplementary Results, section E3**). For a quantitative validation of some biphasic heterogeneous mixtures, the interfacial tension was computed with the GROMACS^{11,12} tool *gmx energy* (see **Supplementary Results, section E5**). For selected homogeneous mixtures, the excess free energy of mixing (ΔG_{ex}) was computed, based on vapor–liquid equilibrium simulations. In this protocol, the systems were converted into a bigger box of $9 \times 9 \times Z \text{ nm}^3$ where Z ranges from ~ 20 to $\sim 30 \text{ nm}$ depending on the liquid, at least 7 nm of which are empty to constitute the vapor phase (see **Supplementary Results, section E4**). The vapor–liquid binary systems were then simulated for at least 2 μs in the NVT ensemble at a temperature between 293.15 K and 308.15 K, depending on the available experimental data. This leads to statistically reliable vapor and liquid densities which can be extracted with the GROMACS^{11,12} tool *gmx density*. The equilibrium densities thus obtained were then used to compute the $\Delta G_{\text{solv},i}$ of the *i*-th component in the binary mixture, according to the scheme of Ben-Naim⁷², given by **Equation 11**:

$$\Delta G_{\text{solv},i} = RT \ln \left(\frac{\rho_{\text{vap},i}}{\rho_{\text{liq},i}} \right) \quad \text{Equation 11}$$

where $\rho_{vap,i}$ and $\rho_{liq,i}$ are the vapor and liquid number densities. ΔG_{ex} can be computed as a function of the molar fraction of the components (x_i) and $\Delta G_{sol,i}$ in the mixture and pure solvent ($\Delta G_{sol,i}^*$), as shown in **Equation 12**:

$$\Delta G_{ex} = x_A(\Delta G_{sol,A} - \Delta G_{sol,A}^*) + x_B(\Delta G_{sol,B} - \Delta G_{sol,B}^*) \quad \text{Equation 12}$$

Miscibility, interfacial tension and excess free energies were only compared with experimental data^{20,75,76}, not with atomistic simulations.

D5) Dimerization PMFs of nucleobases

PMFs were calculated to estimate base pairing free energies of CG models of cytosine and guanine. The PMF profiles were obtained from umbrella sampling (US)⁷⁷ simulations as described for the parametrization of nucleobases in Martini 2^{17,18}. Namely, two bases were placed in a box with about 500 water beads. For the CG systems, windows were spaced 0.05 nm apart from 0.3 to 2.0 nm between the centers of mass of the bases. The distance between the bases was restrained with a harmonic umbrella potential with a force constant of 2,000 kJ·mol⁻¹·nm⁻². Each window was simulated for 100 ns. The nucleobases were constrained to the same plane using position restraints in the direction perpendicular to the reaction coordinate. They were aligned so that the X-axis passed through beads corresponding to the N1 atom of the purines and the N3 atom of the pyrimidines, and the base plane was set as the XY plane. Position restraints with a force constant of 4,000 kJ/(mol·nm²) were used in the Y and Z directions. The free energy profiles were computed using the weighted histogram analysis method (WHAM)⁷⁸ as implemented in the GROMACS^{11,12} tool *gmx wham*, with the statistical error estimated from bootstrapping. For the AA systems, PMFs obtained using CHARMM27^{79,80} and AMBER99sb_parmbsc0⁸¹ were retrieved from Uusitalo *et al.*¹⁷, and hence were obtained as described for the parametrization of the previous Martini models of nucleic acids^{17,18}. Note that we did not attempt to correct the PMFs for the biases introduced by the various constraints, as we only compare different force fields to each other.

D6) Lipid bilayer simulations

We tested 100 different single-component lipid bilayers with Martini 3 and compared the results to Martini 2. The program *insane.py*⁸² was employed to set up the bilayers containing 242 lipids. All bilayers were built in an initial box of 8.8×8.8×8.0 nm³ and solvated with ~3,000 CG water beads. For anionic lipids, the bilayer was neutralized with Na⁺ ions before adding the solvent. The temperature in the simulation box was controlled by a velocity rescale thermostat⁸³ (reference temperature $T_{\text{ref}} = 310$ K, coupling time constant $\tau_T = 1$ ps). Semi-isotropic pressure coupling with a compressibility of $3 \cdot 10^{-4}$ bar⁻¹ was employed. During the equilibration procedure, the pressure was controlled by the Berendsen barostat⁸⁴ (reference pressure $p_{\text{ref}} = 1$ bar; coupling time constant $\tau_p = 1$ ps), while the Parrinello-Rahman barostat⁸⁵ was used during the production ($p_{\text{ref}} = 1$ bar; $\tau_p = 12$ ps). After a steepest decent minimization of 500 steps, the system was equilibrated for 500 ps (time step of 10 fs) and 30 ns (time step of 20 fs) using a leap-frog integrator. Production runs of 1 μ s were performed with a time step of 20 fs. For all systems, we analyzed the average area per lipid, area compressibility, average tail order parameter, and bilayer thickness. All reported errors are standard errors of the mean obtained from block averaging.

A slightly different procedure was used to study the temperature dependence of lipid bilayers properties. The initial conditions were generated using Martinate (see **Supplementary Codes, section H1**) for the desired temperature. The generated initial bilayer patches had sizes 10×10 nm with the bilayer repeat distance of 10 nm. The systems were equilibrated at their respective temperatures for 1 ns prior to production simulations of 1 μ s. The area per lipid from simulations was calculated from the lipid bilayer area by dividing it by the number of lipids per leaflet. The lipid bilayer area was calculated using FatSlim⁸⁶ using phosphate “PO4” beads to construct the surface.

D7) Cation- π interactions and peripheral membrane proteins

The wild-type *Bacillus thuringiensis* phosphatidylinositol-specific phospholipase C (*BtPI-PLC*) structure was taken from previous work^{87,88}, and is a model built from the structure of the Y247S/Y251S mutant (PDB id: 3EA1)⁸⁹ to which tyrosine side chains taken from the W47A/W242A mutant (PDB id: 2OR2)⁹⁰ were added. The protein was placed close

to the membrane surface in the membrane binding orientation using the *insane.py* tool⁸². The distance between the centers of mass of the protein and of the membrane was equal to 6 nm. The membrane binding orientation was obtained from previous atomistic simulations^{87,88}. A pure DMPC bilayer was constructed with 144 lipids in each leaflet. The system was solvated with 6426 CG water molecules and 7 Na⁺ counter ions were added to neutralize the system. A Gō-model was imposed on the protein structures⁹¹. Identical results were obtained using Elastic Networks⁹². The resulting protein-membrane system was briefly minimized for 500 steps of steepest descent. The system was then equilibrated for 1 ns using a time step of 20 fs in the NPT ensemble, followed by a 2 μs simulation in the NPT ensemble using the same time step and semi-isotropic pressure coupling. The temperature was set to 310 K and the pressure to 1 bar. To characterize the binding of the protein to the bilayer, we calculated the depth of insertion of each of the phospholipase's residues in the bilayer below the phosphodiester plane. The depth of insertion is defined as the distance along the membrane normal *z* between the center of mass of the residue and the average phosphodiester plane of the binding leaflet. A positive value means that the residue is above this plane, a negative one that the residue is inserted in the bilayer. Standard deviations indicate the fluctuations of the depth of insertion of a residue in the membrane bound state, and also highlight the stability of the insertion.

D8) Ionic liquids and ion transfer

Ionic liquid/water solution (IL/W) mixtures (~1:1 weight ratio) were built in simulation boxes whose dimensions ranged around ~10x10x13.5 nm³. The IL phase was described by CG models of a quaternary phosphonium cation, trihexyl(tetradecyl) phosphonium ([P₆₆₆₁₄]⁺), in combination with either hexafluorophosphate [PF₆]⁻ or bromide (Br⁻). The water phase was composed of a 200 mM aqueous solution of two salts: NaCl and NaClO₄. In total, 4 systems were tested, considering all combinations of IL/W mixtures: [P₆₆₆₁₄][PF₆]/NaCl(aq), [P₆₆₆₁₄][Br]/NaCl(aq), [P₆₆₆₁₄][PF₆]/NaClO₄(aq) and [P₆₆₆₁₄][Br]/NaClO₄(aq). The GROMACS tools^{11,12} *gmx solvate*, *gmx insert-molecules* and *gmx genion* were used to construct the initial configurations, with all components placed in random initial configurations. After 5,000 steps of minimization and 500 ns of

equilibration (time step of 10 fs), we performed 800 ns of production simulation. Equilibration and production were run at 1 bar and 300 K in the NP_zT ensemble, with X and Y dimensions fixed. PME⁹³ was used to calculate the long-range electrostatic interactions beyond the cut-off. In agreement with both experiments and atomistic simulations, all four systems formed biphasic heterogeneous mixtures^{94,95}. The percentage of anion transfer was computed based on the partial densities using the GROMACS tool^{11,12} *gmx density*. The mean standard errors were obtained from block averaging. The results were compared with the experimental work of Dupont *et al.*⁹⁴ and Naert *et al.* 2018⁹⁶. More information about the CG models and validation of the ionic liquids is presented in **Supplementary Results, section F2**.

D9) Osmotic coefficients of ionic solutions

The osmotic coefficient was computed from simulations adopting the protocol originally proposed by Luo and Roux³¹. A rectangular box was created in which the solute molecules were confined in z-direction by a flat-bottomed potential to the center of the box. At a distance of 2.52078 nm from the center of the box, a harmonic force with a force constant 1000 kJ/nm² was applied to the solute molecules. The box dimensions were taken to be 10.08312 nm in z dimensions and 5.04156 nm in x and y. Previous to each run, a random configuration of solute and solvent molecules was created with polyply (https://github.com/marrink-lab/polyply_1.0) placing solute molecules only in the center of the box and the solvent in the entire box. After energy minimization, this setup was subjected to a 10 ns equilibration using Berendsen barostat⁸⁴. Production simulations were run for 500 ns as previously used for atomistic simulation²⁹ at a pressure of 1 bar and coupling to a temperature of 298.15 K using the velocity rescale algorithm by Bussi *et al.*⁸³ Following Sauter and Grafmuller^{97,98}, the pressure was coupled only in z-dimensions with Parrinello-Rahman pressure coupling⁸⁵. In the original implementation³¹ the simulation was run in an NVT ensemble. PME⁹³ was used in all simulations.

The osmotic pressure was computed from the trajectory by recalculating the total force exerted by the solute particles onto the flat-bottomed potential averaged over the two potentials. Subsequently that force is divided by the xy area of the box. The ensemble average as well as an error were computed from the time-series of the osmotic pressure

values. We verified that our setup yields accurate results by reproducing the osmotic pressure of NaCl at 1 M using CHARMM27 from the original paper³¹. Having obtained the osmotic pressure the osmotic coefficient was computed by dividing the osmotic pressure by the ideal pressure at that given total concentration³¹. We also ran a second replica of each simulation given the importance of replicas in MD simulations⁹⁹. However, as previously observed⁹⁷, the convergence is fast and all values were reproducible within the statistical error of the first run. All statistical errors were less than 1.4%. In total the osmotic coefficient was computed for 10 systems: 5 ion pairs (sodium chloride, potassium chloride, sodium glutamate, lysine chloride, arginine chloride) and 5 zwitterionic molecules (alanine, glycine, valine, serine, threonine) with no net charge. With the exception of valine, all coefficients were calculated at a total molar concentration of 1 M. For the purpose of comparing to previous atomistic simulation data the concentration of valine was 0.6 M. It should be noted that the zwitterionic part of the amino acids is represented by a neutral bead (P6), which means these molecules are neutral in the simulation. More information about osmotic coefficients is presented in **Supplementary Results, section F3**.

D10) Bulk heterojunction morphology: solvent evaporation and annealing

The solvent evaporation protocol follows the one developed by Lee and Pao¹⁰⁰, and refined by Alessandri *et al.*⁵⁶. A mixture of P3HT and PCBM (1:0.8 weight ratio) dispersed in a chlorobenzene solution was placed in a simulation box with dimensions 30×30×88 nm³. The lateral box dimensions were fixed. The solvent was removed every 19 ns of MD simulation until a dried blend was obtained (30×30×~5 nm³). The degree of polymerization of (regioregular) P3HT employed was 24, approximately corresponding to molecular weights of 4,000 Da. After every round of solvent removal, NPT equilibrations followed by production runs (both at 1 bar and 298 K) were performed using the procedure described in detail in Ref.⁵⁶, for a total drying time of around 11 μ s. The final dried configuration was annealed by running MD simulations at higher temperatures, with the temperature being raised gradually to 698 K during 2 μ s (20 ns at 398 K, 20 ns at 498 K, 160 ns at 598 K, and 1.8 μ s at 698 K) and then cooled down by performing 400 ns of MD simulation at 298 K⁵⁶. The simulated diffraction pattern was obtained based on the

distances between all the thiophene centers of mass, as described in Alessandri *et al.*⁵⁶. The scripts used to perform solvent evaporation simulations and to compute the diffraction patterns are available at <https://github.com/ricalessandri/>.

D11) Aedamers: PMF and self-assembly of DAN and NDI tetramers

PMFs were calculated using US⁷⁷, extracting the free energies by WHAM⁷⁸, as implemented by the Grossfield lab¹⁰¹. Pegylated monomers of dialkoxynaphthalene (DAN) and naphthalene diimide (NDI) were placed in a 7.8x7.8x7.8 nm³ simulation box with around 3920 water beads using the GROMACS tools^{11,12} *gmx solvate* and *gmx insert-molecules*. The reaction coordinate was the distance between the centers of aromatic cores; 20-25 windows spaced 0.1 nm apart were used, with a simple harmonic umbrella potential with a force constant of 100 kJ/(mol·nm²). The minimum sampling time for each window was 60 ns. The PMFs were used to calculate free energies of dimerization presented in **Figure 2A** of the main text. Statistical errors were estimated from bootstrapping. The DAN and NDI cores of the pegylated molecules were used to build DAN and NDI tetramers that are connected by charged amide linkers. Experimental work demonstrated that these molecules associate more strongly upon elongation: when longer chains dimerize, they form more DAN-NDI contacts^{102–104}. One DAN and one NDI tetramer were placed in a 10x10x10 nm³ simulation box with 8090 water beads, neutralized with 10 Na⁺ ions. After minimization and equilibration, a total of 100,000 structures sampled at regular intervals from 20 μ s of simulation, taken after the dimer was formed, were clustered. A snapshot of one of the most populated canonically fully alternating structures is presented in the main manuscript. Further details about the CG models, atomistic simulations used as reference and extra details of the analysis performed are described in **Supplementary Results, section F4**.

D12) Water-soluble proteins and polyleucine helices in bilayers

Simulation boxes were built using *insane.py*⁸² and the GROMACS tool^{11,12} *gmx genconf*. For water soluble proteins, 3 systems were tested: the headpiece domain of chicken villin¹⁰⁵, and the modified and mutated cellulose-binding domain from *Cellulomonas fimi* (EXG:CBM)¹⁰⁶. For each system, eight copies of the protein were placed in a 14x14x14

nm³ simulation box, filled with around 26,000 water beads, which correspond to 2 mM protein solutions. These systems were tested in both pure water and 200 mM NaCl. As a model transmembrane (TM) protein, we selected a polyleucine (K₂-L₂₆-K₂). Nine copies of this TM helix were embedded in pure 1,2-dilauroyl-sn-glycero-3-phosphocholine (DLPC) and 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC) bilayers, with an overall box size of 15x15x10 nm³. The simulation systems contained 567 lipids, 14,211 water beads and 150 mM of NaCl. Both soluble and transmembrane protein systems were energy-minimized (500 steps of steepest descent), and equilibrated (1 ns with 10 fs as time step). Production runs of 20 μ s were performed at 310 K and 1 bar. The last 19 μ s were analyzed with the GROMACS tool^{11,12} *gmx cluster*, using the GROMOS algorithm¹⁰⁷. All reported errors are obtained from block averaging.

D13) TM proteins and dimerization PMFs

We computed dimerization free energies for four selected transmembrane (TM) helices: the TM domains of the receptor tyrosine kinases EphA1 (PDB id: 2K1L)¹⁰⁸ and ErbB1 (PDB id: 2M0B)¹⁰⁹; the red blood cell protein glycophorin A (GpA, PDB id: 1AFO)¹¹⁰; as well as the well-known model peptide WALP23^{111,112}. For each of the four peptide dimers, we estimated the dimerization free energies by calculating the PMFs with the US technique and WHAM⁷⁸.

For EphA1 and ErbB1, we followed the setup used in Javanainen *et al.*¹⁶. The simulation models followed experiments^{113,114} as closely as possible: both dimers were embedded in a DLPC bilayer comprised of 400 lipids, solvated by 5,000 water beads, and either ~150 mM (EphA1) or ~500 mM (ErbB1) of NaCl. The systems were generated using *insane.py*⁸². Equilibration simulations of ~50 ns were first performed on a restrained dimer and then on an unrestrained dimer. The 2D inter-peptide center-of-mass (COM) distance was used as a reaction coordinate (ζ). 15 frames were chosen as the umbrella sampling windows, with reference values for ζ ranging from 0.6 to 3.4 nm with a spacing of 0.2 nm. The spring constant of the umbrella potential was set to 400 kJ/(mol·nm²). The initial frames were taken from the work of Javanainen *et al.*¹⁶, also available in the Zenodo repository¹¹⁵, and were built based on the dimer conformations determined by NMR^{108,109}.

GpA simulations were based on the approach of Domanski *et al.*, with the reaction coordinate defined as the difference in distance between helix–helix native contacts of the GXXXG motif (D_{RMS})^{116,117}. GpA was inserted into a POPC bilayer using *g_membed*¹¹⁸. The simulation box contained 25 POPC lipids and over 1,400 coarse-grained water beads. The US simulations were set up with 40 replicas spaced between D_{RMS} 0 and 2.5 nm with sampling umbrella position and spring constants as defined by Domanski *et al.*¹¹⁷. The WHAM method was used to perform the unbiasing of the US trajectories⁷⁸, using the Grossfield lab implementation¹⁰¹. The dimerization PMF obtained was projected onto the radial center-of-mass distance coordinate ζ .

For WALP23, the simulation protocol followed previous calculations carried out with version 2.0 of the Martini force field by Castillo *et al.*¹¹⁹. Single-component POPC and 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC) membranes comprised of 350 lipids were generated using *insane.py*⁸² and solvated by 4400 water particles and 150 mM NaCl. Peptides were embedded in the membranes in a parallel or anti-parallel setting, at distances between 0.6 and 3.5 nm, using the GROMACS tool *gmx insert-molecules*, yielding a total number of 30 starting structures for each system. Lipids overlapping with the peptides were automatically deleted by GROMACS^{11,12} *gmx insert-molecules*. Additionally, a few lipids were deleted to ensure bilayer symmetry and a fixed number of lipids in all systems. Each starting structure was then equilibrated for 200 ns using the 2D inter-peptide COM distance as a harmonic restraint, with a force constant of 1500 kJ/(mol·nm²). The same distance was used as a reaction coordinate ζ in the US production runs. The PMFs were calculated using *gmx wham*¹²⁰ bundled with GROMACS^{11,12}.

The free energies of association were calculated from the PMFs, after adding the Jacobian (also called entropic) correction $RT\ln(r)$ ^{121,122}. Further details about the dimerization free energy calculations are described in **Supplementary Results, section F5**. The temperatures of the peptides, the membrane, and the solvent were maintained separately at 303 K (for EphA1 and ErbB1) or 310 K (for GpA and WALP23). Pressure was controlled semi-isotropically with a reference pressure of 1 bar, compressibility of 3×10^{-4} bar⁻¹, and a time constant of 12 ps. Production runs were carried out for 10 μ s (EphA1 and ErbB1), 12 μ s (WALP23) or 8 μ s (via replica exchange for GpA) for each

umbrella window, using a time step of 20 fs; the initial 10-50 ns were discarded from analyses.

D14) Protein hydration and water flux in FraC

The wild-type Fragaceatoxin C (FraC) structure was taken from the PDB database (4TSY)¹²³. Missing residues at the C-terminus were modeled using VMD⁴⁹. The secondary structure was determined using DSSP^{124,125} and the added residues were manually set to helical. Elastic Networks⁹² were applied to each monomer of the protein complex, with a distance cutoff of 0.8 nm using a force constant of 1000 kJ/(mol·nm²). Additional harmonic bonds were added between residues M48-W149 and T50-N147 of neighboring monomers, mimicking hydrogen bonds between their backbones' amide groups. The bilayer, built using the program *insane.py*⁸², was composed of 50 mol% DOPC and 50 mol% DPSM (N-stearoyl-D-erythro-sphingosylphosphorylcholine). The protein was inserted into the center of the bilayer, with its geometrical principal component aligned with the membrane normal. The system was solvated using water and 150 mM NaCl. The final dimensions of the system were 17 nm in z and 14.5 nm in x and y, containing a total of 335 DOPC, 335 DPSM lipids, 17,248 waters, 162 Na⁺ cations and 218 Cl⁻ anions. All simulations with this system were performed at 310 K and a pressure of 1 bar. Electrostatics were solved using PME⁹³ due to the high amount of charge in the FraC pore. The system was energy minimized using steepest descent for 10,000 steps. Equilibration was performed in three steps. First the system was simulated using a time step of 2 fs and the Berendsen barostat⁸⁴ (semi-isotropic in xy, z) with a coupling time constant of 8 ps, while keeping the protein constrained to its initial position. Second, the protein was unconstrained. Third, a larger time step of 10 fs was used to equilibrate the pressure. After 30,000 steps the final production was started for 1 μ s with a time step of 20 fs, using the Parrinello-Rahman barostat⁸⁵ with a coupling time constant of 12 ps. Hydration was visually inspected using VMD⁴⁹. The flux of particles through the protein pore was calculated using *fluxer.py* (<https://github.com/mnmelo/fluxer>). For the *fluxer* program, the pore was treated as toroidal with a bilayer thickness of 4 nm. The reported flux is the sum of the absolute flux in both z directions, otherwise it would always be zero

at equilibrium. The flux was calculated for water, Na⁺ and Cl⁻ over the complete production run.

D15) A_{2A} receptor and ion binding

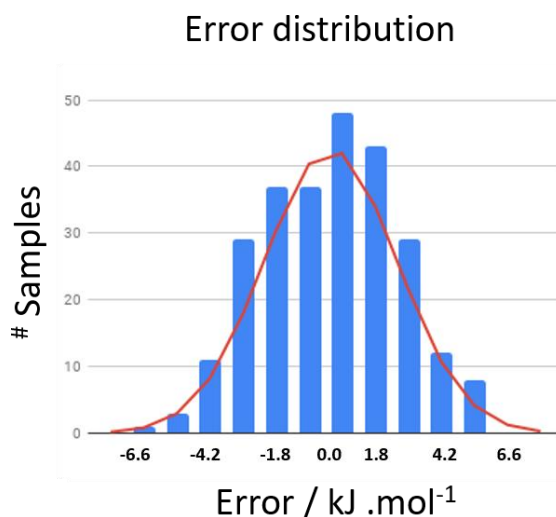
The CG model of the *apo* adenosine A_{2A} receptor (A_{2A}R) was based on the crystal structure (PDB code 3RFM)¹²⁶. Mutations were reverted to the wild type sequence and the missing loop (extracellular loop 2 - ECL2), was modeled using the pdb structure 2YDO as reference¹²⁷. As G protein coupled receptors (GPCRs) are conformationally flexible depending on their environment and association of ligands and other proteins^{128–132}, we used a shorter distance cutoff of 0.7 nm for the Elastic Network⁹², and a force constant of 700 kJ/(mol·nm²) (rather than the standard 500 kJ/(mol·nm²)). Even in this situation, the average RMSD of the backbone in relation to the crystal structure was 3.5 Å, which is in good agreement with atomistic simulations¹³³. To mimic the effect of the hydration in the core and in the buried ligand pocket of A_{2A}R, a tiny CG water bead (representing one to two water molecules) was placed inside the protein, bound to Asn181 and His250. Additional bonds were also added between the residues Thr88-Trp246 and Asn181-His250, representing some water bridges. These four extra interactions were modelled with a harmonic potential using bond distances based on the reference crystal structure and a weak force constant of 500 kJ/(mol·nm²) without excluding the non-bonded interactions. The protonation states of the residues were checked using the H++ server¹³⁴. In accordance with the obtained predictions, Glu13 and His278 were considered charged. Asp52 was also kept charged as it is the main residue involved in the interaction with one Na⁺ ion in the core of this GPCR¹³⁵. The protein was initially placed in a POPC lipid bilayer using the program *insane.py*⁸². CG water particles were also included, and the ionic concentration was set to 150 mM NaCl. As in all other simulations of this work, sodium and chloride ions are represented by TQ5 beads with charge +1 and -1, respectively. The final dimensions of the simulation box were about 12x12x12 nm³. After minimization (500 steps of steepest descent) and 100 ns equilibration (time step of 10 fs), a production run was performed for 5 μs. Both equilibration and production simulations were performed at 310 K and 1 bar using PME⁹³ to calculate the electrostatic interactions.

Supplementary Results E) Validations for small-molecules and solvents

This section presents some of the main validation results obtained for small molecules. All the results presented are compared to experimental data. Computed and experimental values (including references) are also provided in an additional supporting sheet file, which facilitates its usage in future comparisons. The procedures used are described in **Sections D1, D2 and D4**.

E1) Oil/water transfer free energies

The transfer free energy (also called partitioning free energy) has always been one of the main parametrization targets for the Martini force field. An overall view of the results obtained in relation to the experimental references^{21,63–70} was presented in **Figure 1C**. Here, we show some complementary statistical analysis for the same data (258 data points). **Supplementary Figure 9** shows the error distribution of the whole dataset, which presents a Gaussian shape centered at 0.18 kJ/mol, indicating that the dataset has a small bias to be too hydrophilic. The three main partitioning systems used (hexadecane-water, octanol-water and chloroform-water) show similar behavior. **Supplementary Table 26** shows a summary of them.



Supplementary Figure 9: Error distributions for the whole oil-water partitioning dataset.

The mean absolute error (MAE) between experimental and simulation results is smaller for molecules modelled with one single bead (50% of the data) than for bigger molecules (modelled with 2 or more beads). The best results were obtained for water-

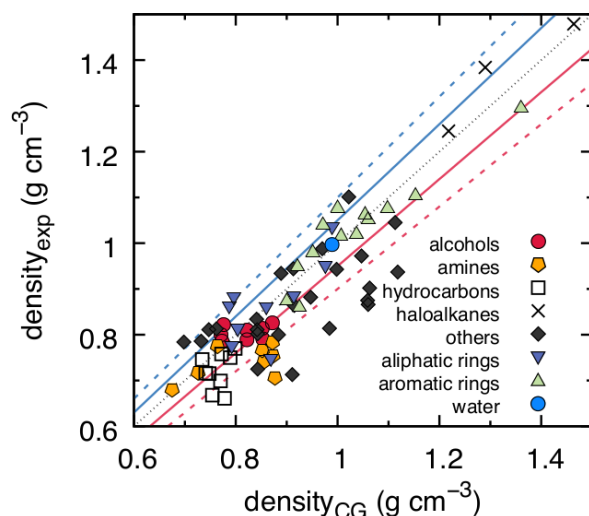
hexadecane partitioning data, showing a MAE of 1.8 kJ/mol and an R^2 of 0.98. Comparing the estimates for water-octanol partitioning performed for Martini 2 by Kanekal and Bereau¹, Martini 3 shows a significant improvement of around 0.6 kJ/mol in the overall MAE, which represents a 23% reduction of the error compared to the previous version.

Supplementary Table 26: Extra information of partitioning free energy.

	#Samples	MAE 1-bead (kJ.mol ⁻¹)	MAE ≥ 2-beads (kJ.mol ⁻¹)	MAE - total (kJ.mol ⁻¹)	R ²
water/hexadecane	36%	1.6	2.0	1.8	0.98
water/octanol	41%	2.0	2.3	2.1	0.93
water/chloroform	33%	2.0	2.2	2.0	0.96
Total	100%	1.8	2.2	2.0	0.95

E2) Mass density

A secondary parametrization target evaluated as well is the mass density, with a correlation plot between experimental¹³⁶ and Martini 3 results shown in **Supplementary Figure 10**.



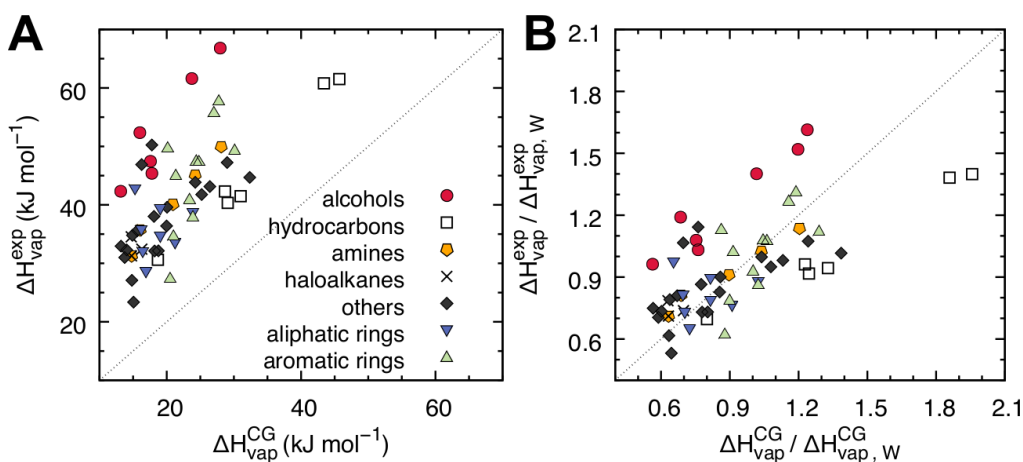
Supplementary Figure 10. Correlation plots between experimental¹³⁶ and Martini 3 densities. The red and blue solid indicate threshold error lines of plus or minus 5% (solid) and 10% (dashed), respectively.

A total of 77 solvents were investigated, with the overall results showing a mean absolute percentage error (MAPE) of 7.3%. On average, Martini 3 CG models display a slightly higher density than experimental references (+4.2%). Aromatic and aliphatic rings

(22 data points) usually perform quite well, with a MAPE of 4.3%. On the other hand, solvents represented by single beads (21 points) have the worst performance, with a MAPE of 8.1%. One exception is water, which shows very good agreement (MAPE of 0.81%). This worse performance for single bead molecules is expected as Martini 3 is based on a building block approach and thus the bead sizes are fixed. Results for density and packing should improve for bigger molecules, as it is easier to obtain the overall shape and symmetry in these cases.

E3) Enthalpy of vaporization

Supplementary Figure 11 shows correlation plots between experimental¹³⁶ and Martini 3 enthalpies of vaporization (ΔH_{vap}). Note that Martini systematically underestimates ΔH_{vap} values^{9,137} due to the limited fluid range of the employed 12-6 LJ potential form (**Supplementary Figure 11A**). A normalization of the data with respect to ΔH_{vap} of water, leads to the correlation plot of **Supplementary Figure 11B**.



Supplementary Figure 11. Strength of solvent-solvent interactions: enthalpies of vaporization. (A) Correlation plot of Martini 3.0 and experimental¹³⁶ ΔH_{vap} data. (B) Same as (A) but normalized with respect to the enthalpy of vaporization of water.

Note that the systematic overestimation of ΔH_{vap} observed for ring compounds in Martini 2² is overcome in Martini 3, with ring models now following the Martini trend (blue and green triangles in **Supplementary Figure 11A**). The underestimation of the enthalpy of vaporization tends to be larger for alcohols and lower for (non-cyclic) hydrocarbons, a feature also observed in Martini 2².

To qualitatively evaluate the miscibility of Martini CG models, classical miscibility tables were computed, with the results presented in **Supplementary Table 27**. In total, 27 solvents were considered in this validation^{138,139}. Considering combinations of all binary mixtures, a total of 351 systems were evaluated.

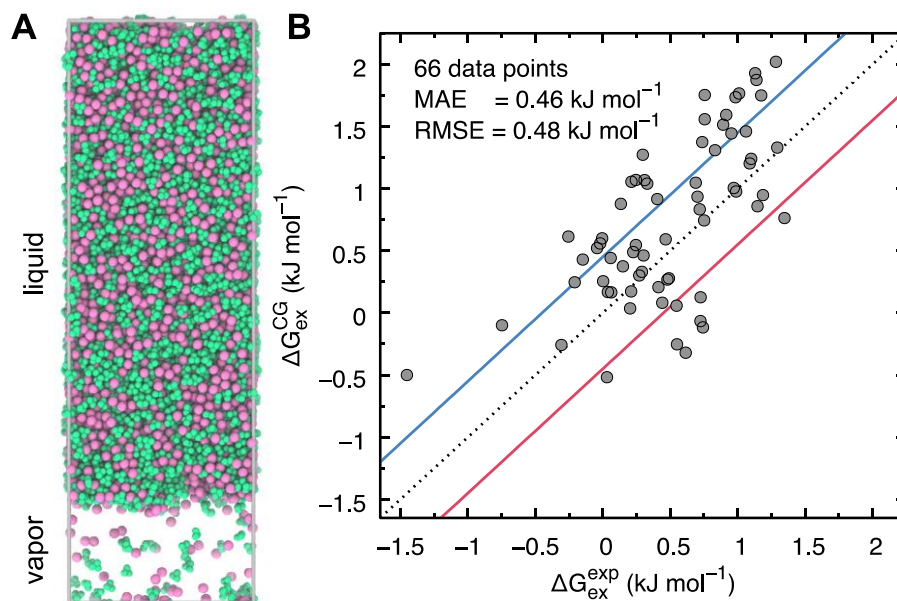
	acetic-acid	acetone	acetonitrile	benzene	n-butanol	butyl-acetate	carbon-tetrachloride	chloroform	cyclohexane	1-2-dichloroethane	dimethylformamide	dimethylsulfoxide	dioxane	ethanol	ethyl-acetate	diethyl-ether	hexane	methanol	2-butanone	n-propanol	isopropanol	diisopropyl-ether	tetrahydrofuran	toluene	trichloroethylene	water	xylene
acetic-acid																											
acetone																											
acetonitrile																											
benzene	X		X																								
n-butanol																											
butyl-acetate	X																										
carbon-tetrachloride	X	X																									
chloroform	X	X																									
cyclohexane	X																										
1-2-dichloroethane																											
dimethylformamide							X		X																		
dimethylsulfoxide				X			X		X																		
dioxane									X																		
ethanol				X			X		X																		
ethyl-acetate																											
diethyl-ether												X															
hexane																	X										
methanol							X		X																		
2-butanone																		X									
n-propanol							X		X									X									
isopropanol				X			X		X									X									
diisopropyl-ether	X	X							X						X			X			X						
tetrahydrofuran	X																										
toluene	X	X		X							X	X		X				X									
trichloroethylene																											
water				X		X	X	X	X						X												

64

or if the pair interactions are involved. Considering the addition of other mixtures evaluated in **sections E5** and **E6**, the predictions improve to 91%

E5) Excess free energy of mixing

Experimental ΔG_{ex} were obtained from the work of Sweere and Fraaije²⁰, who collected the data from the NIST database¹⁴⁰. The selected subset of experimental ΔG_{ex} used in the present work consists of 66 data points; these were chosen based on experimental conditions - temperatures between 293.15 K and 308.15 K, and 1 bar (see the **additional supporting sheet file**) - and based on the availability of Martini 3 models. A rendering of a typical setup used to compute excess free energies (ΔG_{ex}) of mixing is shown in **Supplementary Figure 12A**. Experimental and computed ΔG_{ex} are shown in **Supplementary Figure 12B**. The overall agreement with experimental data is good, as shown by a MAE of 0.46 kJ/mol.

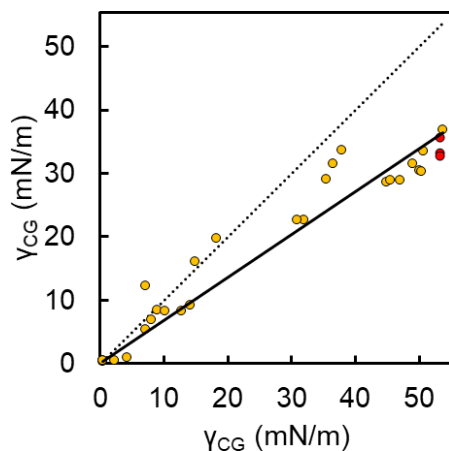


Supplementary Figure 12. Quantitative validation of homogeneous binary mixtures: excess free energy of mixing. **(A)** Rendering of the vapor–liquid equilibrium simulation setup for one of the simulated mixtures, namely the mixture between acetone (pink) and benzene (green). The radii of the Martini beads – regular for acetone and tiny for benzene – are rendered to scale. **(B)** Correlation plot between Martini 3.0 and experimental ΔG_{ex} for binary mixtures. The blue and red solid lines highlight the region with deviations of +0.45 and –0.45 kJ mol^{–1}, respectively. The mean absolute and root mean square errors, MAE and RMSE, respectively, are shown on (B), along with the number of data points plotted.

An error within 0.45 kJ mol⁻¹ of the experimental reference can be considered as good for Martini mixtures. Note that, in the case of ΔG_{ex} , the allowed error cannot be as large as the one considered satisfactory for the partitioning free energies (which have a maximum allowed error of 3.0 kJ mol⁻¹), since the range of ΔG_{ex} spanned by homogeneous mixtures is at most ~3 kJ mol⁻¹, as can be seen from **Supplementary Figure 12B**.

E6) Interfacial tension

In the case of liquid-liquid biphasic heterogeneous mixtures, the interfacial tension was the main target property. The selected subset used in the present work consists of 25 systems. Most of the systems involve oil/water mixtures, for which experimental reference data is available^{141–147}. The tested oil compounds include alkanes, alkenes, aliphatic and aromatic rings, halo compounds, alcohols, ethers, ketones, aldehydes and ethers. The results are presented in **Supplementary Figure 13**.



Supplementary Figure 13: Correlation diagram between the Martini 3 and experimental interfacial tension (γ). Bold black line indicates the linear regression while yellow circles the data points. Red points indicate water/dodecane mixtures, where dodecane was modeled with 3 regular, 4 small or 6 tiny beads.

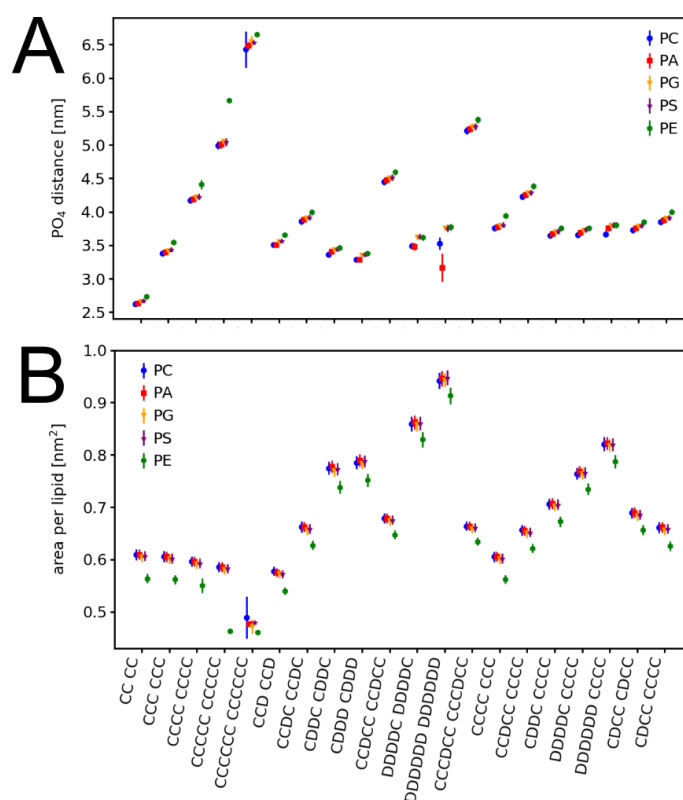
Overall, the trends are in good agreement (R^2 equal to 0.93), although the absolute value of the surface tension is underestimated in mixtures with high difference in hydrophobicity of the solvents - a consequence of the limited fluid range of the LJ potential as discussed before. Importantly, our results show a significant improvement in relation to the previous version of Martini, which was tested in this aspect by Ndao *et al.*⁷⁶ The

data also illustrate how well the beads are balanced across different sizes, as similar water/dodecane interfacial tensions are obtained with dodecane in 3 resolutions (red points in **Supplementary Figure 13**).

Supplementary Results F) Examples of validation for other systems

F1) Bilayer properties of phospholipids in Martini 3.

Lipids were always one of the most accurately modeled systems in Martini, with numerous studies showing good agreement with experiments^{148,149}. One of the great challenges during the Martini 3 development was to improve and develop the beads keeping the accurate properties of the lipids observed in Martini 2. Four bilayer properties were calculated to characterize and compare the different phospholipid types: thickness, area per lipid, tail order parameter, and area compressibility. Martini 3 was directly compared with the previous version, given its known accuracy, as shown in **Supplementary Figure 14** and **15**.



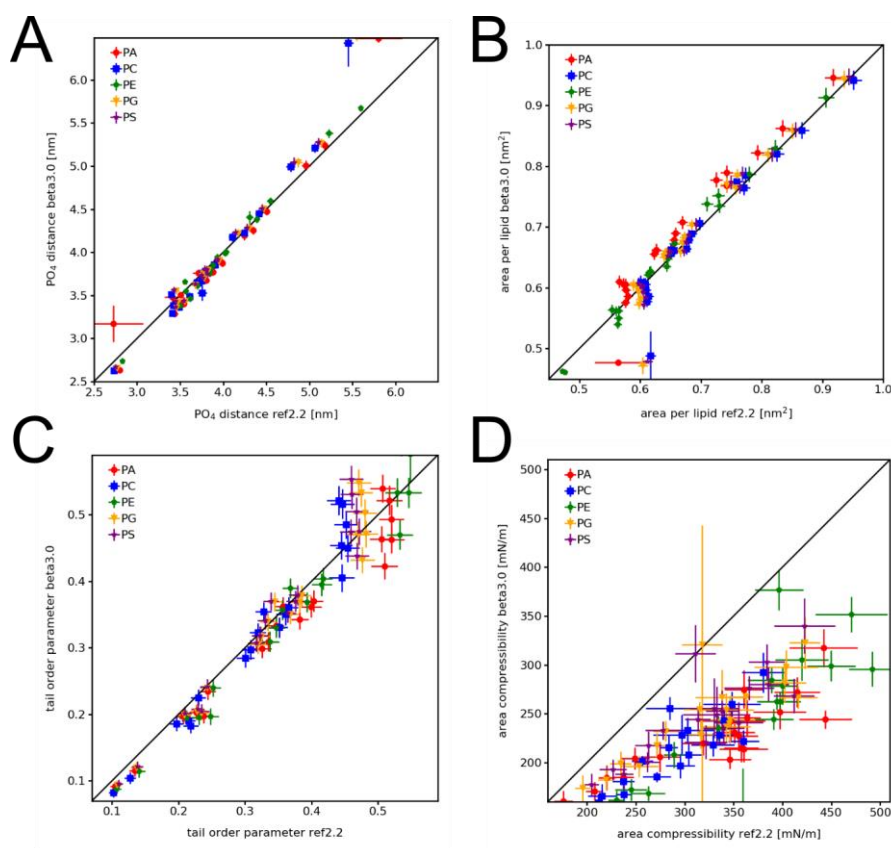
Supplementary Figure 14: (A) The thickness (measured as the distance between the PO₄ beads) and (B) area per lipid (calculated from the average box area in the plane of the bilayer), shown for 100 different lipid types (mean $\pm$ sd). Five different head group types are shown, each with $\times 20$ different acyl chains. Two symbols are used to specify any length of unbranched acyl chains: “C” indicates 4 carbon atoms in a saturated chain and “D” the same chain length with 1–2 cis double bond.

As in Martini 2, the area per lipid was the largest for PC lipids and the smallest for PE lipids in Martini 3 (**Supplementary Figure 14A**), which is in agreement with experimental data¹⁵⁰. PS and PG follow the same trends as Martini 2⁸², with some deviation in the behavior for PA lipids, which have a higher area per lipid in Martini 3. As the protonation state of PA head can play a role in the results, further studies are necessary to indicate in which version the properties are more accurate. As expected, thickness correlates well with the area per lipid, with its values increasing from PC to PE lipids.

The trends for tails also show some similarities between the Martini versions. Increasing the length of saturated acyl chains increases the bilayer thickness, which is in agreement with experimental observations¹⁵¹. Besides, as estimated by experiments^{150,152}, the area per lipid increased significantly with increasing levels of unsaturation (see **Supplementary Figure 14**, where D beads indicate unsaturated bonds). One improvement in relation to Martini 2 is the trends in area per lipid with length of saturated acyl chains. In the previous version, the area per lipid slightly increase with the increase of the saturated chain length⁸². The new version shows the opposite behavior, with a slightly decrease in the area per lipid, which is also observed in experiments^{152,153}.

This different behavior of the lipids with saturated acyl chains reflects the tendency of longer tails in Martini 3 to be in the gel phase at the simulation conditions (310 K and 1 bar). Indeed, some outliers in the correlation plots for thickness and area per lipid (**Supplementary Figures 15A and 15B**) are long saturated tail lipids that adopt a gel phase in Martini 3 (in agreement with experiment), but remain fluid in Martini 2. The difference in trends for saturated tails is also evident in the correlation plot of tail order parameters (**Supplementary Figure 15C**), in the region around the value 0.5. One striking difference in the Martini 3 and 2 models is in the area compressibility modulus (**Supplementary Figure 15D**). Martini 3 models are softer than Martini 2 which seems to give a better agreement with experiments. For instance, Martini 3 estimated an area compressibility for DPPC of 232.9 ± 13.3 mN/m, which is in excellent agreement with the experimental value¹⁵⁴ of 231 ± 20 mN/m.

Understanding the difference in lipid properties between Martini 2 and Martini 3 in full detail is out of the scope of this benchmark, but the comparison between the differences of the models provides some insight. In relation to Martini 2, Martini 3 lipids have: (i) linker and tails with less (self- and cross-) interaction between themselves; (ii) heads with stronger interactions with water and with themselves; (iii) stiffer tails given the new bonded parameters adapted from Carpenter *et al.*¹⁵⁵. Overall, Martini 3 lipids seems to keep the good behavior of Martini 2 lipids, with some improvements in trends for lipids with saturated tails and an overall improvement of area compressibility modulus. Current efforts are directed to further improve the Martini 3 lipidome, and validate other lipid types (e.g., gangliosides, PIs, PIPs, etc).

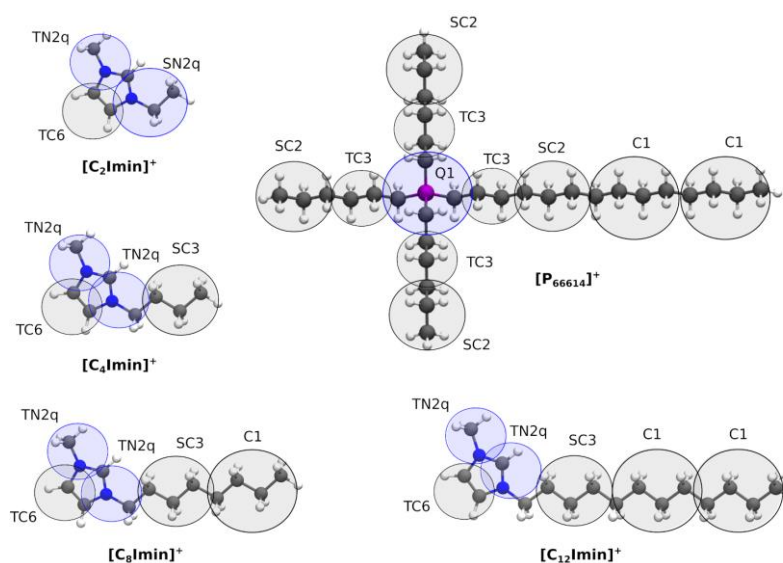


Supplementary Figure 15: Correlation plot of Martini 3.0 and Martini 2.0 thickness (A), area per lipid (B), tail order parameters (C) and area compressibility (D). Overall, thickness and area per lipid are in good agreement between both models. Order parameters show different trends for lipids with saturated tails (around the value 0.5). The area compressibility is reduced in Martini 3 in comparison with Martini 2, indicating that the bilayers are softer in the new version.

F2) Hofmeister series, ionic-liquids, and validation of charged beads.

The Hofmeister series is originally defined as a classification of ions in order of their ability to affect the solubility of proteins³. Nowadays, the concept can be considered extended to other systems¹⁵⁶, including ionic liquids⁹⁴. Here we calibrated and validated the Martini ions in terms of ion transfer between a water phase and an ionic liquid (IL) phase. After bead assignment based on anion exchange simulations, some pure ionic liquids were also tested.

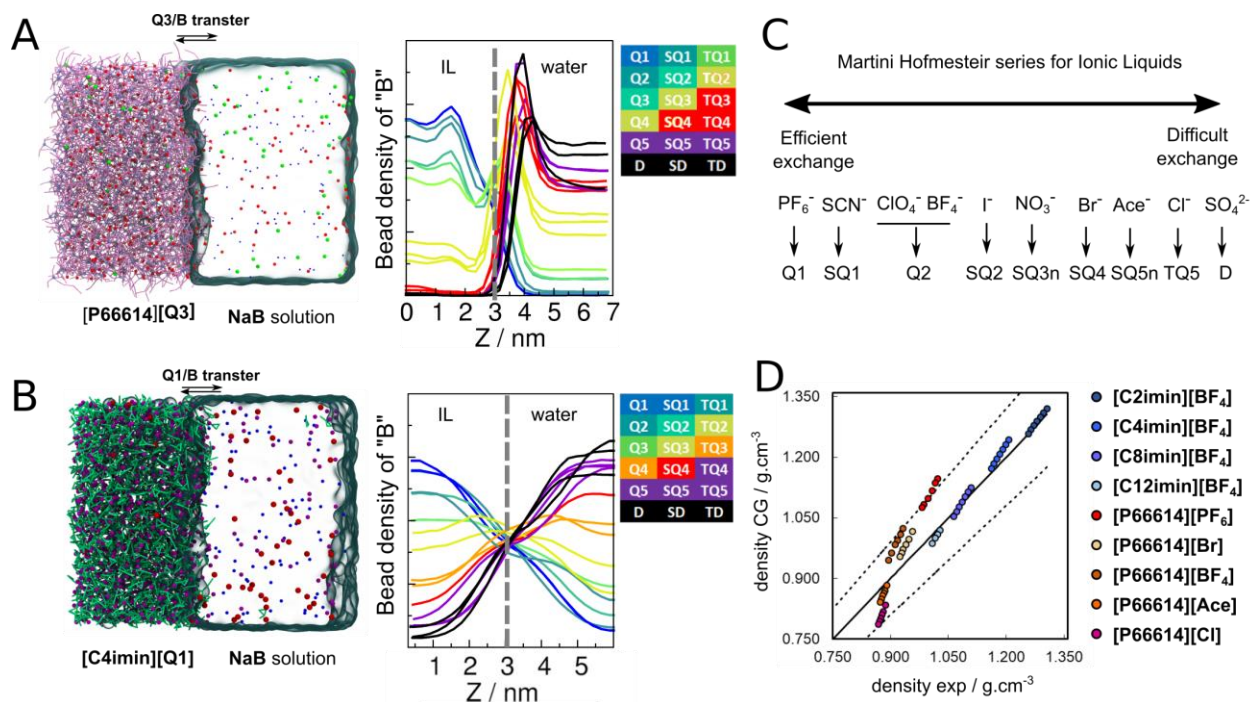
CG IL models: Two IL classes were modelled, based on organophosphonium (in particular, trihexyltetradecylphosphonium $[P_{66614}]^+$) and 1-3-dialkylimidazolium ($[C_n\text{mim}]^+$) cations. The mapping and bead chemical types and sizes are shown in **Supplementary Figure 16**. The $[P_{66614}]^+$ model shows some similarities, in terms of mapping, with the models of Schaeffer *et al.*¹⁵⁷ A central +1 charged Q1 bead represents the tetramethyl phosphonium fragment. The imidazolium charged ring was built on top of the prototype model developed with the open-beta version of Martini 3 (see Vazquez-Salazar *et al.*¹⁵⁸). In the current version, the 1,3 dimethyl imidazolium ring was modelled with three T-beads, using the bead type TN1q for the nitrogen-containing aromatic fragments while the rest of the carbon-based aromatic ring is represented by a TC6 bead. Each TN1q bead has a +0.5 partial charge (see **Section A4**). This template was used for all $[C_n\text{mim}]^+$ cations, except $[C_2\text{mim}]^+$, for which the extra carbon was mimicked by changing one of the nitrogen-containing fragments to SN1q. The alkyl fragments connected to the charged cores of $[P_{66614}]^+$ and $[C_n\text{mim}]^+$ were modelled with C3 type, with size adapted to map the exact length of the alkyl fragment. The remaining saturated alkyl tails of the cations are represented by C1/C2 beads, as typically used for alkanes in Martini 3. Bonded parameters involving the charged fragments are based on atomistic trajectories generated with OPLS parameters adapted by Padua *et al.*¹⁵⁹ to ionic liquids. They were tuned to get the best compromise in terms of bulk density and molecular surface area (see **Section C3**). Bonded parameters of the alkyl chain are based on the models of alkanes of Martini 3. Sodium ion is modelled with charge +1 and TQ5 bead. The small anions were modelled with single beads, with the best bead assignment described in **Supplementary Figure 17C**.



Supplementary Figure 16: CG models of trihexyltetradecylphosphonium and 1,3 dialkylimidazolium cations. The Martini bead types and sizes are indicated. Blue indicates positive charged groups while the gray color indicates the nonpolar groups.

Anion transfer in IL/water biphasic systems: In the presence of inorganic salts, some water-miscible ILs can form stable biphasic systems with water. Importantly, the properties of the IL/aqueous biphasic mixture formed are greatly affected by the salt used⁹⁴. One of these properties is the anion exchange between the IL and the water phase, which will depend on differences in hydrophilicities of the anions. Analogous to classical oil/water partitioning systems, used in the parametrization and bead assignment of neutral beads, anion exchange between IL and water phases was one of the systems used to model charged beads in Martini 3. **Supplementary Figures 17A-17B** summarizes the main results. Initially, two IL/water biphasic systems were tested, $[P_{66614}][Q3]/NaB(aq)$ and $[C_4mim][Q1]/NaB(aq)$. At this point of the Martini 3 development, Q3 and Q1 beads were used as generic anions for the IL phase. The “B” is the anion of the sodium salt, in the water phase, which was defined with all possible charged beads in Martini 3, including divalent D-beads. Details of the simulation are described in **section D8**. The plots in the right panels of **Supplementary Figure 17A-17B** show the bead density of “B” anions in each phase for both IL/water systems. Overall, hydrophilic (or hard) charged beads as D and Q5 type hardly exchange with the anions in the IL phase. On the other hand, soft anions prefer the IL phase. Different ILs showed some differences in the behavior. Hydrophilic ions tends to prefer the interface when the

IL is based on $[P_{66614}]^+$. As $[C_4mim]^+$ is more hydrophilic than $[P_{66614}]^+$, it is notable a slightly higher preference of the hydrophilic ions to migrate to the IL based on $[C_4mim]^+$. Considering the trends observed and comparison with experimental data^{94,96}, it is possible to define a Hofmeister series for the charged-beads, which was used to define the chemical types for a lot of anions, as shown in **Supplementary Figure 17C**. Based on these bead assignments, specific IL/water mixtures were simulated and compared to experimental work of Dupont *et al.*⁹⁴ and Naert *et al.*⁹⁶ (see **Supplementary Figure 2C** of the main manuscript).



Supplementary Figure 17: Ionic liquids and parametrization of charged beads: (A) Snapshot of $[P_{66614}][Q3]/NaB(aq)$ biphasic mixture. Q3 beads is used as generic anions for the IL phase. The "B" is the anion of the sodium salt, in the water phase. The right panel shows a plot with the bead density of B in both phases, with B defined as all possible charged beads. Different colors in the curves indicate the beads with similar anion transfer behavior; (B) Same snapshot and plot presented in (A) but for $[C_4mim][Q1]/NaB(aq)$ biphasic mixture; (C) Bead assignment for anions defined according to the anion exchange simulations.; (D) Correlation plots between experimental and Martini 3 densities for pure IL liquids at different temperatures. The dashed lines indicate threshold error lines of plus or minus 10%.

As the cations used for the salt and IL need to have a big difference in hydrophobicity for the IL/water mixtures to be biphasic, the cation exchange cannot be

used in IL/water systems for cation bead assignment. However, cations do affect the IL solubility in water, which was also considered here for Martini 3 parametrization (in analogy to miscibility, used for neutral molecules). Such aspects will be explored in a future study, dedicated to the parametrization of other ions and ionic liquids⁹⁴. It is worth mentioning that the Martini 3 model of $[C_n\text{mim}]^+$ provides a good example of the usage and the balance achieved with “q” labels, that, in combination with partial charges, allow representation of large charged molecules.

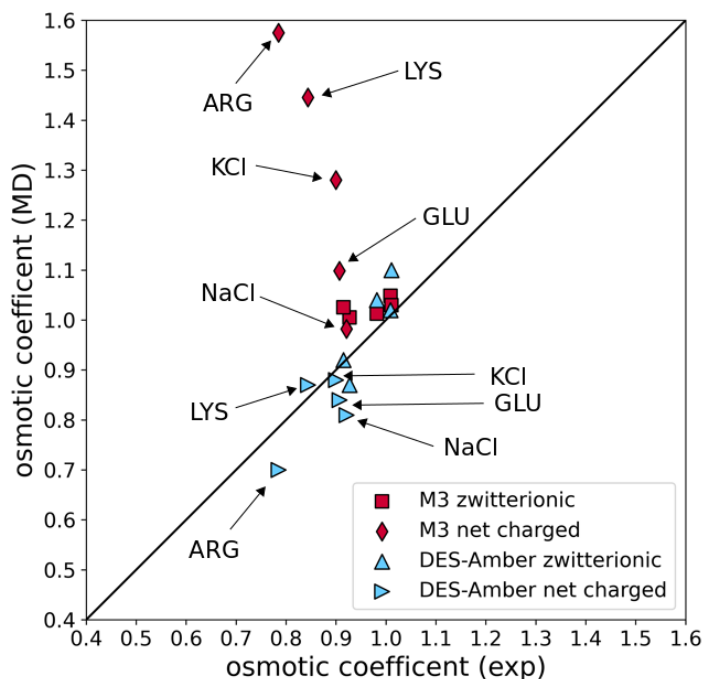
Densities of pure ionic liquids: Pure ionic liquids were also used for validation of the charged beads, based on the same simulation protocol described in Vazquez-Salazar *et al.*¹⁵⁸. Different temperatures at a pressure of 1 bar were tested, spanning from 293 to 363 K (see the additional **Supplementary Sheet File** for more details). Overall, the results of our MD simulations (see **Supplementary Figure 17D**) indicate that the Martini 3 models show accurate densities that correlate well with the experimental data^{160–164}. For imidazolium based ILs, we tested cations with different tail lengths mixed with $[\text{BF}_4]^-$ anion. Densities were only slightly overestimated, with R^2 of 0.98 and MAPE of 2.0%. For organophosphonium based ionic liquids, the same cation - $[\text{P}_{66614}]^+$ – was mixed with different anions: $[\text{PF}_6]^-$, $[\text{Br}]^-$, $[\text{BF}_4]^-$ and $[\text{Cl}]^-$. The results show a R^2 of 0.91 and MAPE of 6.7%. The slightly lower accuracy in the organophosphonium IL case reflects the limits of representing different anions only using single beads in three defined sizes. For instance, in $[\text{P}_{66614}][\text{PF}_6]$ case, a regular Q1 bead is too small to represent the $[\text{PF}_6]^-$ anion, resulting in too high densities (around +10%). On the other hand, a T-bead is too big to represent the $[\text{Cl}]^-$ anion in $[\text{P}_{66614}][\text{Cl}]$, so the densities in this case are underestimated (around -8%).

F3) Osmotic coefficients of ionic solutions

The osmotic pressure is directly related to the chemical potential of the solvent in a mixture. Thus it is a very robust quantity for estimating solubility of molecules for low to moderately high concentrations. For example, it has previously been used as a target quantity to optimize Martini 2 sugars, which were seen to aggregate too much⁹⁸. It has also been used in atomistic force-field optimization^{29,31,97}, because it allows to adjust parameters for ionic species for which no reliable solvation free energies or partition coefficients are available. Experimentally, the osmotic pressure is often not obtained directly but rather the osmotic coefficient is measured. The osmotic coefficient is the ratio of the measured osmotic pressure and the osmotic pressure of an ideal solution at a given concentration. A too large osmotic coefficient is often indicative of an overestimation of solvation, whereas a too low osmotic coefficient indicates unphysical aggregation of the solute^{31,98}.

Supplementary Figure 18 shows the osmotic coefficient of 10 molecules simulated using Martini 3 and a recently published version of Amber called DES-Amber¹⁶⁵ against experimental data^{166–172}. The data is split into zwitterionic molecules and molecules that possess a net charge, including two inorganic salts, NaCl and KCl. For each system, the total concentration of solute species in the water solution was 1.0 M, except for valine, which was simulated at 0.6 M to match the concentration of the experimental reference. Zwitterionic amino acids, with neutral side-chain, possess no net charge and in Martini 3 are represented as completely neutral beads. They are indicated by the red squares in the plot below. The osmotic coefficient of these molecules is overall very close to the experimental coefficients with the largest deviation being observed for serine. In this case, Martini 3 osmotic coefficient was 1.025 whereas experimentally an osmotic coefficient of 0.915 is observed. In general, all osmotic coefficients in Martini 3 are larger than those experimentally observed. This indicates a lack of aggregation and a tendency to overestimate solubility at these higher concentrations. However, in comparison to the atomistic DES-Amber force-field, which has been specifically optimized against this data, for the neutral molecules Martini 3 shows similar deviations. The largest deviation in DES-Amber is observed for valine (1.011 for atomistic model vs. 1.1 in the

experiments). Thus deviations in Martini for this set of molecules are comparable to state of the art atomistic force-fields, and overall not considered problematic.



Supplementary Figure 18: Comparison between the simulated and the experimental osmotic coefficients for neutral zwitterionic amino acids and some inorganic and organic (charged amino acids) salt solutions. Martini 3 (red) and atomistic DES-Amber (blue) are compared in the plot.

The osmotic pressure coefficients for the charged species, which are composed of ion pairs, are worse for both the atomistic force-field and Martini 3. Here the largest deviation for both force-fields is observed for arginine. Interestingly the osmotic coefficient of sodium chloride is actually closer to experiment than for DES-Amber. However, other species such as lysine, glutamate, and potassium chloride perform significantly worse with osmotic coefficients larger by a factor of about 1.2-2.0. However, deviations in the osmotic pressure by this factor range are also not uncommon for atomistic force-fields, which are not optimized against this data. For example, the osmotic pressure of glucose within the GLYCAM06 force-field is also a factor of 2 larger than experiment⁹⁷.

In addition, there are some problems that come inherently with the CG representation of lysine, arginine and glutamate. It is known that arginine can form

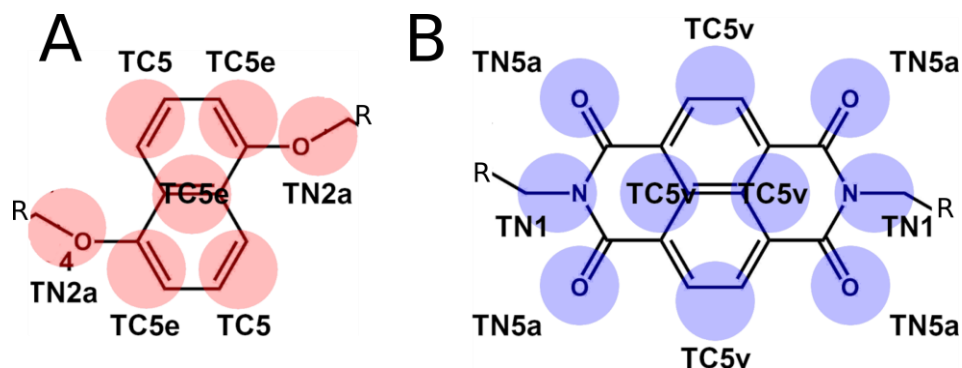
specific stacking interactions^{173,174}. These interactions are probably less well represented in Martini as the whole guanidinium group is represented by a spherical SQ3p bead. On the other hand, for lysine it is expected that the charged side-chain can coordinate with the carboxylic acid of the zwitterionic backbone part of the molecule¹⁷⁰. In the current parametrization, this is counteracted by an angle potential and the fact that the zwitterionic part is not charged. Motivated by this reasoning, we have computed the osmotic coefficient of lysine at 0.5M total concentration, which yields an osmotic pressure coefficient of 1.16 compared to the experimental value of about 0.92. This is already a large improvement indicating that the Martini single amino acids perform better at lower concentrations. In general, a better representation of the zwitterionic part of all single amino-acids, by for example using a polarizable bead is expected to largely improve the osmotic pressure values. Similar strategies can also be applied for ions and charged amino acids, for which polarizable versions also existed in Martini 2^{36,175}.

F4) Aedamers: atomistic simulations, CG models, and additional analysis.

Reference atomistic simulations: Atomistic simulations of aedamers were run with a time-step of 2.5 fs, integrating the equations of motion with the leap-frog algorithm¹⁷⁶ as implemented in GROMACS 2016.x and 2018.x^{11,12}. Temperature and pressure (isotropic coupling to 1 bar) were maintained using the Berendsen coupling schemes⁸⁴, with coupling time constants of 0.3 and 0.5 ps for temperature and pressure, respectively; the compressibility was set to $4.6 \cdot 10^{-5} \text{ bar}^{-1}$. Translation of the center of mass was removed every 10 steps. The interaction parameters for the molecules were based on the GROMOS force fields^{177,178}, with modified charges on the aromatic moieties (see below). For the pegylated aedamers (DANP and NDIP), the 2016H66 version¹⁷⁸ was used because of its superior hydration free energy of ethylene glycol chains. For the amide-linked aedamers (nDAN and nNDI), the G53a6 set¹⁷⁷ was used. For water, the SPC model¹⁷⁹ was used, and counter ions (Na^+) were added to the amide-linked aedamers to neutralize the system; in addition, in some simulations, additional Na^+ and Cl^- ions were added. All bond lengths were constrained using the LINCS algorithm¹⁸⁰, with settings `lincs-order=4` and `lincs-iter=1`. The SETTLE algorithm¹⁸¹ was used to treat SPC water as

rigid. The Verlet update scheme¹⁸² was used with a buffer tolerance of 0.005 kJ.mol⁻¹.ps⁻¹, updating the neighbor list every 10 steps. The cut-off is 1.4 nm for both Lennard-Jones (LJ) and coulomb interactions. The coulomb interactions were treated with the reaction field modification due to Tironi *et al.*¹⁸³ This method alleviates cut-off artifacts by ensuring that both potential and force decay smoothly to zero at the cut-off of 1.4 nm. The parameter for the bulk dielectric constant (epsilon_rf in GROMACS) was set to 62. Partial charges on the DAN and NDI cores were calculated using the quantum chemistry GAMESS-UK package¹⁸⁴ by performing the Dipole Preserving Charge (DPC) analysis⁷ on the charge density calculated *in vacuo* using Density Functional Theory (DFT). The level of DFT employed here used the PBE correlation functional and the PBE0 exchange functional, with 25% Hartree-Fock exchange¹⁸⁵, employing the 6-31g* basis set.

CG aedamer models: The Martini 3 DAN core representation was based on the scheme for naphthalene⁵¹, which is analogous to the scheme described in detail in **Section C4** for 3-methyl-indole, and uses four beads that form a rectangle and three beads constructed from the positions of these four beads. **Supplementary Figure 19** shows the mapping scheme and bead assignment of the DANP molecule. The four corners are formed by the beads marked TC5 and TC5e on the corners of the naphthalene unit. The distances between the adjacent corners are constrained and are taken from the Martini 3 naphthalene model⁵¹. As noted, the model uses the “hinge” construction described in **Section C4** for 3-methylindole. Furthermore, the ether bridged substituents at the 2 and 6 positions of the naphthalene ring (marked TN2a) are also placed by a virtual site construction from the four rectangular core beads, ensuring that there exists a close relationship between the positions of the atoms in an atomistic model and the beads in the Martini 3 model. Note that the masses of the virtual beads are distributed among the actual beads to maintain the total mass of the molecule. By the construction described, the seven beads that make up the DAN core are always close to being in one plane.



Supplementary Figure 19: (A) DAN mapping and bead assignment. The mapping scheme is represented by the circles enveloping the atoms mapped into one bead. The bead types are denoted next to the mapping circles. The central TC5e bead and the two TN2a beads are constructed as virtual sites from the positions of the four beads marked TC5 and TC5e forming a rectangle. **(B)** NDIP mapping and bead assignment. The mapping scheme is represented by the circles enveloping the atoms mapped into one bead. The bead types are denoted next to the mapping circles. The TC5v beads and the two TN1 beads are constructed as virtual sites from the positions of the four beads marked TN5a forming a rectangle.

The NDI core, shown in **Supplementary Figure 19B** as part of the Martini 3 model for the NDIP molecule, is mapped by 10 beads, which are constructed from four beads in a rectangle (marked TN5a), as in the DAN core. The constrained lengths of the edges of the rectangle are larger in NDI than in DAN, reflecting the larger size of NDI. The four beads were chosen to represent the carboxyl groups and the constraint lengths were calculated by mapping the beads to the position of the O-atoms in the atomistic model. As in the DAN base, the four beads are favored to be in a plane by means of an improper type dihedral and one constrained distance along the diagonal is added. The positions of the six other beads defining the NDI core are all calculated from the positions of the four edge beads. Two represent the N-atoms or connections to the side chains (marked TN1), and four represent the central hydrocarbon part of the NDI core (marked TC5v). As in the DAN base, these six beads are technically treated as virtual sites with interactions, and the masses of the atoms that it maps are distributed over the normal beads. For convenience, a further virtual site is added, at the center of the NDI core (calculated as the center of geometry of the four corner beads and not shown in **Supplementary Figure 19B**). This site is only used in interactions in the calculation of PMFs that use the center-center distance as reaction coordinate (or collective variable) and/or for analysis purposes. Finally, for both DAN and NDI cores, non-bonded interaction between core

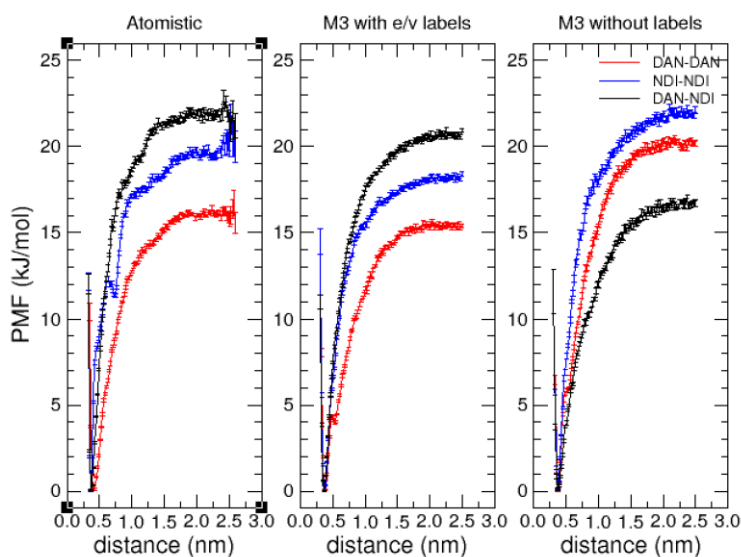
beads are all excluded, otherwise these interfere with the bonded interactions and lead to numerical instabilities, as described in **Section C3**.

Side chains to the DAN and NDI bases are added by defining harmonic bonds and angles, whose parameters were taken from the Martini 3 polyethylene glycol model¹⁸⁶ in the case of DANP and NDIP, and by fine-tuning selected bond length and angle distributions to match mapped distributions from atomistic simulations in the case of the nDAN and nNDI series of molecules. For the nDAN and nNDI molecules, part of the bonded parameters was taken from the protein model, because the amide termini and bridges essentially are amino acids.

Concerning the choice of bead types, the beads of the PEG side chains and amino acid moieties were taken from the Martini 3 PEG and protein models, respectively. Note that the difference in the linker lengths of nDAN and nNDI is reflected by using a T-bead for the shorter nDAN connection and an S-bead for the longer nNDI connection. Bead types for the aromatic cores were fine-tuned by running a series of potential of mean force (PMF) calculations (see below) for the pegylated DAN and NDI homo and hetero pairs. Variations were made in the number of the beads with “e” and “v” labels in the DAN and NDI cores, respectively. Also, the rules for defining the changes in interaction levels upon adding the “e” and “v” labels were experimented with. As is shown in the main text, it was not possible to correctly reflect the enhanced interaction of the hetero dimer compared to the homo dimers using only standard beads, including the “d” and “a” types that are used to reflect enhanced hydrogen-bond donor and acceptor type interactions. Using three “e” labeled beads on the diagonal (the core beads next to the 2,6 substituents and the central bead) for DAN and four “v” labeled beads in the center of the NDI core, resulted in a very satisfactory match of the Martini 3 PMFs to reference GROMOS 2016H66¹⁷⁸ atomistic PMFs, as is shown in **Supplementary Figure 20** in the next section.

Dimerization PMFs of DAN and NDI: Dimerization PMFs calculated for the combinations of the pegylated DAN-DAN, NDI-NDI, and DAN-NDI in three different force fields: atomistic (GROMOS 2016H66¹⁷⁸), Martini 3 without labels, and Martini 3 with e/v labels, are shown in **Supplementary Figure 20**. The depth of the minimum of the PMF,

used as a proxy for relative association strength (**Figure 2A**), is shown in **Supplementary Table 28**.



Supplementary Figure 20: PMFs for pegylated DAN and NDI containing molecules as a function of the distance between the centers of the aromatic groups. In contrast to the M3 model without labels (right), the atomistic GROMOS-based (left) and M3 model with *e/v* labels (middle) reproduce the general behavior that the association free energy increases in the order DAN-DAN (red), NDI-NDI (blue), DAN-NDI (black).

Supplementary Table 28: Dimerization data from PMFs for pegylated aedamers at 298 K.

System	Force field	Min value PMF
DAN-DAN	G2016H66	-16.0 ± 0.1
	M3 no labels	-20.1 ± 0.1
	M3 with labels	-15.4 ± 0.1
NDI-NDI	G2016H66	-19.7 ± 0.1
	M3 no labels	-19.7 ± 0.1
	M3 with labels	-18.2 ± 0.1
DAN-NDI	G2016H66	-21.9 ± 0.1
	M3 no labels	-16.5 ± 0.1
	M3 with labels	-20.7 ± 0.1

F5) Free energies of association of transmembrane proteins

Calculation of standard free energies of association: The free energy of association $\Delta_{ass}G^0$ of transmembrane proteins was obtained from the PMF according to **Equations**

13-14, which are based on the work of Shoup and Szabo¹⁸⁷ and de Jong *et al.*¹⁸⁸, but applying the two-dimensional analogue, i.e., by integrating using a radially symmetric PMF, and dividing by a standard area instead of a standard volume.

$$\Delta_{ass}G^0 = -RT\ln K_{ass} \quad \text{Equation 13}$$

$$K_{ass} = \frac{\int_0^{r_c} 2\pi r e^{-\frac{PMF(r)}{RT}} dr}{A^0} \quad \text{Equation 14}$$

with K_{ass} denoting the association constant, A^0 the standard area, and r_c a cut-off. Before integration, the PMF was aligned such that the average value for data points at distances larger than 2.5 nm was zero. Using **Equation 13**, the standard free energy is calculated from the association constant, using **Equation 14** with a cut-off of 2 nm. Comparison to experiment is difficult because the experimental signal may decay more or less rapidly than the PMF, indicating fewer or more dimers than the PMF suggests. Another reason is that the association constant in **Equation 14** is an apparent equilibrium constant that depends on the detergent (or lipid) concentration. Fleming¹⁸⁹ showed how to connect the apparent association constant to a mole fraction standard free energy, see **Equation 15**.

$$K_X = \frac{X_D}{X_A X_B} = \frac{n_D n_{tot}}{n_A n_B} \approx \frac{n_D n_{Det}}{n_A n_B} = \frac{c_D c_{Det}}{c_A c_B} = K_{app} c_{Det} \quad \text{Equation 15}$$

Here X_J is the mole fraction of species J, n_{tot} is the total number of molecules in the membrane system (dimers D, monomers A and B and detergent (Det), or in our case, lipid (L) molecules), and $c_J = n_J/V_{tot}$. Conversion of amounts to concentrations is based on the assumption that the total number of molecules is approximately equal to the number of detergent molecules (i.e., low concentration of TM helices in the membrane). Following De Jong *et al.*¹⁸⁸, who showed that the association constant that is extracted from the simulation of two molecules in a simulation box, K_{sim} , is the equivalent to the one defined by the law of mass action for a macroscopic but infinitely dilute solution, **Equation 16** shows that in the case of a lipid bilayer, the association constant can be connected to the mole fraction standard if the area per lipid a_L is known.

$$K_{sim} = \frac{n_D}{n_A n_B} \frac{A}{A^0} = \frac{n_D}{n_A n_B} \frac{n_L a_L}{2A^0} = K_X \frac{a_L}{2A^0} \quad \text{Equation 16}$$

The factor 2 follows from the fact that the simulation is in a lipid bilayer, and the total area A is the area of the bilayer system.

The summary of the calculations for all TM helices studied is presented in **Supplementary Table 29** and uses the areas per lipid of the Martini 3 lipids and a standard area of 1 nm². The error bars reflect the standard errors in the PMF, propagated by numerical integration of 1,000 bootstrapped PMFs using the standard errors in the PMFs; the latter are (re)calculated in a convention in which the value of the PMF at the largest distance is assumed zero. This leads to a larger error compared to an often used convention in which cumulative errors are calculated, setting the error in the PMF at the smallest distance equal to zero. It should anyway be realized that the errors reported are statistical errors in the methods for determination of PMF and integration of the PMF, and these are likely to be lower bounds to the uncertainty.

Supplementary Table 29: Dimerization data from PMFs for five transmembrane helix systems, obtained with Martini 2.2 (M22) and Martini 3.0 (M30). Free energies of association are estimated with: the minimum in PMF, standard free energy based on K_{ass} from **Equations 13-15** ($\Delta_{ass}G^0$), and mole fraction standard free energy based on K_X from **Equation 16** ($\Delta_{ass}G_X^0$). The results are compared with experimental data ("exp"), with references indicated in **section D13**.

TM system (T (K), a_L (nm ²))	Minimum in PMF (kJ/mol)		$\Delta_{ass}G^0$ (kJ/mol)		$\Delta_{ass}G_X^0$ (kJ/mol)		$\Delta_{ass}G^0$ (kJ/mol)
	M22	M30	M22	M30	M22	M30	
EphA1/DLPC (303, 0.59)	-34.0 ± 1.0	-17.7 ± 0.6	-35.7 ± 0.2	-19.8 ± 0.1	-36.2 ± 0.2	-20.3 ± 0.1	-15.4±0.5
ErbB1/DLPC (303, 0.59)	-44.8 ± 0.9	-6.1 ± 0.4	-45.5 ± 0.2	-10.0 ± 0.1	-46.0 ± 0.2	-10.4 ± 0.2	-10.5 ± 0.4
GpA/POPC (310, 0.66)	-34.5 ± 0.1	-24.7 ± 0.1	-34.5 ± 0.1	-24.8 ± 0.1	-34.9 ± 0.1	-25.2 ± 0.1	-27.7±14.2
WALP23/POPC (310, 0.66)	-36.0 ± 2.0	-11 ± 1	-35.9 ± 0.6	-12.0 ± 0.3	-36.2 ± 0.6	-12.4 ± 0.3	-12.7 ± 0.7
WALP23/DOPC (310, 0.69)	-39.0 ± 2.0	-10 ± 1	-38.1 ± 0.6	-10.9 ± 0.3	-38.5 ± 0.6	-11.3 ± 0.3	--

Supplementary Results G) Revisiting the Pitfalls of Martini

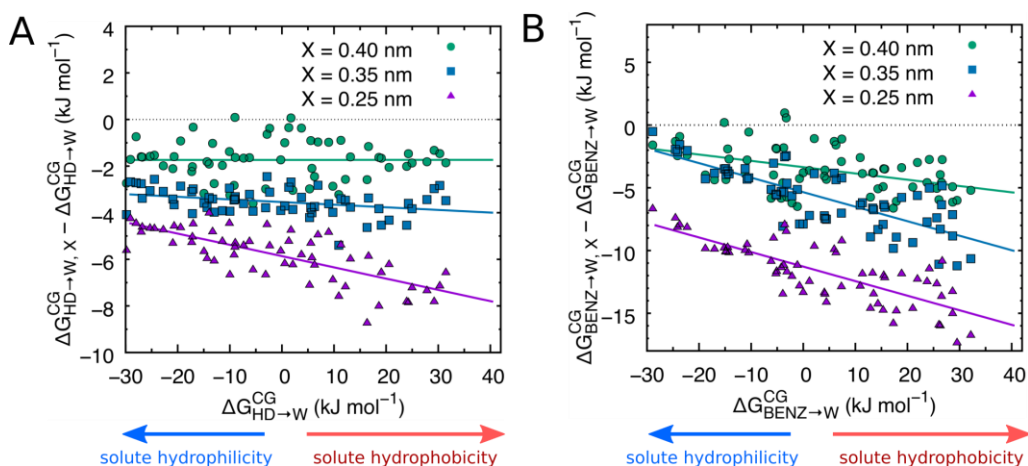
In this section, we present additional quantitative measurements of properties that are known limitations of the Martini force field. Overall, some aspects have clearly improved, as the reported deviations from trends in partitioning free energy² (revisited in **section G1**). Other properties such as the temperature dependence did not change in relation to Martini 2, with the model capturing reasonably well trends in certain thermodynamic and structural properties, but lacking the temperature sensitivity of atomistic models (**sections G2** and **G3**). The entropic driving force underlying the hydrophobic effect is still a limitation in Martini 3 and CG models in general (**section G4**). Below, we provide more details for all of these aspects. The reader should keep these limitations of transferability of Martini 3 in mind when using the new version of the model.

G1) Short bond lengths: effects on oil/water partitioning

The usage of bead sizes in the previous version of Martini was only connected to the topology of the molecules: linear or branched chains were modeled with regular beads while rings used smaller beads. Given that the parametrization at that point was mainly carried out using linear molecules, 4-1 mapping, regular beads, and a number of standard bonded parameters, artifacts were noticed for molecules deviating from these conditions, but using the same building blocks². In particular, Martini 2 models that use shorter bond lengths than the standard values (around 0.47 nm) show deviations in the expected trends of oil/water transfer free energies ($\Delta G_{\text{oil} \rightarrow \text{W}}$). The CG models with short bond lengths (lower than 0.40 nm) usually involve molecules that are overmapped (i.e., use regular beads for 3-1 or 2-1 mapping) and/or branched. Experimental evidence indicates that these smaller molecules are always more hydrophilic than linear molecules with the same chemical group (i.e. same/similar bead chemical type). For water/hexadecane $\Delta G_{\text{oil} \rightarrow \text{W}}$, Martini 2 qualitatively reproduces this trend, but shows substantial deviations in the extremes of the hydrophobicity scale: smaller hydrophobic molecules tend to be too hydrophobic, while small hydrophilic ones too hydrophilic. Quantitative empirical observation indicates the opposite behavior: hydrophobic molecules get from 3.0 to 3.5 kJ/mol more hydrophilic for each aliphatic carbon atom removed, while hydrophilic molecules get a somewhat smaller free energy gain (2–2.5 kJ/mol.)². The discrepancies

related to short bond lengths are drastic when one of the solvent phases consists of a CG model with short bond lengths, such as benzene. In this case, water/benzene $\Delta G_{oil \rightarrow W}$ show the opposite trend in all the studied cases, with smaller molecules more hydrophobic than the bigger ones with the same chemical group. Although benzene is modeled by S-beads in Martini 2¹³⁷, the interaction between R and S beads was the same as the ones used in R-R interactions (see some comments about this feature in **section B1**). More details about these findings and the reasons behind these problems connected to the bond lengths are described in the work of Alessandri *et al.*².

In Martini 3, these discrepancies are largely solved, as shown in **Supplementary Figure 21**. Models with bond lengths shorter than 0.47 nm are more hydrophilic, with the stronger effect in hydrophobic molecules, as observed in experimental data². The key reasons for these improvements in Martini 3 are the new interaction balance of organic beads in different sizes (see **section B2**) and new mapping rules (see **section C1**), which changed the usage of bead sizes: they not only consider the topology, but also consider the number of non-hydrogen atoms mapped by the CG bead. So, in **Supplementary Figure 21**, a two-bead model with bond length 0.47 nm is represented by two R-beads while 0.40, 0.35, and 0.25 nm use R-S, S-S and T-T bead sizes, respectively.



Supplementary Figure 21: Bond lengths and partitioning behavior of molecules upon removal of aliphatic carbon atoms for the same chemical functional group. (A) Water/hexadecane and (B) water/benzene transfer free energies ($\Delta G_{oil \rightarrow W}$) for a Martini two-bead molecule (bond length of 0.47 nm) are plotted against the changes of $\Delta G_{oil \rightarrow W}$ upon bond length reduction to 0.40 nm (green circles), 0.35 nm (blue squares), and 0.25 nm (purple triangles), corresponding to the removal of 1, 2, and 4 aliphatic carbon atoms (that is, to 3.5-1, 3-1, and 2-1 atoms-to-CG sites mapping schemes, respectively).

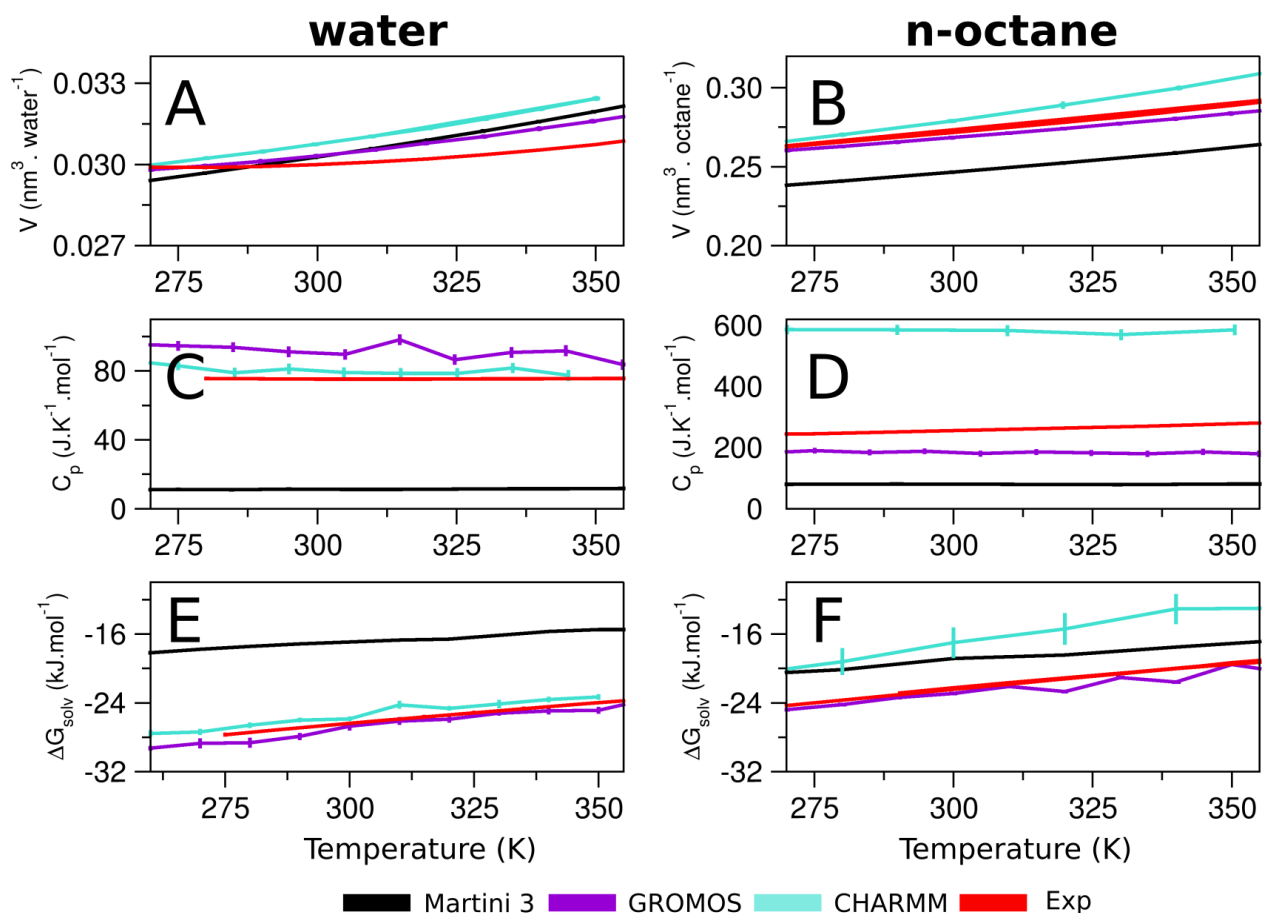
Methods used to compute $\Delta G_{\text{oil} \rightarrow \text{W}}$ are described in **section D1**. The system setup is the same as described in Alessandri *et al.*². Other pitfalls presented in the work of Alessandri *et al.*² were also covered and improved in Martini 3. They are displayed in other parts of this paper. For instance, regarding the artificial free energy barrier and desolvation problem, see **Figure 1D** and its discussion in the main paper. For the excessive self-interactions and vaporization enthalpies of molecules with short bond lengths, see **section E3**. For the excessive aggregation of polyleucines, see **Figure 3D** and its discussion in the main paper.

G2) Temperature dependence of pure solvent systems

In this section, a number of results pertaining to solvent properties in the Martini 3 force field were collected, compared to atomistic models and experimental data and discussed. The main focus will be on the temperature dependence, i.e. the temperature transferability of the models. Properties were calculated for pure water and pure n-octane in the default 4-1 resolutions: W bead for water and C1-C1 for n-octane. The atomistic models SPC¹⁷⁹ and TIP3P¹⁹⁰ for water and the GROMOS 54a7¹⁷⁷ and CHARMM36m⁴⁴ force fields were also considered for comparison. These were simulated with settings commonly used in GROMOS (reaction field electrostatics, coulomb and LJ cutoffs of 1.4 nm) and CHARMM (PME, and LJ cutoff of 1.2 nm, force switch modifier starting from 1.0 nm) model simulations, respectively. The main experimental references were the NIST Chemistry WebBook¹⁹¹ and the work of Banipal *et al.*¹⁹².

Supplementary Figures 22A and 22B show molecular volume in the different models as a function of temperature for water and n-octane, respectively. In the case of the Martini 3 water model, the volume per water bead is divided by 4, for comparison. The volumes of all water models are similar in the region around room temperature (300 K) and match the experiment well. Atomistic and Martini 3 water models show a small discrepancy in the volume heat expansion coefficients (α_p , related to the slope of the curve) in relation to the experimental results. For n-octane, it is observed that the molecular volume in Martini 3 underestimates the volume compared to experimental data (**Supplementary Figure 22B**). It is worth noticing that the volumes of the two atomistic models differ considerably at room temperature, with the GROMOS model being closer

to experiment¹⁹¹. A reduction in epsilon of the LJ potential between the C1 beads of n-octane could in theory improve Martini 3. In relation to Martini 2, this is already the case, with epsilon reduced from 3.50 to 3.39 kJ/mol in the new version. However, preliminary tests performed during the parametrization indicate that reducing the epsilon even more makes alkane systems as n-octane go the gas phase at room temperature. In addition, other systems and properties tend to behave worse, including lipid bilayers. Although molecular volumes are not a perfect match in this case, the volume heat expansion coefficients of the Martini 3 n-octane compare well to that of the GROMOS model and the experimental data.



Supplementary Figure 22: Volume per molecule (V , top panels), heat capacity (C_p , middle panels) and solvation free energy (ΔG_{solv} , bottom panels) as a function of temperature for Martini 3 (black), CHARMM/TIP3P (cyan) and GROMOS/SPC (purple) for two solvents: water (left panels) and n-octane (right panels). Experimental data (red) are also shown.

The (constant pressure) heat capacity (C_p) of water models, which was calculated from finite differences of the enthalpy data, compares best to the experimental one for the TIP3P model as can be seen in **Supplementary Figure 22C**. The heat capacity of the SPC water model is somewhat too large. The heat capacity is considerably smaller for the Martini 3 water model than for the atomistic models and experiment, which is likely due to the fact that Martini water represents 4 water molecules. The behavior of bundles of 4 molecules should be investigated to judge the deviations more carefully. The C_p for n-octane is shown as a function of temperature in **Supplementary Figure 22D**. The results show that all computational models have nearly constant C_p in the temperature range studied, in contrast to the increasing C_p in the experiment. The models show increasing C_p with increasing number of degrees of freedom. The Martini 3 and GROMOS models underestimate the experimental C_p which can be understood in terms of the lower number of particles per molecule of n-octane, but this does not explain the gross overestimation of the C_p of the CHARMM36m model. The latter is explained by the large number of vibrations that contribute R per vibration to the C_p in a classical mechanical treatment by the equipartition theorem. Accounting for the fact that only C-H bond stretches are constraint, the vibrational contribution is approximately 450 J/(mol·K). Subtracting this number takes the C_p close to the experimental value. A recent paper studied n-octane in different atomistic force fields as a function of temperature¹⁹³ and concluded that in general, united-atom models represent it better than all-atom models.

Supplementary Figures 22E and 22F show the solvation Gibbs energy (ΔG_{solv}) of water and octane in the different models as a function of temperature. ΔG_{solv} is the equilibrium work to take one molecule (or bead) from a position in the vapor (non-interacting) to a position in the liquid, see the work of Ben-Naim^{72,194}. The solvation free energies were calculated using a thermodynamic integration⁵⁷ (TI) approach (as described in **section D1**), switching off the intermolecular interactions between the solute and the solvent at constant pressure. For a pure solvent, one solvent molecule (or bead in case of the Martini 3 models) is chosen as the solute. So, the ΔG_{solv} energies for water in water are calculated per bead for the Martini 3 models and per water molecule for the atomistic models. The experimental data are calculated from the vapor pressure of water

(p^*) and the number density of liquid water (ρ_l), according to the expression given by Ben-Naim (**Equation 17**):

$$G_{solv} = -RT \ln \left(\frac{\rho_l}{\rho_v} \right) = -RT \ln \left(\frac{p^*}{k_B T \rho_l} \right) \quad \text{Equation 17}$$

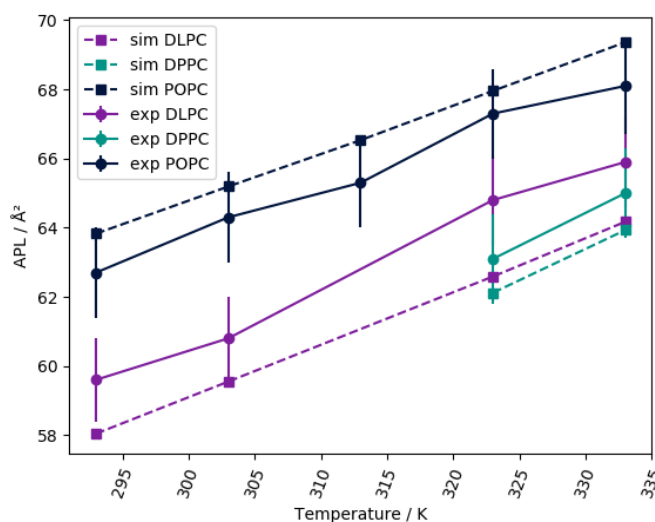
The vapor was treated as a perfect gas, i.e. the interactions of water in the vapor were not taken into account. **Supplementary Figure 22E** shows a clear difference in the ΔG_{solv} between the Martini 3 models and atomistic models. The atomistic models differ somewhat amongst each other, but nicely bracket the experimental data, calculated using compilations provided by NIST¹⁹¹. Proper comparison with the Martini 3 model should consider the enthalpy of groups of 4 water molecules taken from a bulk simulation of atomistic water. The process for the Martini 3 beads represents evaporation of clusters of 4 molecules. These bundles retain cohesive energy and are therefore expected to have a less negative free energy of solvation. Such comparison was done for the bundled water molecule¹⁹⁵, and this comparison for 4-1 mapped water revealed that the solvation free energy of water in water in the original Martini model was very reasonable at room temperature: the excess free energy of the original Martini water in its own liquid at 303 K was reported as 18 kJ/mol, close to the excess free energy of 19 kJ/mol of taking a bundle of 4 SPC water molecules from a liquid consisting of bundled SPC water molecules. See more about the “bundled water model” in the work of Fuhrmans *et al.* 2009¹⁹⁶. Interestingly, the slopes of the free energy of solvation with temperature are similar. Thus, the entropy of solvation, which is the negative slope of the free energy of solvation with temperature is similar for all models. This implies that the differences in the free energy values themselves is predominantly a reflection of the differences in enthalpy per molecule of the models.

Supplementary Figure 22F shows the solvation Gibbs energy of n-octane in n-octane in the different models as a function of temperature. The experimental data¹⁹¹ are calculated using two parameterizations of the Antoine equation¹⁹⁷ for the vapor pressure of pure n-octane. Comparison of the Martini 3 models to the atomistic models shows that the regular Martini 3 model is in between GROMOS and CHARMM. GROMOS clearly presents the best correspondence to the experimental data. In terms of the enthalpy-

entropy balance, the entropies of solvation are similar for all models as can be seen from the similar slopes for all models, although the atomistic force fields are more sensitive to temperature than the Martini 3 models. Thus, as seen for water, the differences in the free energies of solvation are primarily due to the differences in enthalpy of the different models.

G3) Temperature dependence of lipid bilayers

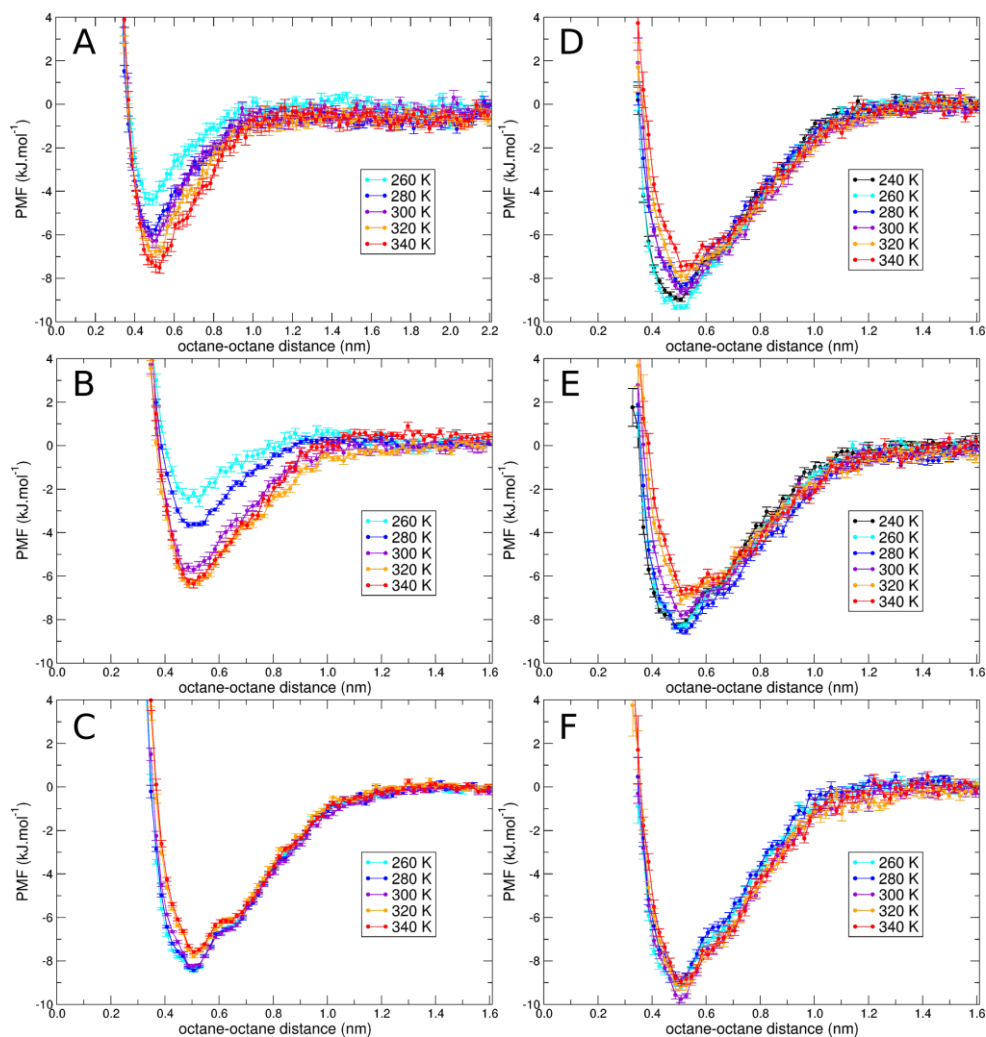
The area per phospholipid is well known to vary with temperature from experiments^{152,198}. We have verified that Martini 3 follows the temperature dependence of this bilayer property by comparing our simulations to the known values of three chosen phospholipid bilayers in **Supplementary Figure 23**. The slopes of the relation between the calculated areas and the temperature are in excellent agreement with experiments. While most values also fall well within the error estimates of the experimentally determined areas, there is still room for improvement involving further optimization of the bonded parameters of the phospholipids. Methods used are described in **section D6**.



Supplementary Figure 23: The area per lipid of pure fluid bilayers composed of three chosen phospholipids (DLPC, DPPC and POPC) from simulations (dashed lines, squares) is plotted as a function of temperature and compared to experimentally determined values^{152,198} (solid line, circles).

G4) Hydrophobic effect and entropy

In this section the hydrophobic interaction is assessed, i.e. the interaction between two hydrophobic molecules (in this case n-octane) in aqueous solution. It is assessed by comparing potentials of mean force (PMFs) between two n-octane molecules in water solvent at different temperatures. The results comparing atomistic and Martini CG models are presented in **Supplementary Figure 24**.



Supplementary Figure 24: Potential mean force (PMF) of octane-octane interaction as a function of temperature for Martini coarse-grained models and atomistic models: (A) GROMOS 54a7 model of n-octane in SPC water, (B) CHARMM36m model of n-octane in TIP3P water, (C) standard (4-to-1) Martini 3 model of n-octane in regular (4-to-1) Martini 3 water, (D) standard (4-to-1) Martini 3 model of n-octane in 3-to-1 mapped Martini 3 water, (E) standard (4-to-1) Martini 3 model of n-octane in 2-to-1 mapped Martini 3 water, (F) Martini 2 model of n-octane in the polarizable water.

It can be seen from **Supplementary Figure 24A** that the interaction between two n-octane molecules in water at the united-atom model level of GROMOS becomes stronger at higher temperature. This is considered a hallmark of the hydrophobic interaction^{199–202}. The effect is strongest at the lower temperatures, and levels off above 300 K. The PMFs for two n-octane molecules in water in the all-atom CHARMM36m force field (**Supplementary Figure 24B**) are comparable to those in the GROMOS 54a7 force field. The binding is stronger at higher temperature. The minima are somewhat deeper at higher temperatures and the temperature effect becomes relatively small above 280 K instead of 300 K. It can be seen in **Supplementary Figure 24C** that in the standard Martini 3 model, the interaction between two n-octane molecules in water is only weakly temperature dependent. The interaction appears to be somewhat weaker at higher temperatures. This is in contradiction to the atomistic reference, showing clearly that Martini 3 cannot properly model the hydrophobic interaction. Direct comparisons in the physiological temperature range at 300 and 320 K shown in **Supplementary Figures 24A–24E**, respectively, shows that the strength of the interaction is 1–2 kJ/mol too strong in the Martini 3 models compared to GROMOS and CHARMM36m, but the balance of interactions leads to very similar behavior in the different water resolutions of Martini 3. The correspondence is also better at 320 K than at 300 K due to the relatively weak temperature dependence in the Martini 3 model versus the atomistic models in which the interaction is stronger at higher temperature.

It can be seen in **Supplementary Figure 24F** that in the polarizable Martini 2 model⁵ the interaction between two n-octane molecules in water is also only weakly temperature dependent. Interestingly, the interaction is strongest at 300 K, and a little weaker both at higher and at lower temperatures. Further improvements in polarizable Martini water models^{5,6} maybe allow future Martini models capture the overall trend observed in atomistic models. Another option would be to develop a Martini model with temperature dependent potentials, as already demonstrated in bottom-up CG models²⁰³.

Lastly, it is worth mentioning that the tests performed here represent the behavior of n-octane in water in the limit of very low dilution. As such, they are only valid for this specific state. At higher concentrations, experiments demonstrate that increasing the

temperature leads to higher miscibility in water/alkane biphasic mixtures, reducing the interfacial tension¹⁴¹ and increasing the solubility of alkanes²⁰⁴ longer than 5 carbon atoms in water. For long chains, the cavity costs win in relation to the hydrophobic effect, explaining this trend²⁰⁵. This is the behavior observed in Martini models, as demonstrated by Ndao et al.⁷⁶ where higher temperatures reduce the interfacial tension and increases the mixing of water/oil biphasic systems. It also explains why trends in membrane bilayers with higher temperatures (**section G3**) are well captured.

Supplementary Codes H) Tools and examples of usage

The Martini 3 force field comes with updates of several of the tools that were developed to assist users in deploying the models, notably *martinize* for coarse graining atomistic structures, and *martinate* for running and managing simulations.

H1) Martinize2

The Martini 3 force field is fully integrated in *martinize2*, which replaces the earlier *martinize* program³⁶ to create simulation input files, both structures and topologies, from atomistic structures. The program and the underlying library *vermouth* are currently being finalized and will be described in detail elsewhere (to be published in Kroon *et al.* 2020). In short, *martinize2* separates the parameters within residues (fragments) from those bridging residues (links) and chooses the correct combinations using graph-based algorithms. This allows it to generate simulation input files for arbitrary polymers (linear, branched, cyclic, etc.) using a single set of simple data files. The program is capable of generating atomistic, united atom and coarse-grain models using a combination of subgraph matching and resolution mapping. It is worth noting that all data files used by the program are human readable and writable.

By separating the data from the algorithms, *martinize2* allows for rapid exploration of simulation parameters, and facilitates the setup of simulations of complex system, which are up until now out of reach. In this work, all protein simulations were set up using *martinize2* and the program was used for effective exploration of protein parameters. The *martinize2* tool and the supporting *vermouth* library can be obtained and installed from the the github repository <https://www.github.com/marrink-lab/vermouth-martinize> or through the PyPi repository using

```
pip install vermouth
```

H2) Martinate

Martinate is a flexible, automated workflow for running Martini based simulations. The program was introduced for facilitating running multiscale simulations²⁰⁶ and was

later used as basis for running high-throughput simulations with the DAFT protocol²⁰⁷. To support deployment of the Martini 3 force field and allow integration in parameter exploration/testing pipelines, several changes were made to the program. In particular, the program can now make use of force field files that are provided by the user, rather than only those distributed with the program.

Martinate is distributed together with the atomistic counterpart *gromit* and can be obtained from github (<https://www.github.com/marrink-lab/gromit>). The program requires a UNIX/Linux environment with GROMACS^{11,12}, *martinize2* and *insane*⁸². The latter is needed for building lipid bilayers as well as for setting up PBC and generating the solvent. The basic usage of *martinate* allows simple setup and running of coarse grain simulations with the Martini force field. The current default is Martini 2.2, which is distributed with the program. In its simplest form it is sufficient to provide the program with a PDB file. If the PDB ID is known, it can be downloaded automatically using the -fetch option:

```
martinate.sh -f 1ubq.pdb -fetch pdb
```

The fetch option also allows downloading membrane proteins from the OPM database²⁰⁸. The *martinate* command-line further allows specifying the membrane to be built by *insane* directly:

```
martinate.sh -f 2oar.pdb -fetch opm --insane-d=5 --insane-l=DPPC --  
insane-sol=W
```

To use *martinate* with *martinize2*, which is necessary for using the Martini 3 force field, the *martinize* executable can be specified (and similarly -insane works for specifying an alternative *insane* executable):

```
martinate.sh -f 2oar.pdb -fetch opm --insane-d=5 --insane-l=DPPC --  
insane-sol=W -martinize /path/to/martinize2
```

As the Martini 3 force field is currently distributed separately, using it with *martinate* additionally requires specifying the location of the force field files and the version, like

```
martinate.sh -f 2oar.pdb -fetch opm --insane-d=5 --insane-l=DPPC --  
insane-sol=W -martinize /path/to/martinize2 -ffdir /path/to/martini3 -  
cg martini304
```

The Martini 3 directory needs to have subdirectories */force_fields* and */mappings* if the input file is to be processed by *martinize2*. Furthermore, topology definitions are required for any lipids, solvent, ions, carbohydrates, etc. that are to be present in the simulation system. The Martini 3 force field comes with possibilities for amending topologies, which are currently not supported by *martinize2*, but which require post-hoc processing. Although this cannot be directly incorporated into the *martinate* workflow, the program does support stopping the protocol at a specific step and continue from that step at a later time. This feature was added to support running part of the workflow locally and finalize on a different machine (cluster), but can be used to make changes to the structure and topology, like generating interactions for GoMartini. The workflow would then look like

```
martinate.sh -f 2oar.pdb -fetch opm --insane-d=5 --insane-l=DPPC --  
insane-sol=W -martinize /path/to/martinize2 -ffdir /path/to/martini3 -  
cg martini304 -stop CG
```

<SET UP GoMartini>

```
martinate.sh -f 2oar.pdb -fetch opm --insane-d=5 --insane-l=DPPC --  
insane-sol=W -martinize /path/to/martinize2 -ffdir /path/to/martini3 -  
cg martini304
```

Provided the structures and topologies are modified in-place, keeping the original names, in the second pass the protocol will verify that up to the generation of the CG

topology all files are present and continue running from the following step. For a complete reference of the options of *martinate*, the full help can be obtained using

```
martinate.sh --help
```

The program allows specifying virtually all options of the downstream programs *martinize* and *insane*, as well as of the GROMACS programs used. The help sections of those programs can be consulted for information on those.

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Epilogue

Recipe of the official Martini 3 cocktail

by Pim W.J.M. Frederix

1. Prepare a cucumber syrup by pureeing a cucumber and adding half-3/4 of its weight in sugar, depending on your sweet tooth. Dissolve and homogenize by boiling briefly.
2. Add 0.5 units of Martini extra dry vermouth and 2 units of good quality vodka to a cocktail shaker with ice cubes. Shake briefly but vigorously.
3. Add 0.5-1 tablespoon of cucumber syrup to the bottom of a Martini glass.
4. Use a bar spoon (or teaspoon) to carefully pour the Martini mixture on top of the cucumber liquid. While phase separation will occur spontaneously (try a concentrated sugar solution and an alcohol solution using the force field), it is easier to keep the phases separated to begin with. The ice should remain in the shaker.
5. Serve with a cocktail pick or rosemary twig decorated with a cucumber, olive and caper bead and watch the Regular, Small and Tiny beads partition over the two phases.

